

Fermi Notes on Physics

Volume Three

Introductory Undergraduate Lectures on Quantum Mechanics

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Lectures on Quantum Mechanics

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Origins of Quantum Theory

1.1 Quantum Physics at School

The quantum physics taught at school rests on a handful of experimental results about light and electrons, together with the relation that connects the energy of light to its frequency. You have met all of this material before, so this section states it briefly and fixes the notation that the course will use for it. The two sections that follow go further into the experiments behind these results. Section 1.2 looks at the spectrum of thermal radiation, where the energy of light was first found to come in packets, and at the scattering of light by electrons, which shows that those packets also carry momentum. Section 1.3 turns to the spectrum of hydrogen, to Bohr's model of the atom, and to the evidence that electrons behave as waves.

Photons

Light of frequency f is absorbed and emitted in packets called photons, each carrying an energy

$$E = hf, \tag{1.1}$$

where h is Planck's constant. Because the wavelength and the frequency of light in a vacuum are related by $c = f\lambda$, the energy of a photon can also be written as $E = hc/\lambda$, so short-wavelength

light, such as ultraviolet light or X-rays, comes in large packets and long-wavelength light, such as radio waves, in small ones.

Energies on the atomic scale are usually quoted in electronvolts. An electronvolt is the energy gained by an electron, of charge $-e$, when it moves through a potential difference of one volt, so that $1 \text{ eV} = e \times 1 \text{ V}$.

The Photoelectric Effect

When light falls on a clean metal surface, electrons can be ejected from it. The experiments on this photoelectric effect give the following results:

1. no electrons are emitted, however intense the light, unless the frequency of the light is above a threshold frequency f_0 that depends on the metal;
2. above the threshold, the maximum kinetic energy of the emitted electrons increases linearly with the frequency and does not depend on the intensity;
3. the number of electrons emitted per second is proportional to the intensity;
4. emission begins as soon as the light is switched on, with no measurable delay, even when the light is very faint.

A wave picture of light, in which the energy arrives spread smoothly over the surface, cannot account for the first, second, or fourth of these results. Einstein explained all four by supposing that each emitted electron has absorbed a single photon. An electron needs at least an energy W , called the work function of the metal, to escape from the surface, and whatever is left over appears as kinetic energy. The most energetic electrons therefore have

$$E_{\text{k,max}} = hf - W, \quad (1.2)$$

which is called Einstein's photoelectric equation. The threshold frequency is $f_0 = W/h$. The maximum kinetic energy is measured by making the collecting electrode negative with respect to the emitting surface and increasing this retarding potential difference until the current stops. If the current stops at the stopping potential

V_s , then $eV_s = E_{k,\max}$, and a graph of V_s against f is a straight line whose gradient is h/e .

Energy Levels and Line Spectra

The electrons in an atom can have only certain energies, called the energy levels of the atom. The lowest is called the ground state and the others are called excited states. An atom can be excited by a collision, for instance with an electron in a discharge tube, or by absorbing a photon, and it returns to lower levels by emitting photons. When an atom drops from a level of energy E_i to a lower level of energy E_f , the photon emitted has

$$hf = E_i - E_f. \quad (1.3)$$

Since the levels are discrete, the light emitted by a hot, low-density gas contains only certain frequencies, and its spectrum is a set of bright lines characteristic of the element. White light passed through a cool gas loses the same frequencies, because the atoms absorb photons whose energies match the differences between their levels, and the spectrum of the transmitted light has dark lines in the same places. The energy needed to remove an electron completely from the ground state is called the ionisation energy.

Matter Waves

De Broglie proposed that a particle of momentum p has a wave associated with it, of wavelength

$$\lambda = \frac{h}{p}, \quad (1.4)$$

which is called the de Broglie wavelength. For an electron accelerated from rest through a potential difference V , the kinetic energy is eV , the momentum is $p = \sqrt{2m_e eV}$, and the wavelength is $h/\sqrt{2m_e eV}$, where m_e is the mass of the electron. The evidence for the wave is diffraction: a beam of electrons passed through a thin film of graphite produces rings on a screen, just as X-rays do, and the radius of the rings decreases as the accelerating voltage is raised, in the way the formula for λ predicts. Light and electrons both show wave properties in some experiments and particle properties in others, and this dual behaviour is called wave-particle duality.

1.2 Thermal Radiation and the Photon

The relation $E = hf$ was first introduced by Planck in 1900, to explain the spectrum of the radiation given off by hot bodies. We begin with that spectrum, since it is where the classical theory of light first failed, and then show that a photon carries momentum as well as energy, and that this momentum can be measured by scattering X-rays from electrons. Blackbody radiation is treated in detail in Section 5.6 of Vol. 2 of the Fermi Notes on Physics, where the spectrum is derived from statistical mechanics.

Blackbody Radiation

Every body at a temperature T emits electromagnetic radiation, and the spectrum of that radiation depends on the body's surface as well as on its temperature. The dependence on the surface disappears for an ideal absorber, a body that absorbs all the radiation falling on it, which is called a black body. A good practical black body is a closed cavity with a small hole in its wall. Radiation entering the hole is reflected many times inside and is almost certainly absorbed before it can find its way out again, and the radiation that leaves through the hole is characteristic of the temperature of the walls alone. You may have seen at school that the power radiated per unit area of a black body is proportional to T^4 , which is Stefan's law, and that the wavelength at which the spectrum peaks is inversely proportional to T , which is Wien's displacement law.

Let $u(f) df$ be the energy per unit volume of the radiation inside the cavity with frequencies between f and $f + df$. The classical theory of the radiation, which treats each standing wave in the cavity as an oscillator with the mean energy $k_B T$ that the equipartition principle of Vol. 2 of the Fermi Notes on Physics assigns to it, where k_B is Boltzmann's constant, gives

$$u(f) = \frac{8\pi f^2}{c^3} k_B T, \quad (1.5)$$

which is called the Rayleigh–Jeans law. The factor $8\pi f^2/c^3$ is the number of standing waves per unit volume per unit frequency, which is counted in Section 5.6 of Vol. 2 in terms of the angular frequency $\omega = 2\pi f$, where it comes to $\omega^2/\pi^2 c^3$ per unit angular frequency, and it grows with frequency because there are more ways of fitting

short waves into a box than long ones. The Rayleigh–Jeans law agrees with measurements at low frequencies. At high frequencies, however, it predicts an energy density that grows without limit, and the total energy $\int_0^\infty u(f) df$ is infinite. The measured spectrum instead rises to a peak and then falls towards zero. This failure of the classical theory at high frequency is called the ultraviolet catastrophe.

Planck found that the measured spectrum is described by

$$u(f) = \frac{8\pi f^2}{c^3} \frac{hf}{e^{hf/k_B T} - 1}, \quad (1.6)$$

which is called Planck’s law. Planck obtained it by supposing that an oscillator of frequency f can hold energy only in whole multiples of hf , which gives the oscillator a mean energy $hf/(e^{hf/k_B T} - 1)$, as worked out in Section 5.2 of Vol. 2 for the vibration of a molecule. When the temperature is low compared with hf/k_B , the chance that an oscillator has even a single unit of energy is small, and the high-frequency oscillators hold almost nothing. This is why the spectrum falls at high frequency.

We can check that Planck’s law reduces to the Rayleigh–Jeans law at low frequency. When $hf \ll k_B T$, the exponential can be expanded as a Taylor series, $e^{hf/k_B T} = 1 + hf/k_B T + \dots$, and keeping only the first two terms gives

$$\frac{hf}{e^{hf/k_B T} - 1} \approx \frac{hf}{hf/k_B T} = k_B T, \quad (1.7)$$

so each oscillator has the classical mean energy, and the Rayleigh–Jeans law is recovered, as Figure 1 shows. In the opposite limit, $hf \gg k_B T$, the 1 in the denominator can be neglected, and $u(f)$ goes as $f^3 e^{-hf/k_B T}$, which falls rapidly as the frequency grows.

Stefan’s Law and Wien’s Law from Planck’s Law

Planck’s law contains both Stefan’s law and Wien’s displacement law. To see how, let $\eta = hf/k_B T$, so that $f = k_B T \eta / h$ and $df = (k_B T / h) d\eta$. The total energy per unit volume is then

$$\int_0^\infty u(f) df = \frac{8\pi(k_B T)^4}{h^3 c^3} \int_0^\infty \frac{\eta^3}{e^\eta - 1} d\eta. \quad (1.8)$$

The integral on the right is a pure number, whose value, $\pi^4/15$,

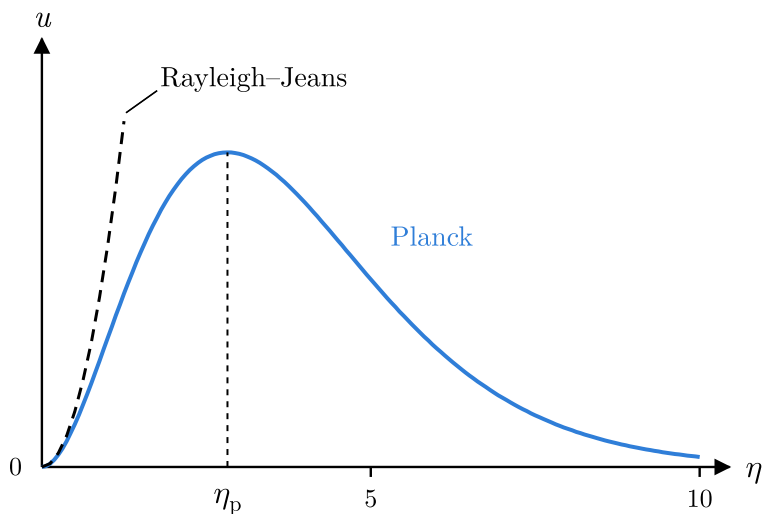


Figure 1. The energy density of blackbody radiation per unit frequency, plotted against $\eta = hf/k_B T$ in units of $8\pi(k_B T)^3/(h^2 c^3)$. The solid curve is Planck's law, $\eta^3/(e^\eta - 1)$, and the dashed curve is the Rayleigh-Jeans law, η^2 , which agrees with Planck's law only at small η .

is worked out in Section 5.5 of Vol. 2, and all the dependence on temperature sits in the factor in front, which goes as T^4 . The power radiated per unit area through the hole in the cavity is $c/4$ times the energy density, as shown in Section 5.6 of Vol. 2, so it too goes as T^4 . This is Stefan's law.

The peak of the spectrum is found by differentiating. In terms of η , the spectrum is proportional to $\eta^3/(e^\eta - 1)$, and setting the derivative of this function to zero gives, by the quotient rule,

$$\frac{3\eta^2(e^\eta - 1) - \eta^3 e^\eta}{(e^\eta - 1)^2} = 0, \quad (1.9)$$

which requires the numerator to vanish, and dividing the numerator by $\eta^2 e^\eta$ gives

$$\eta = 3(1 - e^{-\eta}). \quad (1.10)$$

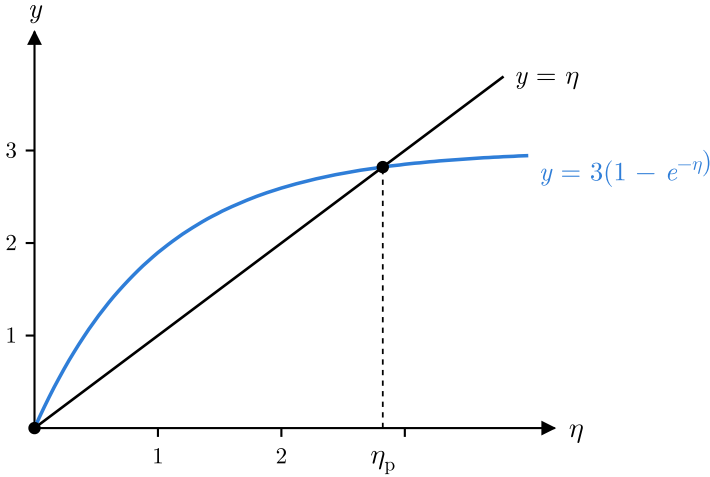


Figure 2. The graphical solution of $\eta = 3(1 - e^{-\eta})$. The line $y = \eta$ and the curve $y = 3(1 - e^{-\eta})$ cross at the origin and at $\eta = \eta_p$, which fixes the position of the peak in Figure 1.

This is a transcendental equation, one in which the unknown appears both by itself and inside a function such as an exponential. An equation of this kind cannot in general be solved by rearranging it, and is usually solved numerically or graphically. Figure 2 shows the graphical solution. The straight line $y = \eta$ and the curve

$y = 3(1 - e^{-\eta})$ cross at $\eta = 0$, which is the minimum of the spectrum at zero frequency, and at one other point, η_p , which is a pure number a little below 3. The frequency at the peak is therefore

$$f_{\max} = \frac{\eta_p k_B T}{h}, \quad (1.11)$$

which is proportional to T , and this is Wien's displacement law for the spectrum per unit frequency. The school form of the law is stated for the spectrum per unit wavelength, which peaks at a different place, and you will work out that form in Problem 2.

The Momentum of a Photon

Light carries momentum as well as energy, and this was known from classical electromagnetism, which gives the momentum of a pulse of light as its energy divided by c . The same relation follows for a single photon from special relativity. A particle of mass m , energy E , and momentum p satisfies

$$E^2 = p^2 c^2 + m^2 c^4, \quad (1.12)$$

which holds at any speed. For a slowly moving particle it reduces to the rest energy mc^2 plus the familiar kinetic energy $p^2/2m$, as you will show in Problem 3. A photon has no mass, so $E = pc$, and its momentum is

$$p = \frac{E}{c} = \frac{hf}{c} = \frac{h}{\lambda}. \quad (1.13)$$

This is the de Broglie relation of Section 1.1 applied to light, so the relation holds for light as well as for matter.

Compton Scattering

In 1923 Compton directed a beam of X-rays of a single wavelength λ at a block of graphite and measured the wavelength of the X-rays scattered through an angle Θ . The scattered beam contained, besides radiation at the original wavelength, radiation at a longer wavelength λ' , and the difference $\lambda' - \lambda$ increased with the angle and did not depend on the material. A classical electromagnetic wave driving an electron makes the electron radiate at the frequency of the wave, so the classical picture produces no such shift.

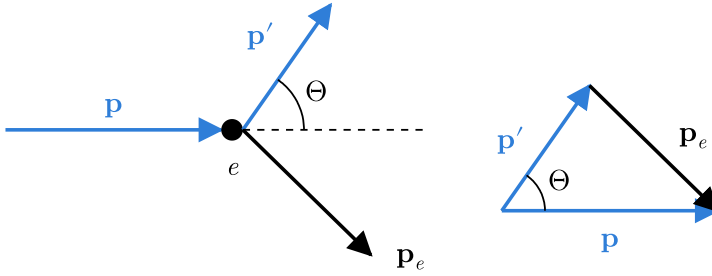


Figure 3. Compton scattering. A photon of momentum $\mathbf{p}$ strikes an electron at rest and is scattered through an angle Θ with momentum $\mathbf{p}'$. The electron recoils with momentum $\mathbf{p}_e$, and the three momenta form the triangle on the right, in which $\mathbf{p}_e = \mathbf{p} - \mathbf{p}'$.

Compton explained the result as a collision between a single photon and a single electron. The outer electrons in graphite are so weakly bound, compared with the energy of an X-ray photon, that we may treat them as free and at rest. The collision is shown in Figure 3. A photon of momentum $\mathbf{p}$ strikes an electron at rest and leaves with momentum $\mathbf{p}'$, at an angle Θ to its original direction, and the electron recoils with momentum $\mathbf{p}_e$. Conservation of momentum gives

$$\mathbf{p}_e = \mathbf{p} - \mathbf{p}', \quad (1.14)$$

and taking the scalar product of each side with itself gives

$$p_e^2 = p^2 + p'^2 - 2pp' \cos \Theta. \quad (1.15)$$

Conservation of energy, with the rest energy of the electron included, gives

$$pc + m_e c^2 = p'c + E_e, \quad (1.16)$$

where $E_e = \sqrt{p_e^2 c^2 + m_e^2 c^4}$ is the energy of the recoiling electron. Rearranging the energy equation to $E_e = (p - p')c + m_e c^2$ and squaring both sides gives

$$p_e^2 c^2 + m_e^2 c^4 = (p - p')^2 c^2 + 2(p - p')m_e c^3 + m_e^2 c^4. \quad (1.17)$$

The terms $m_e^2 c^4$ cancel. Dividing by c^2 and substituting the expression for p_e^2 from the conservation of momentum gives

$$p^2 + p'^2 - 2pp' \cos \Theta = p^2 + p'^2 - 2pp' + 2(p - p')m_e c, \quad (1.18)$$

which can be rewritten as

$$pp'(1 - \cos \Theta) = (p - p')m_e c. \quad (1.19)$$

Dividing both sides by $pp'm_e c$ gives $1/p' - 1/p = (1 - \cos \Theta)/m_e c$. Since $1/p = \lambda/h$ for a photon, this is

$$\lambda' - \lambda = \frac{h}{m_e c} (1 - \cos \Theta), \quad (1.20)$$

which is called the Compton formula. The length $h/m_e c$ is called the Compton wavelength of the electron. The shift is zero in the forward direction and largest, at twice the Compton wavelength, for a photon scattered straight back. It depends only on the angle and on the mass of the particle that does the scattering, which is why it is the same for every material. The unshifted line comes from photons scattered by the inner electrons, which are bound so tightly that the photon in effect recoils from the whole atom, and the mass in the formula is then so large that the shift is too small to see.

Compton's measurements agreed with this formula, and they were among the strongest evidence that the photon carries momentum h/λ and behaves in a collision as a particle does.

1.3 The Bohr Model and Matter Waves

The line spectrum of hydrogen, the simplest atom, was measured with high precision in the nineteenth century, long before anyone could explain it. In this section we first describe the spectrum and the nuclear model of the atom, then work out Bohr's model of hydrogen, and finally look at the experiments showing that electrons behave as waves, which give Bohr's condition a physical meaning.

The Spectrum of Hydrogen

In 1885 Balmer found that the wavelengths of the visible lines of hydrogen fit the formula $\lambda = \lambda_\infty n^2/(n^2 - 4)$ for $n = 3, 4, 5, \dots$, with a single constant λ_∞ , which is the wavelength to which the lines

crowd as n grows. Rydberg later wrote the result in a more general form, which covers the lines that were subsequently found in the ultraviolet and the infrared:

$$\frac{1}{\lambda} = \mathcal{R}_H \left(\frac{1}{n_f^2} - \frac{1}{n_i^2} \right), \quad (1.21)$$

in which n_f and n_i are positive integers with $n_i > n_f$, and $\mathcal{R}_H$ is a constant with the dimensions of an inverse length. The lines with $n_f = 1$ form the Lyman series in the ultraviolet, those with $n_f = 2$ form the Balmer series in the visible, and those with $n_f = 3$ form the Paschen series in the infrared. Multiplying the formula by hc turns it into a statement about photon energies. Each photon energy is the difference between two numbers of the form $-hc\mathcal{R}_H/n^2$, and the relation $hf = E_i - E_f$ of Section 1.1 then says that the energy levels of hydrogen are

$$E_n = -\frac{hc\mathcal{R}_H}{n^2}, \quad n = 1, 2, 3, \dots \quad (1.22)$$

The spectrum fixes the levels only up to a common additive constant, and the constant has been chosen so that the level with $n \rightarrow \infty$, at which the electron is just free of the atom, has zero energy. The problem of the hydrogen spectrum is to explain these levels.

The Nuclear Atom

In 1909 Geiger and Marsden, working under Rutherford, fired alpha particles at a thin gold foil. Most passed through with small deflections, but about one in several thousand was turned through more than a right angle. Rutherford showed in 1911 that a large deflection of this kind needs a very strong electric field, which can only be produced if the positive charge of the atom, and nearly all of its mass, is concentrated in a nucleus much smaller than the atom. The electrons were then pictured as moving around the nucleus, held by the Coulomb attraction, as the planets are held by gravity.

This planetary picture has a serious defect. An electron moving in a circle is accelerating, and classical electromagnetism says that an accelerating charge radiates. The electron would lose energy

continuously, spiral into the nucleus in a small fraction of a second, and emit a continuous spectrum on the way. Real atoms are stable and emit sharp lines.

Bohr's Model

Bohr kept the planetary picture for hydrogen and added the following postulates in 1913:

1. the electron moves in a circular orbit about the nucleus under the Coulomb attraction, obeying Newton's laws;
2. only orbits whose angular momentum is a whole multiple of $\hbar = h/2\pi$ are allowed, $m_e v r = n\hbar$ with $n = 1, 2, 3, \dots$;
3. an electron in an allowed orbit does not radiate, even though it is accelerating;
4. radiation is emitted or absorbed only when the electron jumps between allowed orbits, and the photon carries away or brings in the difference in energy, $hf = E_i - E_f$.

The constant $\hbar$, pronounced “h-bar”, appears so often in quantum mechanics that it is given a separate symbol, and we shall use it from here on.

Let us find the allowed orbits. We treat the nucleus as fixed, which is a valid approximation since the proton is nearly two thousand times heavier than the electron, and return to this point below. An electron of charge $-e$ in a circular orbit of radius r about a proton of charge e has a centripetal acceleration v^2/r , which is supplied by the Coulomb force, so

$$\frac{m_e v^2}{r} = \frac{e^2}{4\pi\epsilon_0 r^2}, \quad (1.23)$$

a relation you should be familiar with from the study of circular orbits at school. Squaring the quantisation condition gives $m_e^2 v^2 r^2 = n^2 \hbar^2$, and eliminating v^2 between this and the force equation gives the radii of the allowed orbits,

$$r_n = n^2 a_0, \quad a_0 = \frac{4\pi\epsilon_0 \hbar^2}{m_e e^2}, \quad (1.24)$$

where a_0 is called the Bohr radius. It is the radius of the smallest orbit, and it sets the size of the hydrogen atom.

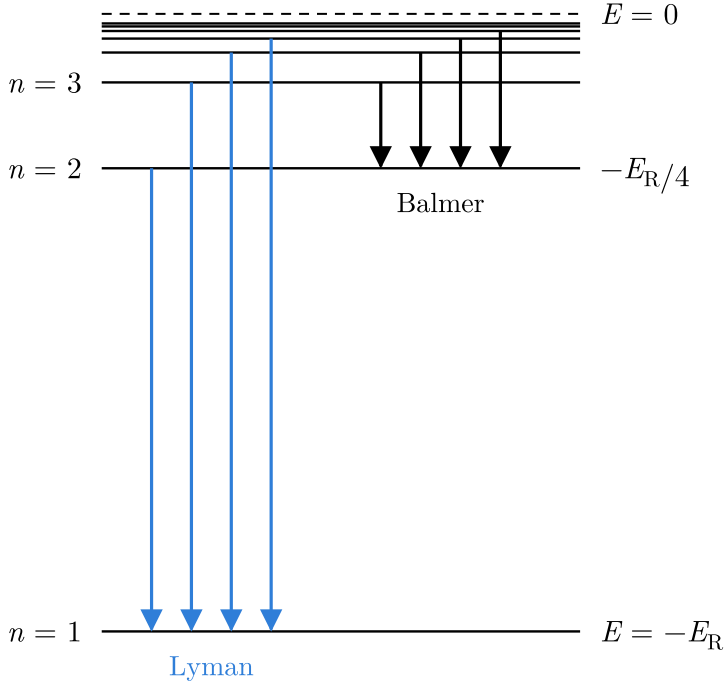


Figure 4. The energy levels $E_n = -E_R/n^2$ of the Bohr model, drawn to scale, with the transitions of the Lyman series, which end on $n = 1$, and of the Balmer series, which end on $n = 2$. The levels crowd together below $E = 0$, which is the energy of an electron that has just escaped.

The energy of the electron is the sum of its kinetic energy and its electrostatic potential energy. The force equation shows that $\frac{1}{2}m_e v^2 = e^2/(8\pi\epsilon_0 r)$, which is minus one half of the potential energy $-e^2/(4\pi\epsilon_0 r)$, so the total energy is

$$E = \frac{1}{2}m_e v^2 - \frac{e^2}{4\pi\epsilon_0 r} = -\frac{e^2}{8\pi\epsilon_0 r}. \quad (1.25)$$

Substituting the allowed radii gives the energy levels,

$$E_n = -\frac{e^2}{8\pi\epsilon_0 a_0} \frac{1}{n^2} = -\frac{E_R}{n^2}, \quad E_R = \frac{m_e e^4}{2(4\pi\epsilon_0)^2 \hbar^2}, \quad (1.26)$$

where E_R is called the Rydberg energy. It is the ionisation energy of hydrogen in its ground state. The levels are drawn in Figure 4. They have the form that the spectrum requires, and comparing them with the levels read from the Rydberg formula gives

$$\mathcal{R}_H = \frac{E_R}{hc} = \frac{m_e e^4}{8\epsilon_0^2 \hbar^3 c}, \quad (1.27)$$

a combination of fundamental constants which agreed with the measured value of $\mathcal{R}_H$ to within the accuracy with which those constants were then known. This agreement was the principal evidence in favour of the model.

Bohr's model can also be applied to a hydrogen-like ion, which is a nucleus of charge Ze with a single electron, such as singly ionised helium. Replacing e^2 by Ze^2 in the force equation shows that the radii become $n^2 a_0/Z$ and the energies $-Z^2 E_R/n^2$, and the spectrum of singly ionised helium does show lines at four times the hydrogen photon energies.

The Mass of the Nucleus

The nucleus is not infinitely heavy, and the electron and the nucleus both move about their common centre of mass. In Vol. 1 of the Fermi Notes on Physics, the relative motion of two bodies of masses m_1 and m_2 was shown to be that of a single body of mass

$$m_r = \frac{m_1 m_2}{m_1 + m_2}, \quad (1.28)$$

which is called the reduced mass, moving about a fixed centre. Applying Bohr's quantisation condition to the total angular momentum about the centre of mass, and repeating the calculation above,

replaces m_e by the reduced mass of the electron and the nucleus in both a_0 and E_R , as you will check in Problem 2. The correction is small, since the reduced mass differs from m_e by about one part in two thousand for hydrogen, but it is easily measured. Deuterium, whose nucleus is about twice as heavy as the proton, has a slightly larger reduced mass, so its lines are shifted slightly towards shorter wavelengths, and it was by this shift that deuterium was discovered in 1931.

Matter Waves and Bohr's Condition

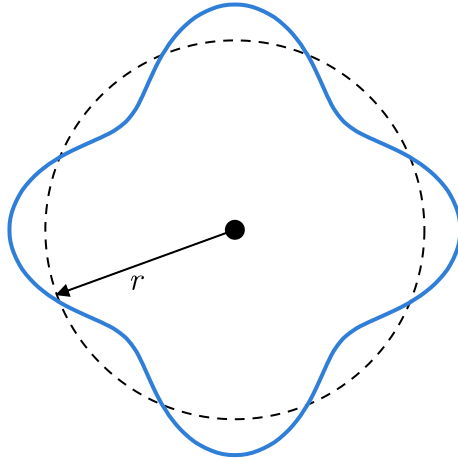


Figure 5. A standing wave with four wavelengths fitted around a circular orbit, drawn as a displacement from the dashed circle. De Broglie's requirement that the wave join onto itself after one circuit is equivalent to Bohr's condition $m_e v r = n\hbar$.

De Broglie gave Bohr's condition a physical meaning. If the electron in its orbit has a wave associated with it, of wavelength h/p , the wave must join smoothly onto itself after going once round the orbit. The circumference must therefore contain a whole number of wavelengths,

$$2\pi r = n\lambda = \frac{nh}{m_e v}, \quad (1.29)$$

which can be rewritten as $m_e v r = n\hbar$, and this is Bohr's condition. Figure 5 shows such a standing wave for $n = 4$.

Electron Diffraction

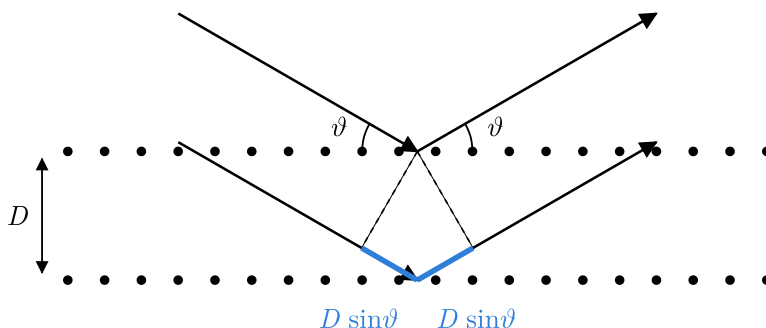


Figure 6. Reflection of waves from two successive planes of atoms a distance D apart, at a glancing angle ϑ . The dashed lines are perpendicular to the rays, and the wave reflected from the lower plane travels the extra distance marked, $2D \sin \vartheta$ in all.

The wave nature of electrons was confirmed in 1927 by Davisson and Germer, who fired a beam of electrons of known energy at a single crystal of nickel and measured the number of electrons scattered in different directions. At certain angles and energies the scattered beam was strong, and at others it was weak, in the same way that X-rays are reflected by a crystal. The atoms of a crystal lie in parallel planes, and waves reflected from successive planes a distance D apart reinforce one another when the extra distance travelled by the wave reflected from the deeper plane is a whole number of wavelengths. For waves meeting the planes at a glancing angle ϑ , as in Figure 6, this extra distance is $2D \sin \vartheta$, and the condition for a strong reflected beam is

$$2D \sin \vartheta = N\lambda, \quad N = 1, 2, 3, \dots, \quad (1.30)$$

which is called the Bragg condition, the integer N being the order of the reflection. With D known from X-ray measurements on the same crystal, the wavelengths found from the strong electron beams agreed with h/p . In the same year G. P. Thomson passed

electrons through thin metal foils and obtained diffraction rings like those given by X-rays, which is the same kind of experiment as the diffraction by a graphite film described at school.

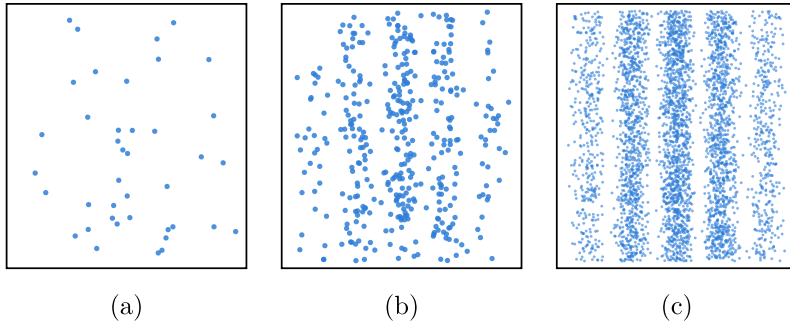


Figure 7. The build-up of the pattern on the screen in the double-slit experiment, with electrons arriving one at a time. Each dot marks the arrival of one electron. (a) After a few dozen electrons the arrivals look random. (b) After a few hundred the fringes begin to appear. (c) After some thousands the bright and dark fringes are clear.

The most direct demonstration of the wave behaviour of electrons is the double-slit experiment. When a beam of electrons is sent through two narrow slits, the electrons arrive at a screen beyond them one at a time, each making a single small spot, as a particle would. As the spots accumulate, however, they build up the pattern of bright and dark fringes that a wave passing through the two slits would produce, as shown in Figure 7, and the pattern forms even when the beam is so weak that only one electron is in the apparatus at any time. If either slit is closed, the fringes disappear. The experiment was first carried out with electrons in 1961 by Jöns-son, and with electrons arriving one at a time by Merli, Missiroli, and Pozzi in 1974 and by Tonomura and his collaborators in 1989. Neither a picture of the electron as a particle nor a picture of it as a classical wave accounts for both features of the result, the arrival of each electron at a single point and the build-up of fringes.

The Wilson–Sommerfeld Rule

Bohr's condition was generalised by Wilson and Sommerfeld in 1915 to any system whose motion is periodic. For a coordinate q and its conjugate momentum p , as defined in the Hamiltonian formulation of Vol. 1 of the Fermi Notes on Physics, the rule is that the integral of p over one complete cycle of the motion is a whole multiple of h ,

$$\oint p \, dq = nh. \quad (1.31)$$

For a circular orbit we may take q to be the angle ϕ around the orbit, whose momentum is the angular momentum $m_e v r$, which is constant, and the integral over one revolution is $2\pi m_e v r$, so the rule gives Bohr's condition again. The rule can also be applied to the harmonic oscillator, and doing so gives energies $E = nhf$, where f is the frequency of the oscillator, which are the energies Planck assigned to his oscillators, as you will show in Problem 4.

The quantisation conditions of Bohr, Wilson, and Sommerfeld are together called the old quantum theory. They had real successes, but they are conditions imposed on classical mechanics from outside rather than a theory, and they fail in several ways. They give no way of calculating the intensities of spectral lines. They do not work for helium or any other atom with more than one electron. They assign the ground state of hydrogen an angular momentum $\hbar$, which is found by experiment to be zero, and they cannot account for the splitting of spectral lines in a magnetic field in many atoms.

Problems

§1.1 Quantum Physics at School

1. Show that the energy of a photon of wavelength λ , expressed in electronvolts, is $hc/(e\lambda)$ when λ is in metres.
2. Light of frequency f_1 falling on a metal gives a stopping potential V_1 , and light of frequency f_2 gives V_2 . Find Planck's constant and the work function of the metal in terms of these four quantities and e .
3. An atom has three energy levels, $E_1 < E_2 < E_3$. List the frequencies of the lines in its emission spectrum, and show that

one of them is the sum of the other two.

4. Show that the de Broglie wavelength of a particle of mass m and kinetic energy E_k , moving slowly compared with light, is $h/\sqrt{2mE_k}$, and find the ratio of the wavelengths of an electron and a proton that have the same kinetic energy.

§1.2 Thermal Radiation and the Photon

1. Show that in the limit $hf \gg k_B T$ Planck's law becomes $u(f) = (8\pi hf^3/c^3) e^{-hf/k_B T}$, and find the frequency at which this approximate form has its maximum. Compare it with the peak of Planck's law found in the text.
2. The spectrum per unit wavelength, $u_\lambda(\lambda)$, is defined so that $u_\lambda(\lambda) |d\lambda|$ is the energy per unit volume in the same range as $u(f) df$. Using $f = c/\lambda$, show that $u_\lambda(\lambda) = (8\pi hc/\lambda^5)/(e^{hc/\lambda k_B T} - 1)$, and show that its maximum is at $\lambda_{\max} = hc/(\zeta_p k_B T)$, where ζ_p is the non-zero root of $\zeta = 5(1 - e^{-\zeta})$. Explain why $\lambda_{\max}$ is different from $c/f_{\max}$.
3. Show that for a particle of mass m with $pc \ll mc^2$, the relation $E^2 = p^2 c^2 + m^2 c^4$ gives $E \approx mc^2 + p^2/2m$.
4. A photon of wavelength λ is scattered straight back by a free electron at rest. Show that the kinetic energy of the recoiling electron is $2hc\lambda_C/[\lambda(\lambda + 2\lambda_C)]$, where $\lambda_C = h/m_e c$ is the Compton wavelength.
5. Explain why visible light shows no measurable Compton shift when it is scattered, even though the shift $\lambda' - \lambda$ does not depend on the wavelength.

§1.3 The Bohr Model and Matter Waves

1. Show that the speed of the electron in the n th Bohr orbit of hydrogen is $v_n = e^2/(4\pi\epsilon_0 \hbar n)$, and that the ratio v_1/c is a pure number. Explain why this number being small justifies the use of non-relativistic mechanics in the model.
2. Let the nucleus have mass M and the electron mass m_e , both moving in circles about their common centre of mass with the same angular velocity ω and a separation r . Show that the total

angular momentum is $m_r \omega r^2$, where m_r is the reduced mass, and that requiring it to equal $n\hbar$ gives energies $-(m_r/m_e)E_R/n^2$.

3. Find the ratio of the longest and shortest wavelengths in the Balmer series of hydrogen.
4. A particle of mass m moves in one dimension in the potential $\frac{1}{2}m\omega^2 x^2$ with energy E . Show that its path in the plane of x and p is an ellipse with semi-axes $\sqrt{2E/m\omega^2}$ and $\sqrt{2mE}$, and use the Wilson–Sommerfeld rule, together with the area of an ellipse, to show that $E = n\hbar\omega$.
5. In a Davisson–Germer experiment, electrons accelerated through a potential difference V give a first-order reflection at glancing angle ϑ from planes of spacing D . Find V in terms of D , ϑ , and fundamental constants.

Wave Mechanics in One Dimension

2.1 The Schrödinger Equation and the Wave Function

In this chapter we set up the equation that governs the matter waves of Chapter 1 and solve it for a particle that moves along a line. We begin with the equation itself and with the meaning of its solutions. In the sections that follow we solve it first for a particle confined to a box, then for particles held by wells of finite depth, then for a free particle and for a particle meeting a barrier, and finally for a particle in a harmonic potential.

A Wave Equation for Matter

The waves we shall need depend on position and time together, and we will differentiate them with respect to one variable while holding the other fixed. The result is called a partial derivative. For a function $\Psi(x, t)$, the partial derivative $\partial\Psi/\partial x$ is found by treating t as a constant and differentiating with respect to x in the ordinary way, and $\partial\Psi/\partial t$ is found by treating x as a constant.

Let us look for an equation whose solutions include the de Broglie wave of a free particle. A wave of wave number k and angular

frequency ω travelling in the positive x -direction can be written as

$$\Psi(x, t) = A e^{i(kx - \omega t)}, \quad (2.1)$$

where A is a constant. The de Broglie relation $\lambda = h/p$, with $k = 2\pi/\lambda$, gives the momentum of the particle as $p = \hbar k$, and the relation $E = hf$, with $\omega = 2\pi f$, gives its energy as $E = \hbar\omega$. For a free particle of mass m moving slowly compared with light, $E = p^2/2m$, and the wave number and the frequency must therefore be related by

$$\hbar\omega = \frac{\hbar^2 k^2}{2m}. \quad (2.2)$$

Differentiating the plane wave gives $\partial\Psi/\partial t = -i\omega\Psi$ and $\partial^2\Psi/\partial x^2 = -k^2\Psi$. Multiplying the first of these by $i\hbar$ and the second by $-\hbar^2/2m$, and using the relation between ω and k , we find that the plane wave satisfies

$$i\hbar \frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2\Psi}{\partial x^2}. \quad (2.3)$$

This equation is linear, which means that if Ψ_1 and Ψ_2 are solutions, then so is $c_1\Psi_1 + c_2\Psi_2$ for any constants c_1 and c_2 . Linearity is essential, since the electrons in the double-slit experiment of Section 1.3 interfere, and interference happens when waves are added. The equation also contains the first derivative with respect to time, so the state of the wave at one instant fixes its state at every later instant. These requirements force the wave to be complex. A real wave such as $\cos(kx - \omega t)$ cannot satisfy an equation of this kind, because its first derivative with respect to time is a sine, while its second derivative with respect to position is a cosine, and a sine is not a constant multiple of a cosine. For the complex exponential, the first derivative with respect to time and the second derivative with respect to position are both constant multiples of the exponential itself.

When the particle moves in a potential $V(x)$, its energy is $p^2/2m + V$, and the natural generalisation is

$$i\hbar \frac{\partial\Psi}{\partial t} = -\frac{\hbar^2}{2m} \frac{\partial^2\Psi}{\partial x^2} + V(x)\Psi, \quad (2.4)$$

which is called the Schrödinger equation, after Schrödinger, who published it in 1926. The argument we have given makes the equa-

tion plausible, although it falls short of a derivation. The equation is a new law of physics, in the same way that Newton's second law is, and it is justified by the agreement of its predictions with experiment. Its solution $\Psi(x, t)$ is called the wave function of the particle.

The Statistical Interpretation

A wave function is spread out in space, while an electron striking a screen makes a single small spot. Born proposed in 1926 that the wave function gives the probability of finding the particle at each place, in the following sense: if the position of the particle is measured at time t , the probability of finding it between x and $x + dx$ is

$$|\Psi(x, t)|^2 dx. \quad (2.5)$$

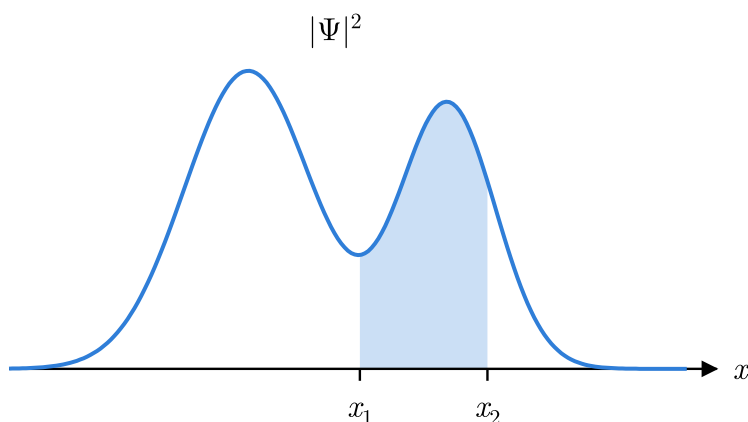


Figure 8. A probability density $|\Psi(x, t)|^2$ at one instant. The probability of finding the particle between $x = x_1$ and $x = x_2$ is the shaded area, and the total area under the curve is one.

The quantity $|\Psi|^2 = \Psi^* \Psi$ is therefore a probability density, and the probability of finding the particle between $x = x_1$ and $x = x_2$ is the area under the graph of $|\Psi|^2$ between those points, as shown in Figure 8. The particle must be found somewhere, so the total

probability is one,

$$\int_{-\infty}^{\infty} |\Psi(x, t)|^2 dx = 1, \quad (2.6)$$

and a wave function that satisfies this condition is said to be normalised. Since the Schrödinger equation is linear, a solution multiplied by any constant is again a solution, and the constant can be chosen to normalise it. This can be done only if the integral of $|\Psi|^2$ is finite, which requires Ψ to fall to zero as $x \rightarrow \pm\infty$, and wave functions that do not fall off in this way do not describe a single particle.

The interpretation explains the double-slit experiment of Section 1.3. Let Ψ_1 be the wave that reaches a point on the screen through the first slit and Ψ_2 the wave that reaches it through the second. With both slits open the wave is $\Psi_1 + \Psi_2$, and the probability density is

$$|\Psi_1 + \Psi_2|^2 = |\Psi_1|^2 + |\Psi_2|^2 + 2 \operatorname{Re}(\Psi_1^* \Psi_2). \quad (2.7)$$

The first two terms are the densities for each slit with the other closed, and the last term, which is positive at some points on the screen and negative at others, produces the fringes. Each electron arrives at a single point, and the fringes appear only in the distribution of many arrivals.

Conservation of Probability

Normalisation raises the following question: since the wave function changes in time, does a wave function that is normalised at one instant stay normalised at later instants? To answer this question, let $P(x, t) = |\Psi|^2$ be the probability density. By the product rule,

$$\frac{\partial P}{\partial t} = \Psi^* \frac{\partial \Psi}{\partial t} + \Psi \frac{\partial \Psi^*}{\partial t}. \quad (2.8)$$

The Schrödinger equation gives $\partial \Psi / \partial t$, and its complex conjugate, with V real, gives

$$\frac{\partial \Psi^*}{\partial t} = -\frac{i\hbar}{2m} \frac{\partial^2 \Psi^*}{\partial x^2} + \frac{i}{\hbar} V \Psi^*, \quad (2.9)$$

while the equation itself can be rewritten as $\partial \Psi / \partial t = (i\hbar/2m) \partial^2 \Psi / \partial x^2 - (i/\hbar) V \Psi$. Substituting these two time deriva-

tives into the expression for $\partial P/\partial t$, the terms containing V cancel, and we are left with

$$\frac{\partial P}{\partial t} = \frac{i\hbar}{2m} \left(\Psi^* \frac{\partial^2 \Psi}{\partial x^2} - \Psi \frac{\partial^2 \Psi^*}{\partial x^2} \right) = \frac{\partial}{\partial x} \left[\frac{i\hbar}{2m} \left(\Psi^* \frac{\partial \Psi}{\partial x} - \Psi \frac{\partial \Psi^*}{\partial x} \right) \right], \quad (2.10)$$

where the second form can be checked by differentiating the bracket with the product rule, since the terms $(\partial \Psi^*/\partial x)(\partial \Psi/\partial x)$ cancel. The quantity in the round brackets, $\Psi^* \partial \Psi/\partial x - \Psi \partial \Psi^*/\partial x$, is the difference between a complex number and its conjugate, which is $2i$ times its imaginary part. Let

$$j(x, t) = \frac{\hbar}{m} \operatorname{Im} \left(\Psi^* \frac{\partial \Psi}{\partial x} \right), \quad (2.11)$$

which is called the probability current. The result then reads

$$\frac{\partial P}{\partial t} = -\frac{\partial j}{\partial x}. \quad (2.12)$$

This equation has the same form as the equation for the conservation of charge or of the mass of a fluid. Integrating it between $x = x_1$ and $x = x_2$ gives

$$\frac{d}{dt} \int_{x_1}^{x_2} P dx = j(x_1, t) - j(x_2, t), \quad (2.13)$$

so the probability of finding the particle between x_1 and x_2 changes only because probability flows in at one end or out at the other, and j is the rate at which it flows past a point in the positive x -direction. Letting $x_1 \rightarrow -\infty$ and $x_2 \rightarrow \infty$, where Ψ and therefore j vanish, shows that the total probability does not change. A wave function normalised at one instant stays normalised.

For the plane wave $A e^{i(kx - \omega t)}$ we have $\Psi^* \partial \Psi/\partial x = ik|A|^2$, so $j = \hbar k|A|^2/m$. This is the density $|A|^2$ multiplied by the velocity p/m of the particle, as it would be for a stream of particles.

Expectation Values

The mean value of the position, over many measurements made on particles all prepared with the same wave function, is

$$\langle x \rangle = \int_{-\infty}^{\infty} x |\Psi(x, t)|^2 dx, \quad (2.14)$$

which is called the expectation value of x . It is the mean of a probability distribution, as in school statistics, and a single measurement need not give this value.

Let us find how $\langle x \rangle$ changes in time. Differentiating under the integral sign, and using the conservation law above, gives

$$\frac{d\langle x \rangle}{dt} = \int_{-\infty}^{\infty} x \frac{\partial P}{\partial t} dx = - \int_{-\infty}^{\infty} x \frac{\partial j}{\partial x} dx = \int_{-\infty}^{\infty} j dx, \quad (2.15)$$

where the last step is an integration by parts, and the boundary term $[xj]$ vanishes because Ψ falls to zero at large $|x|$. Multiplying by m and substituting the definition of j gives

$$m \frac{d\langle x \rangle}{dt} = \hbar \operatorname{Im} \int_{-\infty}^{\infty} \Psi^* \frac{\partial \Psi}{\partial x} dx. \quad (2.16)$$

The integral here has no real part. Its real part is half the integral of $\Psi^* \partial \Psi / \partial x + \Psi \partial \Psi^* / \partial x$, which by the product rule is the integral of $\partial |\Psi|^2 / \partial x$, and this vanishes because $|\Psi|^2$ is zero at both ends. The imaginary part of a purely imaginary number is $-i$ times the number itself, so

$$m \frac{d\langle x \rangle}{dt} = \int_{-\infty}^{\infty} \Psi^* \left(-i\hbar \frac{\partial}{\partial x} \right) \Psi dx. \quad (2.17)$$

In classical mechanics the left-hand side, the mass multiplied by the rate of change of the mean position, would be the momentum of the particle, and we take the integral on the right-hand side as the definition of $\langle p \rangle$. The expectation value of the position is found by multiplying Ψ by x , sandwiching the result between Ψ^* and the integral sign, and integrating. The expectation value of the momentum is found in the same way, with the multiplication by x replaced by the operation $-i\hbar \partial / \partial x$. An instruction of this kind, which turns one function into another, is called an operator, and we write operators with a hat:

$$\hat{x} = x, \quad \hat{p} = -i\hbar \frac{\partial}{\partial x}. \quad (2.18)$$

The general principle, which is part of the statistical interpretation, is that any quantity $Q(x, p)$ has the expectation value

$$\langle Q \rangle = \int_{-\infty}^{\infty} \Psi^* \hat{Q} \Psi dx, \quad (2.19)$$

where the operator $\hat{Q}$ is obtained by replacing p with $\hat{p}$ in the expression for Q . The operator for the energy, $p^2/2m + V(x)$, which is

$$\hat{H} = \frac{\hat{p}^2}{2m} + V(x) = -\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x^2} + V(x), \quad (2.20)$$

is called the Hamiltonian. The name was met in Vol. 1 of the Fermi Notes on Physics, where the Hamiltonian is the energy written in terms of positions and momenta. In this notation the Schrödinger equation is simply

$$i\hbar \frac{\partial \Psi}{\partial t} = \hat{H} \Psi. \quad (2.21)$$

The spread of the measured values is described by the variance. For the position it is

$$\sigma_x^2 = \langle x^2 \rangle - \langle x \rangle^2, \quad (2.22)$$

and σ_x is the standard deviation, with σ_p defined in the same way for the momentum.

A Wave Function That Decays Exponentially

Let us consider an example. Suppose that at some instant a particle has the wave function

$$\Psi(x) = A e^{-\kappa|x|}, \quad (2.23)$$

where κ is a positive constant with the dimensions of an inverse length. The function is even, and the integral of $|\Psi|^2$ over all x is twice the integral over positive x , so the normalisation condition is

$$2|A|^2 \int_0^\infty e^{-2\kappa x} dx = \frac{|A|^2}{\kappa} = 1, \quad (2.24)$$

and we may take $A = \sqrt{\kappa}$. Since $|\Psi|^2$ is even, $\langle x \rangle = 0$. The mean square position is $\langle x^2 \rangle = 2\kappa \int_0^\infty x^2 e^{-2\kappa x} dx$. Integrating by parts twice, each time integrating the exponential and differentiating the power of x , gives

$$\begin{aligned} \int_0^\infty x^2 e^{-2\kappa x} dx &= \frac{2}{2\kappa} \int_0^\infty x e^{-2\kappa x} dx \\ &= \frac{2}{(2\kappa)^2} \int_0^\infty e^{-2\kappa x} dx = \frac{2}{(2\kappa)^3}, \end{aligned} \quad (2.25)$$

where the boundary terms vanish at both limits. Therefore $\langle x^2 \rangle = 1/(2\kappa^2)$, and the standard deviation is $\sigma_x = 1/(\sqrt{2}\kappa)$, so the particle is found within a distance of order $1/\kappa$ of the origin. The probability that it is found further than $1/\kappa$ from the origin is

$$2\kappa \int_{1/\kappa}^{\infty} e^{-2\kappa x} dx = e^{-2}. \quad (2.26)$$

Since Ψ is real, $\Psi d\Psi/dx = \frac{1}{2} d\Psi^2/dx$, whose integral vanishes, and therefore $\langle p \rangle = 0$. This holds for any real, normalisable wave function.

2.2 Stationary States and the Infinite Square Well

The Schrödinger equation is a partial differential equation, and for most potentials its solutions cannot be written down directly. When the potential does not depend on time, however, the equation can be reduced to an ordinary differential equation in x , and the general solution can be built from the solutions of that simpler equation. This section shows how the reduction is done, and then applies it to a particle confined between two impenetrable walls.

Separation of Variables

Let us look for solutions of the Schrödinger equation that are a product of a function of x and a function of t ,

$$\Psi(x, t) = \psi(x) \Phi(t). \quad (2.27)$$

This method is called separation of variables. Substituting the product into the Schrödinger equation, and dividing both sides by $\psi\Phi$, gives

$$\frac{i\hbar}{\Phi} \frac{d\Phi}{dt} = \frac{1}{\psi} \left(-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V\psi \right). \quad (2.28)$$

The left-hand side depends on t alone and the right-hand side on x alone. If we change t while keeping x fixed, the right-hand side does not change, so the left-hand side cannot change either, and a similar argument holds for x . The left-hand and right-hand sides must therefore equal the same constant, which we call E . The

equation for Φ is $d\Phi/dt = -(iE/\hbar)\Phi$, whose solution is

$$\Phi(t) = e^{-iEt/\hbar}, \quad (2.29)$$

where any constant factor has been absorbed into ψ . The equation for ψ is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi, \quad (2.30)$$

which is called the time-independent Schrödinger equation. In operator notation it reads $\hat{H}\psi = E\psi$. An equation of this form, in which an operator acting on a function returns the same function multiplied by a constant, is called an eigenvalue equation. The function is called an eigenfunction of the operator and the constant is called the corresponding eigenvalue.

The separated solutions are stationary. The probability density is $|\psi(x)e^{-iEt/\hbar}|^2 = |\psi(x)|^2$, since the exponential has unit modulus, and it does not change in time. The same cancellation happens in every expectation value, so nothing measurable about the particle changes, and the separated solutions are called stationary states for this reason.

The stationary states also have a definite energy. The expectation value of the energy is $\langle H \rangle = \int \psi^* \hat{H} \psi dx = E \int |\psi|^2 dx = E$. Applying $\hat{H}$ twice gives $\hat{H}^2\psi = E^2\psi$, so $\langle H^2 \rangle = E^2$ and the variance $\langle H^2 \rangle - \langle H \rangle^2$ is zero. Every measurement of the energy of a particle in a stationary state gives the value E .

Finally, the stationary states can be combined into more general solutions. The time-independent equation usually has acceptable solutions only for particular values of E , which we label $E_1, E_2, E_3, \dots$, with eigenfunctions $\psi_1, \psi_2, \psi_3, \dots$. Since the Schrödinger equation is linear, any combination

$$\Psi(x, t) = \sum_n c_n \psi_n(x) e^{-iE_n t/\hbar} \quad (2.31)$$

with constant coefficients c_n is also a solution. For the square well treated below, every solution can be written in this way.

The Infinite Square Well

Let the potential be

$$V(x) = \begin{cases} 0, & 0 < x < w, \\ \infty, & \text{otherwise.} \end{cases} \quad (2.32)$$

A particle in this potential moves freely between two walls a distance w apart and cannot escape past them. This is a model of an electron confined to a small region, such as a thin layer of one semiconductor sandwiched between two others. The particle cannot be found where the potential is infinite, so $\psi = 0$ outside the well. Inside the well $V = 0$, and the time-independent equation becomes

$$\frac{d^2\psi}{dx^2} = -k^2\psi, \quad k = \frac{\sqrt{2mE}}{\hbar}, \quad (2.33)$$

where we have assumed that $E > 0$, since you will show in Problem 1 that there are no solutions with $E \leq 0$. This is the equation of simple harmonic motion, with x in place of the time, and its general solution is

$$\psi(x) = A \sin kx + B \cos kx, \quad (2.34)$$

where A and B are constants to be determined. The wave function must be continuous, for reasons that are given in Section 2.3, so it must vanish at both walls. The condition $\psi(0) = 0$ gives $B = 0$, and the condition $\psi(w) = 0$ then gives $A \sin kw = 0$. Taking $A = 0$ would leave no wave function, so $\sin kw = 0$, and therefore

$$k_n = \frac{n\pi}{w}, \quad n = 1, 2, 3, \dots \quad (2.35)$$

The value $n = 0$ gives $\psi = 0$ again, and negative values of n give the same functions multiplied by -1 , which describe the same states. The allowed energies are

$$E_n = \frac{\hbar^2 k_n^2}{2m} = \frac{n^2 \pi^2 \hbar^2}{2mw^2}. \quad (2.36)$$

The energy is quantised, and the quantisation comes from the requirement that the wave fit between the walls. Normalising the wave functions requires $|A|^2 \int_0^w \sin^2(n\pi x/w) dx = 1$. Writing $\sin^2 \theta = \frac{1}{2}(1 - \cos 2\theta)$, the integral of the cosine over a whole number of periods vanishes, the integral of $\sin^2(n\pi x/w)$ is $w/2$, and we

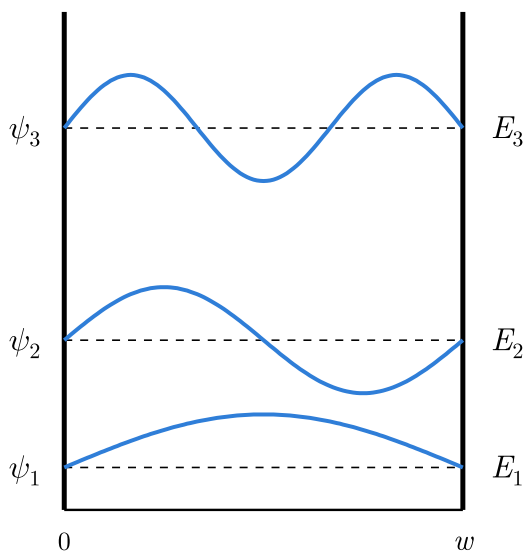


Figure 9. The first three stationary states of the infinite square well, each drawn on a baseline at the height of its energy. The energies are in the ratio $1 : 4 : 9$, and the n th state has $n - 1$ nodes inside the well.

may take $A = \sqrt{2/w}$. The stationary states are therefore

$$\psi_n(x) = \sqrt{\frac{2}{w}} \sin \frac{n\pi x}{w}, \quad 0 < x < w, \quad (2.37)$$

and the first three of them are drawn in Figure 9.

The states have several notable features. The lowest energy, E_1 , is greater than zero, so the particle can never be at rest at the bottom of the well. Each state fits n half-wavelengths between the walls, as a string fixed at both ends does in its n th harmonic, and the n th state has $n - 1$ nodes, which are points inside the well where ψ vanishes. The states are alternately even and odd about the centre of the well.

Orthogonality

The stationary states of the well satisfy

$$\int_0^w \psi_{n'}(x) \psi_n(x) dx = \delta_{nn'}, \quad (2.38)$$

where $\delta_{nn'}$, which is called the Kronecker delta, is 1 when $n' = n$ and 0 otherwise. For $n' = n$ this is the normalisation we have just carried out. For $n' \neq n$ we use the identity $\sin \theta_1 \sin \theta_2 = \frac{1}{2}[\cos(\theta_1 - \theta_2) - \cos(\theta_1 + \theta_2)]$, which follows from the formulas for the cosine of a sum and of a difference, and which gives

$$\int_0^w \psi_{n'} \psi_n dx = \frac{1}{w} \int_0^w \left[\cos \frac{(n' - n)\pi x}{w} - \cos \frac{(n' + n)\pi x}{w} \right] dx. \quad (2.39)$$

Since $n' - n$ and $n' + n$ are non-zero integers, each cosine runs through a whole number of half-periods between 0 and w , and each integrates to a sine that vanishes at both limits. A pair of functions whose product integrates to zero is said to be orthogonal, by analogy with two vectors whose scalar product is zero, and a set of normalised functions that are orthogonal to one another is said to be orthonormal.

Expanding an Arbitrary State

Let $f(x)$ be any reasonable function on $0 < x < w$ that vanishes at both ends. A theorem of Fourier analysis, which we quote here

since its proof is beyond this course, states that f can be written as a sum of sines,

$$f(x) = \sum_{n=1}^{\infty} c_n \psi_n(x), \quad (2.40)$$

which is called a Fourier sine series. A set of functions from which every function of this kind can be built is said to be complete. The coefficients are found from the orthogonality of the ψ_n . Multiplying both sides by $\psi_{n'}$ and integrating over the well, every term of the sum vanishes except the one with $n = n'$, and therefore

$$c_{n'} = \int_0^w \psi_{n'}(x) f(x) dx. \quad (2.41)$$

This result allows us to find the motion of a particle in the well from any starting wave function. Given the wave function $\Psi(x, 0)$ at $t = 0$, we find its coefficients c_n from this formula, and the wave function at any later time is

$$\Psi(x, t) = \sum_n c_n \psi_n(x) e^{-iE_n t/\hbar}, \quad (2.42)$$

since this sum satisfies the Schrödinger equation and agrees with $\Psi(x, 0)$ at $t = 0$.

The coefficients have a physical meaning. Substituting the sum into the normalisation condition, and using the orthogonality of the ψ_n to remove every cross term, gives

$$\begin{aligned} \int_0^w |\Psi|^2 dx &= \sum_{n'} \sum_n c_{n'}^* c_n e^{i(E_{n'} - E_n)t/\hbar} \int_0^w \psi_{n'} \psi_n dx \\ &= \sum_n |c_n|^2 = 1. \end{aligned} \quad (2.43)$$

A similar calculation, using $\hat{H}\psi_n = E_n\psi_n$, gives the expectation value of the energy,

$$\langle H \rangle = \sum_n |c_n|^2 E_n. \quad (2.44)$$

These results are what we would expect if $|c_n|^2$ were the probability that a measurement of the energy gives the value E_n , and this is what the statistical interpretation asserts. A measurement of the energy can only give one of the eigenvalues E_n , and the probability

of obtaining E_n is $|c_n|^2$. The probabilities do not change in time, since the factor $e^{-iE_n t/\hbar}$ has unit modulus.

A Parabolic Initial State

As an example, suppose that a particle in the well starts with the wave function

$$\Psi(x, 0) = A x(w - x), \quad 0 < x < w, \quad (2.45)$$

which vanishes at both walls and has a single hump, like the ground state. The normalisation integral is $|A|^2 \int_0^w x^2(w-x)^2 dx = |A|^2 w^5/30$, which is found by expanding the bracket and integrating term by term, so $A = \sqrt{30/w^5}$. The coefficients are

$$c_n = \sqrt{\frac{2}{w}} \sqrt{\frac{30}{w^5}} \int_0^w x(w-x) \sin \frac{n\pi x}{w} dx. \quad (2.46)$$

Substituting $u = n\pi x/w$ turns the integral into $(w/n\pi)^3 \int_0^{n\pi} u(n\pi - u) \sin u du$. Integrating by parts twice, each time integrating the trigonometric factor, gives

$$\begin{aligned} \int_0^{n\pi} u(n\pi - u) \sin u du &= \left[-u(n\pi - u) \cos u \right]_0^{n\pi} \\ &\quad + \left[(n\pi - 2u) \sin u \right]_0^{n\pi} + 2 \int_0^{n\pi} \sin u du \\ &= 2[1 - (-1)^n], \end{aligned} \quad (2.47)$$

since the first two brackets vanish at both limits and $\cos n\pi = (-1)^n$. Therefore

$$c_n = \frac{4\sqrt{15}}{(n\pi)^3} [1 - (-1)^n], \quad (2.48)$$

which is zero for even n and $8\sqrt{15}/(n\pi)^3$ for odd n . The coefficients with even n vanish because the initial wave function is even about the centre of the well, while the states with even n are odd about it. The probability that a measurement of the energy gives E_1 is $|c_1|^2 = 960/\pi^6$, which is very close to one, and the probabilities for higher states fall off as $1/n^6$. The initial state looks very like the ground state, and the expansion confirms that it consists almost entirely of the ground state.

The expectation value of the energy is found most easily from the wave function directly. Inside the well $\hat{H} = -(\hbar^2/2m) d^2/dx^2$, and $d^2[x(w-x)]/dx^2 = -2$, so

$$\langle H \rangle = \frac{\hbar^2}{m} |A|^2 \int_0^w x(w-x) dx = \frac{\hbar^2}{m} \frac{30}{w^5} \frac{w^3}{6} = \frac{5\hbar^2}{mw^2}. \quad (2.49)$$

This is slightly larger than $E_1 = \pi^2 \hbar^2 / 2mw^2$, since $\pi^2/2$ is a little less than 5, and you will show in Problem 3 that $\langle H \rangle$ can never be smaller than E_1 .

A Superposition of Two States

As another example, consider a particle whose wave function at $t = 0$ is an equal mixture of the two lowest states,

$$\Psi(x, 0) = \frac{1}{\sqrt{2}} [\psi_1(x) + \psi_2(x)]. \quad (2.50)$$

The coefficients are $c_1 = c_2 = 1/\sqrt{2}$, and the wave function at time t is $[\psi_1 e^{-iE_1 t/\hbar} + \psi_2 e^{-iE_2 t/\hbar}]/\sqrt{2}$. Its probability density is

$$|\Psi(x, t)|^2 = \frac{1}{2} [\psi_1^2 + \psi_2^2 + 2\psi_1 \psi_2 \cos \omega_{21} t], \quad (2.51)$$

$$\omega_{21} = \frac{E_2 - E_1}{\hbar} = \frac{3\pi^2 \hbar}{2mw^2},$$

where the cross terms have been combined using $e^{i\theta} + e^{-i\theta} = 2 \cos \theta$. A superposition of states of different energy is not stationary, and its density oscillates at a frequency set by the difference of the energies. The expectation value of the position is

$$\langle x \rangle = \frac{1}{2} \int_0^w x (\psi_1^2 + \psi_2^2) dx + \cos \omega_{21} t \int_0^w x \psi_1 \psi_2 dx. \quad (2.52)$$

Each of ψ_1^2 and ψ_2^2 is symmetric about the centre of the well, so $x\psi_1^2$ and $x\psi_2^2$ each integrate to $w/2$, the first integral is w , and the first term, which carries the factor of one half, is $w/2$. For the second integral, the product identity used above gives $\psi_1 \psi_2 = (1/w)[\cos(\pi x/w) - \cos(3\pi x/w)]$, and integrating by parts gives, for any k with $\sin kw = 0$,

$$\int_0^w x \cos kx dx = \left[\frac{x \sin kx}{k} \right]_0^w - \int_0^w \frac{\sin kx}{k} dx = \frac{\cos kw - 1}{k^2}. \quad (2.53)$$

With $k = \pi/w$ and $k = 3\pi/w$ this is $-2w^2/\pi^2$ and $-2w^2/9\pi^2$, and therefore

$$\langle x \rangle = \frac{w}{2} - \frac{16w}{9\pi^2} \cos \omega_{21} t. \quad (2.54)$$

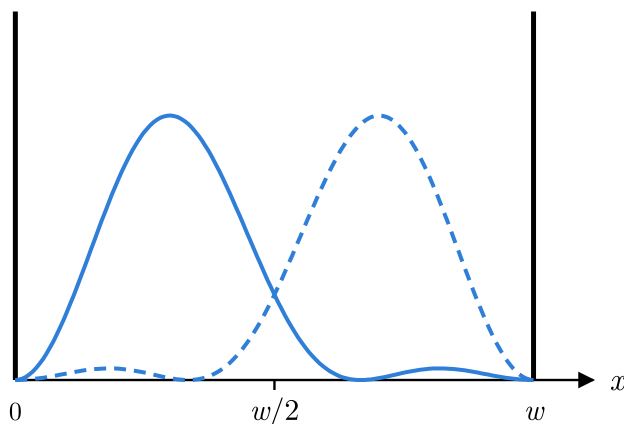


Figure 10. The probability density of the equal superposition of ψ_1 and ψ_2 at $t = 0$ (solid) and half a period later, at $t = \pi/\omega_{21}$ (dashed). The density moves from one side of the well to the other and back with angular frequency ω_{21} .

The particle moves back and forth across the well, as shown in Figure 10, with an amplitude a little less than a fifth of the width of the well.

2.3 Bound States in Wells of Finite Depth

The walls of the infinite square well are an idealisation, and a real particle is held by a potential of finite depth, which it can partly penetrate. This section works out the conditions a wave function must meet where the potential changes suddenly, and then finds the bound states of two simple wells, the delta-function well and the finite square well.

Conditions on the Wave Function

A state whose energy is less than the potential at large distances on both sides, so that a classical particle with that energy could

never escape, is called a bound state. For bound states the time-independent equation has acceptable solutions only for particular energies, as it did in the infinite well, and the requirement that picks out those energies is that the wave function be normalisable.

In a region where the potential is constant and larger than E , which a classical particle could not enter, the time-independent equation reads

$$\frac{d^2\psi}{dx^2} = \kappa^2\psi, \quad \kappa = \frac{\sqrt{2m(V-E)}}{\hbar}, \quad (2.55)$$

and its solutions are $e^{\kappa x}$ and $e^{-\kappa x}$. A region of this kind is called classically forbidden. In a forbidden region extending to $x = +\infty$, the growing exponential $e^{\kappa x}$ would make ψ impossible to normalise, so only $e^{-\kappa x}$ is allowed there, and in a forbidden region extending to $x = -\infty$ only $e^{\kappa x}$ is allowed. The wave function therefore decays into a forbidden region, over a distance of order $1/\kappa$, and does not vanish in it.

Where the potential jumps from one value to another, the solutions on either side must be joined. The wave function is continuous, since a jump in ψ would make $d\psi/dx$ infinite at that point, and $d^2\psi/dx^2$ more singular still, and the Schrödinger equation could then not be satisfied with a potential that is merely finite. To find the condition on the derivative, we integrate the time-independent equation over a short interval from $x_0 - \epsilon$ to $x_0 + \epsilon$, which gives

$$-\frac{\hbar^2}{2m} \left[\frac{d\psi}{dx} \Big|_{x_0+\epsilon} - \frac{d\psi}{dx} \Big|_{x_0-\epsilon} \right] = \int_{x_0-\epsilon}^{x_0+\epsilon} [E - V(x)] \psi dx. \quad (2.56)$$

If V is finite, the integral on the right tends to zero as $\epsilon \rightarrow 0$, and so $d\psi/dx$ is continuous as well. At a wall where the potential is infinite, as in Section 2.2, the derivative can jump, and only the continuity of ψ is required.

The Dirac Delta Function

A potential that forms a very tall and very narrow spike is often most simply described by a delta function, and a very deep and very narrow well by its negative. Let $\delta_\epsilon(x)$ be a rectangle of width ϵ and height $1/\epsilon$, centred on the origin, as in Figure 11, so that its area is one whatever the value of ϵ . As ϵ is made smaller the

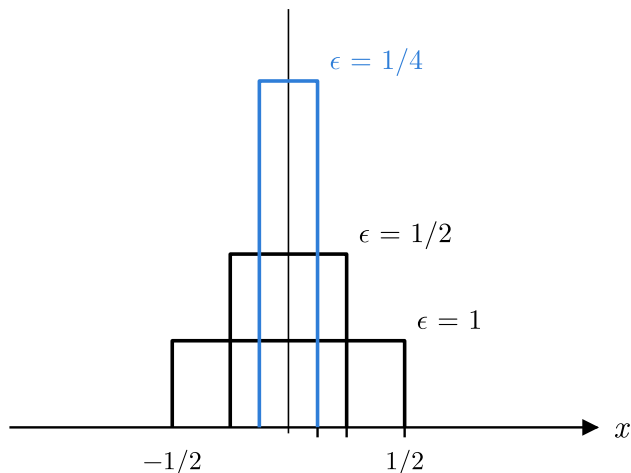


Figure 11. Rectangles $\delta_\epsilon(x)$ of width ϵ and height $1/\epsilon$, drawn for three values of ϵ . Each has unit area, and the delta function is the limit as $\epsilon \rightarrow 0$.

rectangle becomes narrower and taller, and the limit is called the Dirac delta function,

$$\delta(x) = \lim_{\epsilon \rightarrow 0} \delta_\epsilon(x). \quad (2.57)$$

It is zero everywhere except at the origin, and the area under it is one. The delta function is not a function in the ordinary sense, since its value at the origin is infinite, and it is only used inside integrals, where the limit is taken after the integral has been done.

The property that makes the delta function useful follows from the definition. For any continuous function f ,

$$\int_{-\infty}^{\infty} f(x) \delta_\epsilon(x - x_0) dx = \frac{1}{\epsilon} \int_{x_0 - \epsilon/2}^{x_0 + \epsilon/2} f(x) dx, \quad (2.58)$$

which is the mean value of f over an interval of width ϵ about x_0 . As $\epsilon \rightarrow 0$ that mean value tends to $f(x_0)$, and therefore

$$\int_{-\infty}^{\infty} f(x) \delta(x - x_0) dx = f(x_0). \quad (2.59)$$

The Delta-Function Well

We now take a particle in the potential

$$V(x) = -\alpha \delta(x), \quad (2.60)$$

where α is a positive constant, which is called the strength of the well. This is a simple model of an attraction whose range is much shorter than the wavelength of the particle. The potential vanishes everywhere except at the origin, so a bound state must have $E < 0$, and on either side of the origin the region is classically forbidden, with $\kappa = \sqrt{-2mE}/\hbar$. The normalisable solutions are

$$\psi(x) = \begin{cases} Ae^{\kappa x}, & x < 0, \\ Be^{-\kappa x}, & x > 0, \end{cases} \quad (2.61)$$

and continuity at the origin requires $B = A$, so $\psi = Ae^{-\kappa|x|}$. The derivative jumps at the origin, from κA just to the left to $-\kappa A$ just to the right. Integrating the time-independent equation across the origin, as above, and using the property of the delta function, gives

$$-\frac{\hbar^2}{2m}(-2\kappa A) - \alpha\psi(0) = 0, \quad (2.62)$$

where the term containing E has vanished as the interval shrinks. Since $\psi(0) = A$, this gives $\hbar^2\kappa/m = \alpha$, or

$$\kappa = \frac{m\alpha}{\hbar^2}, \quad E = -\frac{\hbar^2\kappa^2}{2m} = -\frac{m\alpha^2}{2\hbar^2}. \quad (2.63)$$

The delta-function well has exactly one bound state, whatever its strength. The wave function is the one we normalised in the example of Section 2.1, with $A = \sqrt{\kappa}$, and it is drawn in Figure 12. The stronger the well, the larger κ is, and the more tightly the particle is held near the origin.

The Finite Square Well

The finite square well has the potential

$$V(x) = \begin{cases} -V_0, & |x| < w/2, \\ 0, & |x| > w/2, \end{cases} \quad (2.64)$$

where $V_0 > 0$, so that the well has depth V_0 and width w . A bound state has $-V_0 < E < 0$. The upper limit is the definition of a bound state in this potential. The lower limit holds because the energy of

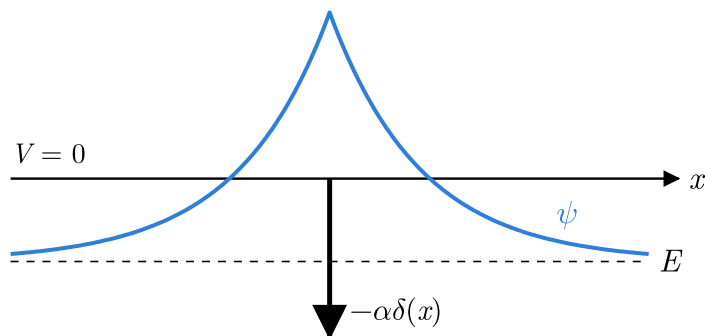


Figure 12. The delta-function well, drawn as a downward spike at the origin, and its single bound state $\psi(x) = \sqrt{\kappa} e^{-\kappa|x|}$, drawn on a baseline at the height of the energy $E = -m\alpha^2/2\hbar^2$. The derivative of the wave function jumps at the origin.

a stationary state is $\langle H \rangle = \langle p^2 \rangle / 2m + \langle V \rangle$, in which $\langle p^2 \rangle$ is never negative, as you showed in Problem 4 of Section 2.1, and $\langle V \rangle$ cannot be less than $-V_0$, the smallest value the potential takes. Inside the well the kinetic energy $E + V_0$ is positive, and the time-independent equation reads $\psi'' = -k^2\psi$, with

$$k = \frac{\sqrt{2m(E + V_0)}}{\hbar}, \quad (2.65)$$

while outside the well the region is classically forbidden, with $\kappa = \sqrt{-2mE}/\hbar$. The two constants are related by

$$k^2 + \kappa^2 = \frac{2mV_0}{\hbar^2}. \quad (2.66)$$

The potential is even, $V(-x) = V(x)$. If $\psi(x)$ solves the time-independent equation, then so does $\psi(-x)$, with the same energy, since changing x to $-x$ does not change the second derivative. Their sum $\psi(x) + \psi(-x)$ and their difference $\psi(x) - \psi(-x)$ are then solutions too, the sum being even and the difference odd, so we may look for even and odd solutions separately. We work out the even solutions here and leave the odd ones as a problem.

An even solution has the form

$$\psi(x) = \begin{cases} C \cos kx, & |x| < w/2, \\ Ge^{-\kappa|x|}, & |x| > w/2, \end{cases} \quad (2.67)$$

and since it is even we need only match at $x = w/2$. Continuity of ψ and of ψ' there gives

$$C \cos \frac{kw}{2} = Ge^{-\kappa w/2}, \quad -kC \sin \frac{kw}{2} = -\kappa Ge^{-\kappa w/2}. \quad (2.68)$$

Dividing the second equation by the first gives $\kappa = k \tan(kw/2)$. To simplify the subsequent expressions, let

$$z = \frac{kw}{2}, \quad z_0 = \frac{w}{2\hbar} \sqrt{2mV_0}, \quad (2.69)$$

so that the relation between k and κ gives $\kappa w/2 = \sqrt{z_0^2 - z^2}$. The condition for an even bound state becomes

$$\tan z = \sqrt{\frac{z_0^2}{z^2} - 1}. \quad (2.70)$$

The number z_0 combines the depth and the width of the well, and it is the only property of the well that enters the condition. The equation is transcendental, as the equation for the peak of the Planck spectrum was in Section 1.2, and we solve it graphically. Figure 13 shows both sides of the equation plotted against z . The right-hand side starts at infinity when z is small, falls, and reaches zero at $z = z_0$. The left-hand side has a branch rising from zero at each multiple of π , and every branch that starts below z_0 crosses the right-hand side once. Each crossing gives an allowed value of z , and hence of k and of the energy $E = \hbar^2 k^2/2m - V_0$.

There are two conclusions that we can draw directly from the graph.

1. There is always at least one even bound state, however shallow or narrow the well, since the first branch of $\tan z$ starts at zero, where the right-hand side is positive, and rises without limit as z approaches $\pi/2$, while the right-hand side falls to zero at $z = z_0$, so the two curves must cross before z reaches the smaller of z_0 and $\pi/2$.
2. The number of even bound states is the number of multiples of π , counting zero, that are less than z_0 , so a deeper or wider well holds more bound states.

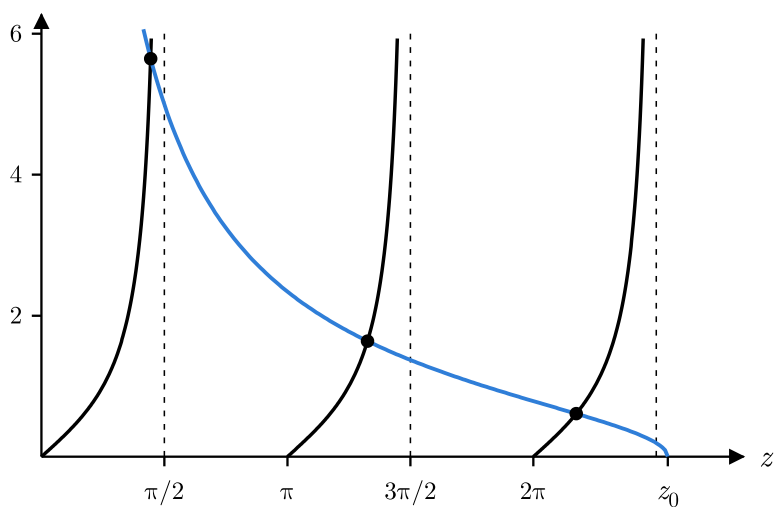


Figure 13. The graphical solution of $\tan z = \sqrt{z_0^2/z^2 - 1}$, drawn for $z_0 = 8$. The branches of $\tan z$ are solid and the right-hand side is the accent curve. Each crossing, marked with a dot, gives an even bound state; here there are three.

For a very deep or very wide well, z_0 is large, the right-hand side is very large for all the crossings with z well below z_0 , and those crossings lie close to the places where $\tan z$ becomes infinite, $z = \pi/2, 3\pi/2, 5\pi/2, \dots$. There $kw = n\pi$ with n odd, and

$$E + V_0 \approx \frac{n^2 \pi^2 \hbar^2}{2mw^2}, \quad (2.71)$$

which gives the energies of the infinite square well of the same width, measured from the bottom of the well, for odd n . The odd states supply the energies for even n , as you will show in Problem 2.

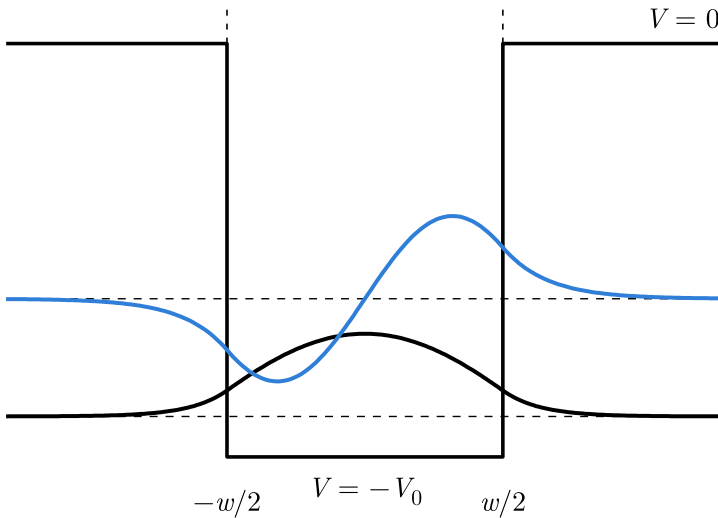


Figure 14. The ground state (solid) and the first excited state (accent) of a finite square well with $z_0 = 4$, computed from the matching conditions and drawn on baselines at their energies. The dashed lines mark the walls, and both wave functions decay exponentially outside them.

The wave functions of the two lowest states of a finite well are drawn in Figure 14. Inside the well they look like those of the infinite well, but they do not vanish at the walls, and they decay exponentially outside. The particle is found in the classically forbidden region with a probability that is small but not zero, and this is one of the features of quantum mechanics that has no classical counterpart. Since the ground state of the finite well does not have to vanish at the walls, its wavelength inside the well is longer than that of the ground state of the infinite well of the same width, and its energy

above the bottom of the well is correspondingly lower. As the well is made deeper, κ grows and the decay outside the walls becomes steeper, so the values of the wave function at the walls are pushed towards zero while the wave function stays continuous there, and its slope at the walls tends to a jump. This is the reason, promised in Section 2.2, that the wave function of the infinite square well must vanish at the walls even though its derivative jumps there.

2.4 The Free Particle and Wave Packets

A particle on which no force acts has $V = 0$ everywhere, and it would seem to be the simplest case of all. Its stationary states, however, cannot be normalised, and a real free particle has to be described by a combination of them. This section builds such a combination, called a wave packet, and follows its motion.

Plane Waves

With $V = 0$ the time-independent equation is $\psi'' = -k^2\psi$ with $k = \sqrt{2mE}/\hbar$, and the solutions can be written as e^{ikx} , with k now allowed to be positive or negative. The corresponding stationary states are

$$\Psi_k(x, t) = A e^{i(kx - \hbar k^2 t / 2m)}, \quad (2.72)$$

which are the plane waves of Section 2.1, travelling to the right for $k > 0$ and to the left for $k < 0$, with energy $E = \hbar^2 k^2 / 2m$ and momentum $\hbar k$. There is no condition at any boundary, so every real k is allowed and the energy is not quantised.

The plane waves have two features which show that they cannot by themselves describe a real particle. The first is that $|\Psi_k|^2 = |A|^2$ is the same everywhere, so the integral of $|\Psi_k|^2$ over all x is infinite and no choice of A normalises the wave. The second is that a crest of the wave moves at the speed $\omega/k = \hbar k / 2m$, which is called the phase velocity, and this is half the classical speed $p/m = \hbar k / m$ of a particle with the same momentum. Adding together plane waves with a range of wave numbers removes both difficulties: the sum can be normalised, and, as we show below, the envelope of the sum moves at the classical speed, although the crests within it still move at the phase velocity.

Fourier Transforms

In Section 2.2 a function on the interval $0 < x < w$ was written as a sum of sine waves whose wave numbers were spaced by π/w . As the interval is made longer, the spacing between the wave numbers shrinks, and in the limit of an infinite interval the sum becomes an integral. The result is called Fourier's theorem. It states that a reasonable function $f(x)$ whose square can be integrated over all x can be written as

$$f(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(k) e^{ikx} dk, \quad (2.73)$$

where the function $g(k)$ is given by

$$g(k) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} f(x) e^{-ikx} dx. \quad (2.74)$$

The function g is called the Fourier transform of f , and f is called the inverse Fourier transform of g . A second theorem of Fourier analysis states that

$$\int_{-\infty}^{\infty} |f(x)|^2 dx = \int_{-\infty}^{\infty} |g(k)|^2 dk, \quad (2.75)$$

which is called Plancherel's theorem. We quote Fourier's theorem and Plancherel's theorem, since their proofs are beyond this course. The function $g(k)$ gives the amount of each plane wave e^{ikx} contained in f , in the same way as the coefficient c_n gives the amount of each sine function ψ_n contained in the sine series of Section 2.2.

The Fourier transform gives the general solution for a free particle. If the wave function at $t = 0$ is $\Psi(x, 0)$, with Fourier transform $g(k)$, then

$$\Psi(x, t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} g(k) e^{i(kx - \hbar k^2 t / 2m)} dk, \quad (2.76)$$

since each plane wave in the integral satisfies the Schrödinger equation, the integral is a continuous version of the sum $\sum_n c_n \psi_n e^{-iE_n t / \hbar}$, and it agrees with $\Psi(x, 0)$ at $t = 0$. A wave function built in this way is called a wave packet. In this integral the function $g(k)$ takes the place of the coefficients c_n of that sum, and the integral over k takes the place of the sum over n . Plancherel's theorem shows that g is normalised whenever Ψ is, and the statistical interpretation takes $|g(k)|^2 dk$ to be the probability

that a measurement of the momentum gives a value between $\hbar k$ and $\hbar(k + dk)$. Each plane wave only acquires a phase factor of unit modulus as time passes, so this distribution of momenta does not change, as we would expect for a particle on which no force acts.

The Gaussian Integral

The wave packet we shall study has the shape of the normal distribution of school statistics, and to work with it we need the Gaussian integral,

$$\int_{-\infty}^{\infty} e^{-\beta u^2} du = \sqrt{\frac{\pi}{\beta}}, \quad (2.77)$$

where β is a positive constant. This integral is worked out in Section 1.2 of Vol. 2 of the Fermi Notes on Physics, by squaring it and slicing the plane into rings.

We shall also need the integral with a linear term in the exponent. Completing the square gives $-\beta u^2 + bu = -\beta(u - b/2\beta)^2 + b^2/4\beta$, and shifting the variable of integration by $b/2\beta$ does not change an integral over all u , so

$$\int_{-\infty}^{\infty} e^{-\beta u^2 + bu} du = \sqrt{\frac{\pi}{\beta}} e^{b^2/4\beta}. \quad (2.78)$$

We shall use this result with b imaginary, and later with β complex, with a positive real part. For imaginary b the shift of the variable would be through an imaginary amount, so the argument above does not apply, and the result has to be found in another way. Let $b = ik$ with k real, and let

$$\mathcal{J}(k) = \int_{-\infty}^{\infty} e^{-\beta u^2 + iku} du. \quad (2.79)$$

Differentiating under the integral sign with respect to k , and writing $u e^{-\beta u^2}$ as $-(1/2\beta)$ times the derivative of $e^{-\beta u^2}$, gives

$$\frac{d\mathcal{J}}{dk} = \int_{-\infty}^{\infty} iu e^{-\beta u^2} e^{iku} du = -\frac{i}{2\beta} \int_{-\infty}^{\infty} \frac{d e^{-\beta u^2}}{du} e^{iku} du. \quad (2.80)$$

Integrating by parts, with the boundary terms vanishing because $e^{-\beta u^2}$ does, turns the last integral into $-ik \mathcal{J}(k)$, so that $d\mathcal{J}/dk =$

$-(k/2\beta)\mathcal{J}$. The solution of this equation with $\mathcal{J}(0) = \sqrt{\pi/\beta}$ is

$$\mathcal{J}(k) = \sqrt{\frac{\pi}{\beta}} e^{-k^2/4\beta}, \quad (2.81)$$

which is the formula above with $b = ik$, since then $b^2 = -k^2$. The formula also remains true for complex β with a positive real part. Proving this requires the theory of functions of a complex variable, which is beyond the scope of this course.

A Gaussian Wave Packet

Suppose now that the wave function of a free particle at $t = 0$ is

$$\Psi(x, 0) = (2\pi\sigma^2)^{-1/4} e^{-x^2/4\sigma^2} e^{ik_0x}, \quad (2.82)$$

where σ and k_0 are constants. The factor e^{ik_0x} has unit modulus, so the probability density is

$$|\Psi(x, 0)|^2 = \frac{1}{\sqrt{2\pi}\sigma} e^{-x^2/2\sigma^2}, \quad (2.83)$$

which is the normal distribution with mean zero and standard deviation σ . The Gaussian integral with $\beta = 1/2\sigma^2$ confirms that its integral is one. The particle is therefore localised near the origin with $\sigma_x = \sigma$, and its mean momentum is $\hbar k_0$, as you showed in Problem 3 of Section 2.1.

The Fourier transform of the wave function is

$$g(k) = \frac{(2\pi\sigma^2)^{-1/4}}{\sqrt{2\pi}} \int_{-\infty}^{\infty} e^{-x^2/4\sigma^2 - i(k-k_0)x} dx, \quad (2.84)$$

which is the integral above with $\beta = 1/4\sigma^2$ and $b = -i(k - k_0)$. Since $b^2/4\beta = -\sigma^2(k - k_0)^2$, this gives

$$g(k) = \left(\frac{2\sigma^2}{\pi}\right)^{1/4} e^{-\sigma^2(k-k_0)^2}. \quad (2.85)$$

The distribution of momenta is therefore

$$|g(k)|^2 = \sqrt{\frac{2}{\pi}} \sigma e^{-2\sigma^2(k-k_0)^2}, \quad (2.86)$$

which is also a normal distribution, with mean k_0 and standard deviation $1/2\sigma$ in k . The standard deviation of the momentum is

therefore $\sigma_p = \hbar/2\sigma$, and

$$\sigma_x \sigma_p = \frac{\hbar}{2}. \quad (2.87)$$

The narrower the packet is in position, the broader it is in momentum. This is a property of Fourier transforms, since a narrow packet can only be built by adding waves with a wide range of wave numbers, which interfere destructively everywhere except in a small region.

The Motion of the Packet

The wave function at time t is $(2\pi)^{-1/2}$ times the integral over k of $g(k) e^{i(kx - \hbar k^2 t/2m)}$. Let $u = k - k_0$, and let $v_0 = \hbar k_0/m$. The exponent in the integrand can then be rearranged as

$$\begin{aligned} & -\sigma^2 u^2 + i(k_0 + u)x - \frac{i\hbar t}{2m} (k_0 + u)^2 \\ & = -\left(\sigma^2 + \frac{i\hbar t}{2m}\right) u^2 + i(x - v_0 t)u + i\left(k_0 x - \frac{\hbar k_0^2 t}{2m}\right). \end{aligned} \quad (2.88)$$

The last term does not depend on u and comes outside the integral as a factor of unit modulus. The remaining integral is the Gaussian integral with $\beta = \sigma^2 + i\hbar t/2m$ and $b = i(x - v_0 t)$, and it gives

$$\Psi(x, t) = \left(\frac{\sigma^2}{2\pi}\right)^{1/4} \frac{1}{\sqrt{\beta}} e^{-(x - v_0 t)^2/4\beta} e^{i(k_0 x - \hbar k_0^2 t/2m)}. \quad (2.89)$$

To find the probability density we need the modulus of $e^{-(x - v_0 t)^2/4\beta}$. Multiplying the numerator and the denominator of $1/\beta$ by the complex conjugate of β gives $1/\beta = \beta^*/|\beta|^2$, whose real part is $\sigma^2/|\beta|^2$, and $|\beta|^2 = \sigma^4 + \hbar^2 t^2/4m^2$. The modulus of an exponential depends only on the real part of its exponent, so

$$\begin{aligned} |\Psi(x, t)|^2 &= \frac{1}{\sqrt{2\pi} \sigma(t)} e^{-(x - v_0 t)^2/2\sigma(t)^2}, \\ \sigma(t) &= \sigma \sqrt{1 + \left(\frac{\hbar t}{2m\sigma^2}\right)^2}, \end{aligned} \quad (2.90)$$

where the factor in front is the squared modulus of $(\sigma^2/2\pi)^{1/4} \beta^{-1/2}$, which is $\sigma/(\sqrt{2\pi} |\beta|)$, and $|\beta| = \sigma \sigma(t)$. The density is therefore normalised at every time, as Section 2.1 showed it must be. This result

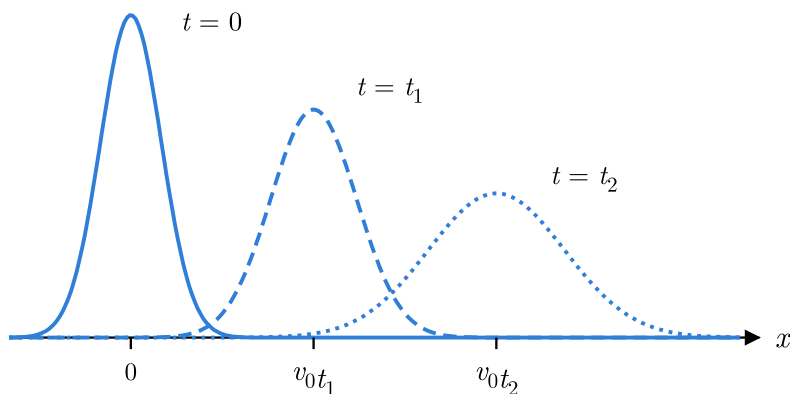


Figure 15. The probability density of a Gaussian wave packet at $t = 0$ and at two later times. The centre moves at $v_0 = \hbar k_0/m$, the width grows as $\sigma(t)$, and the area under each curve remains one. The times are chosen so that $\hbar t/2m\sigma^2 = 0, 1$, and 2 .

is drawn in Figure 15. The packet remains a normal distribution at all times, and there are two changes as time passes.

The centre of the packet moves at the speed $v_0 = \hbar k_0/m = p/m$, which is the classical velocity of a particle with the mean momentum. It is the speed at which the envelope of the packet moves, which is called the group velocity, and it is different from the phase velocity of the waves of which the packet is made. The group velocity is $d\omega/dk$ evaluated at k_0 , which for $\omega = \hbar k^2/2m$ is $\hbar k_0/m$, and in Problem 1 you will see why a group of waves moves at this speed.

The packet also spreads. For times short compared with $2m\sigma^2/\hbar$ its width hardly changes, while for much longer times $\sigma(t) \approx \hbar t/2m\sigma = \sigma_p t/m$. The spread of velocities in the packet is σ_p/m , and at long times the width grows as it would for a group of classical particles leaving the origin with this spread of velocities. A packet that starts narrow has a broad spread of momenta, and it therefore spreads quickly.

2.5 Potential Steps and Tunnelling

A particle whose energy is greater than the potential at large distances is not bound, and it can come in from far away, meet a change in the potential, and travel off again. States of this kind are called scattering states. For them the energy is not quantised, and the questions of interest are how likely the particle is to be reflected and how likely it is to be transmitted. We answer these questions first for a sudden step in the potential and then for a rectangular barrier, where quantum mechanics allows a particle to pass through a region that is forbidden to it classically.

Reflection and Transmission

A particle in a scattering state is described by a wave packet, as in Section 2.4, but when the packet is long compared with the size of the region where the potential changes, it contains a narrow range of wave numbers, and we may work with a single plane wave of definite energy instead. A wave Ae^{ikx} coming in from the left meets the potential, and part of it is reflected as Be^{-ikx} and part transmitted beyond. As you showed in Problem 2 of Section 2.1, the incoming and reflected waves carry separate probability currents, $\hbar k|A|^2/m$ and $-\hbar k|B|^2/m$. The probability that the particle is reflected is the ratio of the magnitude of the reflected current to the incident current, and the probability that it is transmitted is the ratio of the transmitted current to the incident current. We write these probabilities as P_R and P_T . Probability is conserved, so $P_R + P_T = 1$.

The Potential Step

Let the potential be

$$V(x) = \begin{cases} 0, & x < 0, \\ V_0, & x > 0, \end{cases} \quad (2.91)$$

with $V_0 > 0$, and let a particle of energy $E > V_0$ approach from the left. A classical particle would always pass the step, losing kinetic energy as it did so. On the left the solution is

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \quad k = \frac{\sqrt{2mE}}{\hbar}, \quad (2.92)$$

and on the right, where only a transmitted wave travels, it is

$$\psi(x) = Ce^{ik'x}, \quad k' = \frac{\sqrt{2m(E - V_0)}}{\hbar}. \quad (2.93)$$

The continuity of ψ and of ψ' at the origin gives

$$A + B = C, \quad k(A - B) = k'C. \quad (2.94)$$

Solving for B and C in terms of A gives

$$B = \frac{k - k'}{k + k'} A, \quad C = \frac{2k}{k + k'} A. \quad (2.95)$$

The transmitted current is $\hbar k' |C|^2 / m$, so the reflection and transmission probabilities are

$$P_R = \left(\frac{k - k'}{k + k'} \right)^2, \quad P_T = \frac{k'}{k} \frac{|C|^2}{|A|^2} = \frac{4kk'}{(k + k')^2}, \quad (2.96)$$

and you will check in Problem 1 that they add to one. The particle is reflected with a probability that is not zero, even though it has more than enough energy to pass the step. The reflection is weak when E is much larger than V_0 , since k' is then close to k , and it becomes complete as E falls to V_0 , where $k' \rightarrow 0$. The factor k'/k in the transmission probability is needed because the transmitted wave travels at a different speed from the incident one, and it is the current, the product of the density and the speed, that measures the flow of probability.

Now let $E < V_0$, so that the region $x > 0$ is classically forbidden. On the right the solution that does not grow with x is $Ce^{-\kappa x}$ with $\kappa = \sqrt{2m(V_0 - E)}/\hbar$, and the matching conditions become $A + B = C$ and $ik(A - B) = -\kappa C$. They give

$$\frac{B}{A} = \frac{k - i\kappa}{k + i\kappa}, \quad (2.97)$$

which is a ratio of a complex number to its conjugate, and so has modulus one. Every particle is reflected, as it would be classically. The wave function nevertheless penetrates into the forbidden region, decaying over a distance of order $1/\kappa$, as shown in Figure 16. The real exponential $Ce^{-\kappa x}$ multiplied by a constant phase carries no current, which agrees with $P_R = 1$.

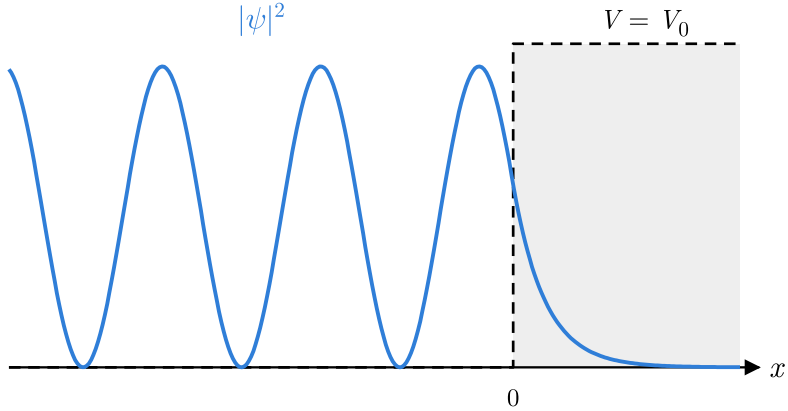


Figure 16. The probability density $|\psi|^2$ for a particle with $E < V_0$ incident on a potential step from the left. On the left the incident and reflected waves interfere to give a standing pattern. On the right, in the classically forbidden region, the density decays exponentially over a distance of order $1/\kappa$.

The Rectangular Barrier

Let the potential be

$$V(x) = \begin{cases} V_0, & 0 < x < w, \\ 0, & \text{otherwise,} \end{cases} \quad (2.98)$$

with $V_0 > 0$, and let a particle of energy $E < V_0$ approach from the left. Classically the particle would be reflected with certainty. In the three regions the solutions are

$$\psi(x) = \begin{cases} Ae^{ikx} + Be^{-ikx}, & x < 0, \\ Ce^{\kappa x} + Ge^{-\kappa x}, & 0 < x < w, \\ Fe^{ikx}, & x > w, \end{cases} \quad (2.99)$$

with $k = \sqrt{2mE}/\hbar$ and $\kappa = \sqrt{2m(V_0 - E)}/\hbar$ as before. The growing and the decaying exponentials are both allowed inside the barrier, since it has a finite width. The transmitted wave on the right has the same wave number as the incident one, so $P_T = |F|^2/|A|^2$, and we need the ratio F/A .

We work from right to left. Matching ψ and ψ' at $x = w$ gives

$$Ce^{\kappa w} + Ge^{-\kappa w} = Fe^{ikw}, \quad \kappa(Ce^{\kappa w} - Ge^{-\kappa w}) = ikFe^{ikw}, \quad (2.100)$$

and adding and subtracting the first equation and the second divided by κ gives

$$C = \frac{1}{2} \left(1 + \frac{ik}{\kappa}\right) Fe^{ikw} e^{-\kappa w}, \quad G = \frac{1}{2} \left(1 - \frac{ik}{\kappa}\right) Fe^{ikw} e^{\kappa w}. \quad (2.101)$$

Matching at $x = 0$ gives $A + B = C + G$ and $ik(A - B) = \kappa(C - G)$. Adding the first of these to the second divided by ik eliminates B :

$$2A = \left(1 - \frac{i\kappa}{k}\right) C + \left(1 + \frac{i\kappa}{k}\right) G. \quad (2.102)$$

Substituting C and G requires the products

$$\begin{aligned} \left(1 + \frac{ik}{\kappa}\right) \left(1 - \frac{i\kappa}{k}\right) &= 2 + i \left(\frac{k}{\kappa} - \frac{\kappa}{k}\right), \\ \left(1 - \frac{ik}{\kappa}\right) \left(1 + \frac{i\kappa}{k}\right) &= 2 - i \left(\frac{k}{\kappa} - \frac{\kappa}{k}\right), \end{aligned} \quad (2.103)$$

and collecting the terms in $e^{\kappa w}$ and $e^{-\kappa w}$ into the hyperbolic functions $\cosh \kappa w = \frac{1}{2}(e^{\kappa w} + e^{-\kappa w})$ and $\sinh \kappa w = \frac{1}{2}(e^{\kappa w} - e^{-\kappa w})$ gives

$$A = Fe^{ikw} \left[\cosh \kappa w + \frac{i(\kappa^2 - k^2)}{2k\kappa} \sinh \kappa w \right]. \quad (2.104)$$

The squared modulus of the bracket is $\cosh^2 \kappa w + [(\kappa^2 - k^2)/2k\kappa]^2 \sinh^2 \kappa w$. Using $\cosh^2 = 1 + \sinh^2$ and the identity $(\kappa^2 - k^2)^2 + 4k^2\kappa^2 = (\kappa^2 + k^2)^2$, this becomes $1 + [(\kappa^2 + k^2)/2k\kappa]^2 \sinh^2 \kappa w$. Finally, $\kappa^2 + k^2 = 2mV_0/\hbar^2$ and $k^2\kappa^2 = 4m^2E(V_0 - E)/\hbar^4$, so the transmission probability is

$$P_T = \left[1 + \frac{V_0^2 \sinh^2 \kappa w}{4E(V_0 - E)} \right]^{-1}. \quad (2.105)$$

The transmission probability is not zero. A particle can pass through a barrier that it does not have the energy to cross, and this effect is called tunnelling. The probability density inside the barrier, shown in Figure 17, decays but does not reach zero, and what remains at the far side continues as a transmitted wave of constant density.

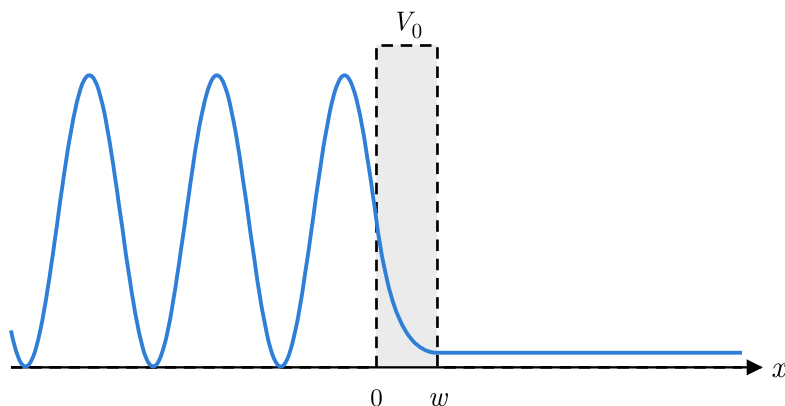


Figure 17. The probability density for a particle of energy $E < V_0$ incident on a rectangular barrier of width w , computed from the solution in the text. The density decays through the barrier and emerges on the right as a transmitted wave of constant density, while on the left the incident and reflected waves interfere.

For a barrier that is thick compared with $1/\kappa$, so that $\kappa w \gg 1$, we have $\sinh \kappa w \approx \frac{1}{2}e^{\kappa w}$, the second term in the bracket is much larger than one, and

$$P_T \approx \frac{16E(V_0 - E)}{V_0^2} e^{-2\kappa w}. \quad (2.106)$$

The factor in front is never larger than four, and the exponential dominates. Doubling the width of the barrier squares the exponential factor, and the dependence on $\kappa = \sqrt{2m(V_0 - E)}/\hbar$ means that a heavier particle, or one further below the top of the barrier, tunnels far less readily. For energies above the barrier the transmission probability is found in the same way, and you will show in Problem 4 that it reaches one at certain energies, as Figure 18 shows.

Examples of Tunnelling

Tunnelling explains alpha decay. An alpha particle inside a heavy nucleus is held by the strong nuclear force, and outside the nucleus it is repelled by the Coulomb force of the remaining protons. The

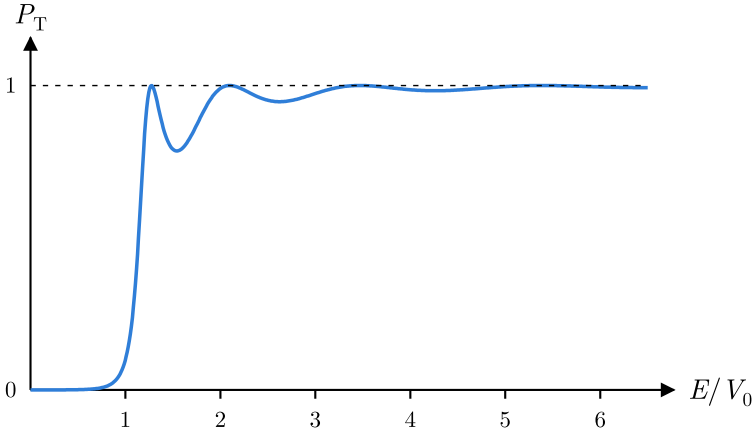


Figure 18. The transmission probability P_T through a rectangular barrier, plotted against E/V_0 , for a barrier with $w\sqrt{2mV_0}/\hbar = 6$. Below the top of the barrier the particle tunnels with a small probability. Above it the transmission is not always complete, and it reaches one only at the energies found in Problem 4.

potential energy therefore rises to a peak at the surface of the nucleus and falls away beyond it, and the alpha particle has an energy below the peak. Classically it could never escape. Gamow, and independently Gurney and Condon, showed in 1928 that the alpha particle tunnels through the Coulomb barrier with a small probability each time it strikes the surface. Since the probability depends exponentially on the width and height of the barrier, a modest change in the energy of the alpha particle changes the lifetime of the nucleus by a large factor, and this accounts for the observed relation between the energies of alpha particles and the half-lives of the nuclei that emit them, which span more than twenty orders of magnitude.

Tunnelling is also the basis of the scanning tunnelling microscope, built by Binnig and Rohrer in 1981. A sharp metal tip is held a very small distance above a conducting surface, with a small potential difference between them. The gap is a barrier for the electrons in the metal, and a current flows only by tunnelling. Since the current depends exponentially on the width of the gap, a change in

the height of the tip by a small fraction of the diameter of an atom changes the current measurably. Moving the tip across the surface while adjusting its height to keep the current constant traces out the positions of the individual atoms of the surface.

2.6 The Harmonic Oscillator

The harmonic oscillator is the last of the one-dimensional systems in this chapter, and one of the most widely applied. We solve it in two ways. The first uses operators, and the second solves the differential equation directly by a power series.

Small Oscillations

Near a minimum at x_0 , any smooth potential can be expanded as a Taylor series,

$$V(x) = V(x_0) + V'(x_0)(x - x_0) + \frac{1}{2}V''(x_0)(x - x_0)^2 + \dots, \quad (2.107)$$

in which $V'(x_0) = 0$ because x_0 is a minimum. The constant $V(x_0)$ only shifts the zero of energy, and for small displacements the cubic and higher terms can be neglected, so the potential is that of a harmonic oscillator, as in the treatment of small oscillations in Vol. 1 of the Fermi Notes on Physics. Choosing the origin at x_0 and writing $V''(x_0) = m\omega^2$, where ω is the angular frequency of the classical oscillation, the potential is

$$V(x) = \frac{1}{2}m\omega^2x^2. \quad (2.108)$$

The vibrations of the atoms in a diatomic molecule about their equilibrium separation, and of the atoms of a crystal about their lattice sites, are described in this way, as long as the vibrations do not grow too large. The time-independent Schrödinger equation is

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + \frac{1}{2}m\omega^2x^2\psi = E\psi. \quad (2.109)$$

The Commutator of Position and Momentum

Operators act on functions, and two operators applied one after the other give another operator. The order matters. If $f(x)$ is any

function, then

$$\hat{x}\hat{p}f = -i\hbar x \frac{df}{dx}, \quad \hat{p}\hat{x}f = -i\hbar \frac{d}{dx}(xf) = -i\hbar f - i\hbar x \frac{df}{dx}, \quad (2.110)$$

where the second result uses the product rule. The difference between applying two operators in one order and in the other is called their commutator,

$$[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}, \quad (2.111)$$

and the expressions for $\hat{x}\hat{p}f$ and $\hat{p}\hat{x}f$ above show that $[\hat{x}, \hat{p}]f = i\hbar f$ for every f . Since the function is arbitrary, we write

$$[\hat{x}, \hat{p}] = i\hbar, \quad (2.112)$$

which is called the canonical commutation relation. The function f was needed only to keep track of what the derivative acts on, and the relation is a statement about the operators themselves.

Ladder Operators

The Hamiltonian of the oscillator is $\hat{H} = (\hat{p}^2 + m^2\omega^2\hat{x}^2)/2m$. If $\hat{p}$ and $\hat{x}$ were numbers, the sum of their squares could be factorised as $(i\hat{p} + m\omega\hat{x})(-i\hat{p} + m\omega\hat{x})$, and this suggests that we define the operators

$$\hat{a}_{\pm} = \frac{1}{\sqrt{2\hbar m\omega}} (\mp i\hat{p} + m\omega\hat{x}). \quad (2.113)$$

Since the operators do not commute, the product differs from the sum of squares. Multiplying out,

$$\hat{a}_-\hat{a}_+ = \frac{1}{2\hbar m\omega} [\hat{p}^2 + m^2\omega^2\hat{x}^2 - im\omega(\hat{x}\hat{p} - \hat{p}\hat{x})] = \frac{1}{\hbar\omega}\hat{H} + \frac{1}{2}, \quad (2.114)$$

where the canonical commutation relation has been used in the last step. A similar calculation gives $\hat{a}_+\hat{a}_- = \hat{H}/\hbar\omega - \frac{1}{2}$. Subtracting the second result from the first gives $[\hat{a}_-, \hat{a}_+] = 1$, and the Hamiltonian can be written in either of the forms

$$\hat{H} = \hbar\omega \left(\hat{a}_+\hat{a}_- + \frac{1}{2} \right) = \hbar\omega \left(\hat{a}_-\hat{a}_+ - \frac{1}{2} \right). \quad (2.115)$$

Now suppose that ψ is a solution of the time-independent equation with energy E . Then $\hat{a}_+\psi$ is also a solution, with energy $E + \hbar\omega$.

To see this, we apply $\hat{H}$ in its first form to $\hat{a}_+\psi$:

$$\begin{aligned}\hat{H}\hat{a}_+\psi &= \hbar\omega\left(\hat{a}_+\hat{a}_-\hat{a}_+ + \frac{1}{2}\hat{a}_+\right)\psi = \hat{a}_+\hbar\omega\left(\hat{a}_-\hat{a}_+ + \frac{1}{2}\right)\psi \\ &= \hat{a}_+(\hat{H} + \hbar\omega)\psi = (E + \hbar\omega)\hat{a}_+\psi,\end{aligned}\quad (2.116)$$

where the second form of $\hat{H}$ was used to write $\hbar\omega(\hat{a}_-\hat{a}_+ + \frac{1}{2}) = \hat{H} + \hbar\omega$. The same argument shows that $\hat{a}_-\psi$ is a solution with energy $E - \hbar\omega$. The operator $\hat{a}_+$ is called the raising operator and $\hat{a}_-$ the lowering operator, and together they are called ladder operators, since they move a state up or down a ladder of energies spaced by $\hbar\omega$.

The ladder cannot go down for ever. The energy of any state of the oscillator is the sum of $\langle p^2 \rangle / 2m$ and $\frac{1}{2}m\omega^2\langle x^2 \rangle$, and neither of these can be negative, since $\langle x^2 \rangle$ is the integral of $x^2|\psi|^2$, and $\langle p^2 \rangle$ is the integral of $\hbar^2|d\psi/dx|^2$, as you showed in Problem 4 of Section 2.1. There must therefore be a lowest state ψ_0 , and applying the lowering operator to it must give a function that is not an acceptable state, which can only be zero:

$$\hat{a}_-\psi_0 = 0. \quad (2.117)$$

With $\hat{p} = -i\hbar d/dx$ the lowering operator is $(\hbar d/dx + m\omega x)/\sqrt{2\hbar m\omega}$, so this condition is the first-order equation

$$\frac{d\psi_0}{dx} = -\frac{m\omega}{\hbar}x\psi_0, \quad (2.118)$$

which can be solved by dividing both sides by ψ_0 and integrating with respect to x , and gives

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-m\omega x^2/2\hbar}, \quad (2.119)$$

where the constant has been fixed by the Gaussian integral of Section 2.4 with $\beta = m\omega/\hbar$. The energy of the ground state follows from the first form of the Hamiltonian, since $\hat{H}\psi_0 = \hbar\omega(\hat{a}_+\hat{a}_-\psi_0 + \frac{1}{2}\psi_0) = \frac{1}{2}\hbar\omega\psi_0$. Applying the raising operator repeatedly then gives the whole spectrum,

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \quad n = 0, 1, 2, \dots, \quad (2.120)$$

with ψ_n proportional to $(\hat{a}_+)^n\psi_0$. Every stationary state lies on this ladder. If ψ is any solution, applying the lowering operator to

it repeatedly must eventually give zero, since otherwise the energy would fall below zero, and the last function before zero satisfies $\hat{a}_-\psi_0 = 0$, whose only solution is the Gaussian above. The energy of ψ therefore exceeds $\frac{1}{2}\hbar\omega$ by a whole number of steps of $\hbar\omega$. The levels are equally spaced, which is the property Planck assumed for his oscillators, and the lowest has an energy $\frac{1}{2}\hbar\omega$ above the bottom of the potential, which is called the zero-point energy. A particle resting at the bottom of the well would have a definite position and a definite momentum, and the ground state has neither; its wave function is spread over a region of width of order $\sqrt{\hbar/m\omega}$, and its kinetic and potential energies cannot both vanish. The zero-point energy is measurable. The vibrations of the molecules H_2 and D_2 have different zero-point energies, since the frequency depends on the masses of the atoms, and the two molecules have different dissociation energies as a result.

Normalising the Excited States

The functions $(\hat{a}_+)^n\psi_0$ are not normalised, and to normalise them we need the following property of the ladder operators. For any functions f and g that vanish at infinity,

$$\int_{-\infty}^{\infty} f^* (\hat{a}_{\pm} g) dx = \int_{-\infty}^{\infty} (\hat{a}_{\mp} f)^* g dx, \quad (2.121)$$

which you will prove in Problem 6 by integrating by parts. Since $\hat{a}_+\psi_n$ has the energy E_{n+1} , it is a multiple of the normalised state ψ_{n+1} , and the squared modulus of that multiple is found by using the property above with $f = \psi_n$ and $g = \hat{a}_+\psi_n$, together with the result $\hat{a}_-\hat{a}_+\psi_n = (E_n/\hbar\omega + \frac{1}{2})\psi_n = (n+1)\psi_n$:

$$\int_{-\infty}^{\infty} (\hat{a}_+\psi_n)^* (\hat{a}_+\psi_n) dx = \int_{-\infty}^{\infty} \psi_n^* (\hat{a}_-\hat{a}_+\psi_n) dx = n+1. \quad (2.122)$$

The phase of ψ_{n+1} can be chosen so that the multiple is real and positive, which gives $\hat{a}_+\psi_n = \sqrt{n+1}\psi_{n+1}$. A similar calculation gives $\hat{a}_-\psi_n = \sqrt{n}\psi_{n-1}$, and therefore

$$\begin{aligned} \hat{a}_+\psi_n &= \sqrt{n+1}\psi_{n+1}, & \hat{a}_-\psi_n &= \sqrt{n}\psi_{n-1}, \\ \psi_n &= \frac{1}{\sqrt{n!}} (\hat{a}_+)^n \psi_0. \end{aligned} \quad (2.123)$$

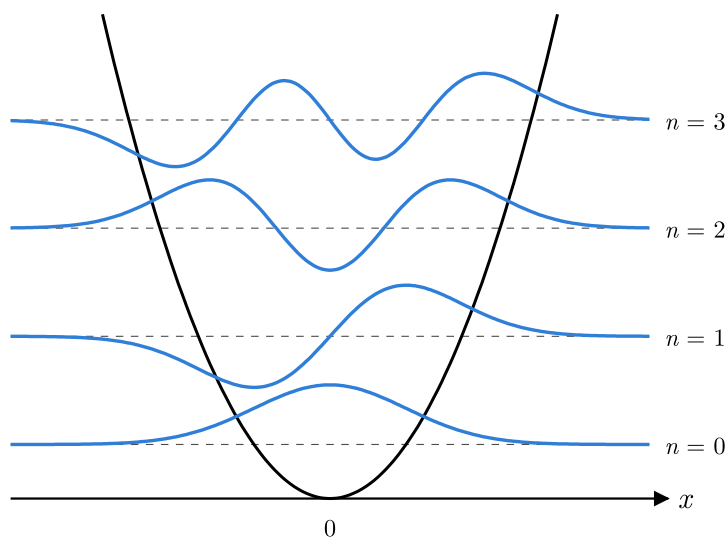


Figure 19. The potential $\frac{1}{2}m\omega^2x^2$ and the first four stationary states of the harmonic oscillator, each drawn on a baseline at the height of its energy $(n + \frac{1}{2})\hbar\omega$. The n th state has n nodes, and each wave function extends a little beyond the points where its baseline meets the potential.

As an example, the first excited state is

$$\psi_1 = \hat{a}_+ \psi_0 = \frac{1}{\sqrt{2\hbar m\omega}} \left(-\hbar \frac{d}{dx} + m\omega x \right) \psi_0 = \sqrt{\frac{2m\omega}{\hbar}} x \psi_0, \quad (2.124)$$

since $-\hbar d\psi_0/dx = m\omega x\psi_0$. The first four states are drawn in Figure 19.

The same property of the ladder operators shows that states of different energy are orthogonal. Applying it to $\int \psi_{n'}^* (\hat{a}_+ \hat{a}_- \psi_n) dx$ and using $\hat{a}_+ \hat{a}_- \psi_n = n\psi_n$ gives

$$n \int \psi_{n'}^* \psi_n dx = \int (\hat{a}_- \psi_{n'})^* (\hat{a}_- \psi_n) dx = n' \int \psi_{n'}^* \psi_n dx, \quad (2.125)$$

where the last step uses the same property to move the lowering operator from ψ_n onto $\psi_{n'}$, as a raising operator, together with $\hat{a}_+ \hat{a}_- \psi_{n'} = n'\psi_{n'}$. Therefore $(n - n') \int \psi_{n'}^* \psi_n dx = 0$, and the integral vanishes when $n' \neq n$.

The Power-Series Method

The same energies can be found by solving the differential equation directly. Let $\xi = \sqrt{m\omega/\hbar} x$ and $K = 2E/\hbar\omega$, so that the time-independent equation becomes

$$\frac{d^2\psi}{d\xi^2} = (\xi^2 - K) \psi. \quad (2.126)$$

For large $|\xi|$ the term K is negligible compared with ξ^2 , and the equation is satisfied approximately by $e^{-\xi^2/2}$, whose second derivative is $(\xi^2 - 1)e^{-\xi^2/2}$. This suggests that we write

$$\psi(\xi) = h(\xi) e^{-\xi^2/2} \quad (2.127)$$

and look for an equation for h . By the product rule, $\psi' = (h' - \xi h)e^{-\xi^2/2}$ and $\psi'' = (h'' - 2\xi h' + (\xi^2 - 1)h)e^{-\xi^2/2}$, where primes now denote derivatives with respect to ξ , and substituting these gives

$$\frac{d^2 h}{d\xi^2} - 2\xi \frac{dh}{d\xi} + (K - 1)h = 0, \quad (2.128)$$

which is called Hermite's equation. We solve it by writing h as a power series,

$$h(\xi) = \sum_{j=0}^{\infty} a_j \xi^j. \quad (2.129)$$

This method is called the power-series method, and it works for many of the differential equations one encounters in physics. The series is substituted into the equation, and the coefficients of each power of ξ are collected. A power series that equals zero for every ξ must have every coefficient zero, since otherwise the lowest power with a non-zero coefficient would dominate for small ξ , and this condition gives equations for the a_j . Differentiating term by term gives $h' = \sum_j j a_j \xi^{j-1}$ and $h'' = \sum_j j(j-1) a_j \xi^{j-2}$. In the second series we shift the label of summation by two, so that it reads $\sum_j (j+2)(j+1) a_{j+2} \xi^j$, and Hermite's equation becomes

$$\sum_{j=0}^{\infty} [(j+1)(j+2)a_{j+2} - 2ja_j + (K-1)a_j] \xi^j = 0. \quad (2.130)$$

Setting each coefficient to zero gives the recursion relation

$$a_{j+2} = \frac{2j+1-K}{(j+1)(j+2)} a_j. \quad (2.131)$$

Starting from a_0 , the relation fixes $a_2, a_4, \dots$, and starting from a_1 it fixes $a_3, a_5, \dots$, so the general solution is the sum of an even series and an odd series with two arbitrary constants, as a second-order equation requires.

For most values of K the series does not end, and then the solution is not acceptable. For large j the recursion relation gives $a_{j+2} \approx (2/j) a_j$. The series for $e^{\xi^2} = \sum_i \xi^{2i}/i!$ has the same property, since the ratio of the coefficients of ξ^{2i+2} and ξ^{2i} is $1/(i+1)$, which is approximately $2/j$ with $j = 2i$. A series that does not end therefore grows like e^{ξ^2} at large ξ , and $\psi = h e^{-\xi^2/2}$ grows like $e^{\xi^2/2}$, which cannot be normalised. The series must end, which happens when the numerator of the recursion relation vanishes for some $j = n$, that is when

$$K = 2n + 1, \quad E = \left(n + \frac{1}{2}\right) \hbar \omega, \quad (2.132)$$

with the other series, the odd one if n is even and the even one if n is odd, set to zero. This is the same spectrum that the ladder operators gave. For each n the function h is a polynomial of degree n . With the overall constant chosen so that the coefficient of ξ^n is 2^n , these polynomials are called the Hermite polynomials $H_n(\xi)$,

and the first four are

$$H_0 = 1, \quad H_1 = 2\xi, \quad H_2 = 4\xi^2 - 2, \quad H_3 = 8\xi^3 - 12\xi. \quad (2.133)$$

For example, with $n = 2$ we have $K = 5$ and $a_2 = (1 - 5)a_0/2 = -2a_0$, and choosing $a_0 = -2$ gives H_2 . The normalised stationary states are

$$\psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} \frac{1}{\sqrt{2^n n!}} H_n(\xi) e^{-\xi^2/2}, \quad (2.134)$$

which agree with the states given by the ladder operators, as you will check in Problem 2 for $n = 1$ and $n = 2$.

Comparison with the Classical Oscillator

A classical oscillator of amplitude x_A moves as $x = x_A \sin \omega t$ and spends most of its time near the ends of its swing, where it moves slowly. The time it spends in an interval dx during one half-period π/ω is $dx/|v|$, where $|v| = \omega\sqrt{x_A^2 - x^2}$, so the probability of finding it in dx at a random instant is

$$P_{\text{cl}}(x) dx = \frac{\omega}{\pi} \frac{dx}{|v|} = \frac{dx}{\pi\sqrt{x_A^2 - x^2}}, \quad |x| < x_A. \quad (2.135)$$

Figure 20 compares this distribution with $|\psi_{20}|^2$, taking the amplitude for which the classical energy $\frac{1}{2}m\omega^2 x_A^2$ equals E_{20} . The quantum density oscillates rapidly, but its average over a few oscillations follows the classical curve, rising towards the turning points. The quantum density also extends a little beyond the turning points, into the classically forbidden region. For the ground state the quantum and the classical densities are quite different, since $|\psi_0|^2$ is largest at the centre, where the classical particle is least likely to be. The requirement that quantum results approach classical ones for highly excited states is called the correspondence principle, and it was used by Bohr as a guide in constructing the old quantum theory.

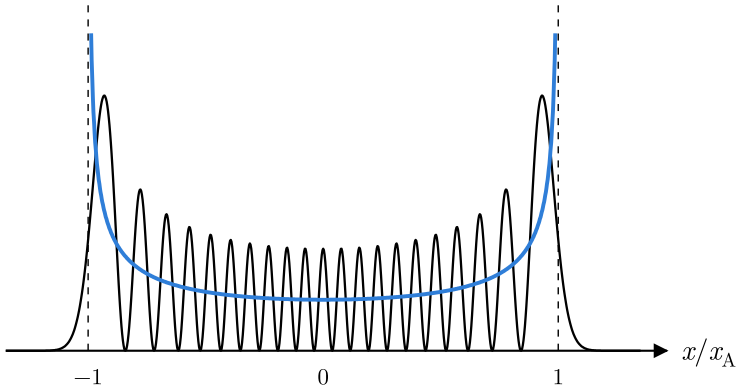


Figure 20. The probability density $|\psi_{20}|^2$ of the twentieth excited state of the oscillator (solid), and the classical probability density $P_{\text{cl}}(x)$ for an oscillator of the same energy (accent), plotted against x/x_A . The dashed lines mark the classical turning points.

Problems

§2.1 The Schrödinger Equation and the Wave Function

1. Show that if Ψ_1 and Ψ_2 satisfy the Schrödinger equation, so does $c_1\Psi_1 + c_2\Psi_2$. Show also that $\cos(kx - \omega t)$ does not satisfy the free Schrödinger equation for any real, non-zero k and ω .
2. Show that the probability current of $\Psi = Ae^{i(kx - \omega t)} + Be^{-i(kx + \omega t)}$, with k real, is $j = (\hbar k/m)(|A|^2 - |B|^2)$, so that waves travelling in opposite directions contribute separately to the current.
3. Let $\psi(x)$ be a real, normalised function, and let $\Psi(x) = \psi(x)e^{ik_0x}$ with k_0 real. Show that Ψ is normalised and that $\langle p \rangle = \hbar k_0$.
4. Show that $\langle p^2 \rangle = \hbar^2 \int_{-\infty}^{\infty} |\partial\Psi/\partial x|^2 dx$ for any normalisable Ψ , and hence that $\langle p^2 \rangle$ is never negative.

§2.2 Stationary States and the Infinite Square Well

1. Show that the time-independent Schrödinger equation for the infinite square well has no solution other than $\psi = 0$ when $E < 0$ or $E = 0$.
2. Show that in the state ψ_n of the infinite square well $\langle x \rangle = w/2$ and $\langle x^2 \rangle = w^2(\frac{1}{3} - \frac{1}{2n^2\pi^2})$. Find σ_x and describe what happens to it as n becomes large.
3. Using $\langle H \rangle = \sum_n |c_n|^2 E_n$ and $\sum_n |c_n|^2 = 1$, show that $\langle H \rangle \geq E_1$ for any state of the infinite square well.
4. A particle in the well has $\Psi(x, 0) = A[\psi_1(x) + \psi_2(x) + \psi_3(x)]$. Find A , the probability of each possible result of a measurement of the energy, and $\langle H \rangle$ in terms of E_1 .
5. Show that $\langle p \rangle = 0$ and $\langle p^2 \rangle = 2mE_n$ in the stationary state ψ_n of the well, and explain the second result in terms of the energy of the particle.
6. For the superposition of ψ_1 and ψ_2 treated in this section, find $\langle p \rangle$ from $m d\langle x \rangle/dt$.

§2.3 Bound States in Wells of Finite Depth

1. Show that the odd bound states of the finite square well satisfy $-\cot z = \sqrt{z_0^2/z^2 - 1}$, and that there are none unless $z_0 > \pi/2$.
2. Show that for a very deep well the odd states have $E + V_0 \approx n^2\pi^2\hbar^2/2mw^2$ with n even.
3. Show that $\delta(cx) = \delta(x)/|c|$ for any non-zero constant c , in the sense that $\delta(cx)$ and $\delta(x)/|c|$ give the same result when multiplied by a continuous function and integrated.
4. For the bound state of the delta-function well, find the probability that the particle is found in the region $|x| < 1/\kappa$, and find σ_x in terms of m , α , and $\hbar$.
5. A particle moves in the potential $V(x) = -\alpha[\delta(x - w/2) + \delta(x + w/2)]$, which consists of two delta-function wells a distance w apart. Let $\cosh u = \frac{1}{2}(e^u + e^{-u})$, $\sinh u = \frac{1}{2}(e^u - e^{-u})$, and $\tanh u = \sinh u / \cosh u$. Show that an even bound state has

the form $C \cosh \kappa x$ between the wells and $Ge^{-\kappa|x|}$ outside them, and that κ satisfies $\kappa[1 + \tanh(\kappa w/2)] = 2m\alpha/\hbar^2$. Show that when the wells are far apart the energy tends to that of a single delta-function well.

§2.4 The Free Particle and Wave Packets

1. A pair of plane waves has wave numbers $k_0 \pm \Delta k$ and angular frequencies $\omega_0 \pm \Delta\omega$. Show that the sum of the two waves is $2 \cos(\Delta k x - \Delta\omega t) e^{i(k_0 x - \omega_0 t)}$, and explain why the envelope moves at the speed $\Delta\omega/\Delta k$, which tends to $d\omega/dk$ as $\Delta k \rightarrow 0$.
2. Find the Fourier transform of $\Psi(x) = \sqrt{\kappa} e^{-\kappa|x|}$, the bound state of the delta-function well, and show that $g(k) = \sqrt{2/\pi} \kappa^{3/2}/(\kappa^2 + k^2)$.
3. Show directly, using the Gaussian integral, that $\int |g(k)|^2 dk = 1$ for the Gaussian packet of this section.
4. Find the time at which the width of a Gaussian packet has doubled, in terms of m , σ , and $\hbar$.
5. Show that $\langle x \rangle = v_0 t$ for the Gaussian packet, and that $m d\langle x \rangle/dt$ equals the mean momentum found from $|g(k)|^2$.

§2.5 Potential Steps and Tunnelling

1. Show that the reflection and transmission probabilities for the potential step with $E > V_0$ add to one.
2. A particle of energy E meets a downward step, $V = 0$ for $x < 0$ and $V = -V_1$ for $x > 0$, with $V_1 > 0$. Show that it is reflected with probability $[(k - k'')/(k + k'')]^2$, where $k'' = \sqrt{2m(E + V_1)}/\hbar$, although a classical particle would never be reflected.
3. A particle of energy $E > 0$ meets the delta-function barrier $V(x) = \alpha \delta(x)$ with $\alpha > 0$. Show that $P_T = [1 + m\alpha^2/(2\hbar^2 E)]^{-1}$.
4. For a rectangular barrier with $E > V_0$, show that the transmission probability is $P_T = [1 + V_0^2 \sin^2(k'w)/(4E(E - V_0))]^{-1}$, where $k' = \sqrt{2m(E - V_0)}/\hbar$, and find the energies at which

$P_T = 1$. Explain these energies in terms of the wavelength of the particle inside the barrier.

5. Show that P_T for the rectangular barrier with $E < V_0$ tends to $1/[1 + mV_0^2w^2/(2\hbar^2E)]$ when $w \rightarrow 0$ with V_0w held fixed, and compare this with Problem 3.

§2.6 The Harmonic Oscillator

1. Show that $\hat{x} = \sqrt{\hbar/2m\omega}(\hat{a}_+ + \hat{a}_-)$ and $\hat{p} = i\sqrt{\hbar m\omega/2}(\hat{a}_+ - \hat{a}_-)$. Use these results to show that in the state ψ_n , $\langle x \rangle = \langle p \rangle = 0$, $\langle x^2 \rangle = (n + \frac{1}{2})\hbar/m\omega$, and $\langle p^2 \rangle = (n + \frac{1}{2})\hbar m\omega$. Hence show that the mean kinetic and potential energies are each $E_n/2$, and find $\sigma_x\sigma_p$.
2. Find ψ_2 by applying the raising operator to ψ_1 , and check that ψ_1 and ψ_2 agree with the formula in terms of Hermite polynomials.
3. Show that the probability that a particle in the ground state of the oscillator is found beyond the classical turning points is $(2/\sqrt{\pi}) \int_1^\infty e^{-y^2} dy$. Integrals of this kind are usually evaluated numerically.
4. A particle moves in the potential $V = \frac{1}{2}m\omega^2x^2$ for $x > 0$, with an impenetrable wall at $x = 0$. Find its allowed energies.
5. Show that for $n = 0$ the power-series method gives ψ_0 as found with the ladder operators.
6. Prove the property of the ladder operators used in this section, $\int f^*(\hat{a}_\pm g) dx = \int (\hat{a}_\mp f)^* g dx$, for functions that vanish at infinity.

The Formalism of Quantum Mechanics

3.1 Operators, Commutators, and the Uncertainty Relations

The examples of Chapter 2 share a common structure. Each observable quantity was represented by an operator, the energies of the stationary states were the eigenvalues of the Hamiltonian, the eigenfunctions were orthogonal, and the commutator of position and momentum entered the treatment of the oscillator. In this chapter we show why these features hold generally, and we derive from them a general relation between the spreads of two observables, of which the result $\sigma_x \sigma_p = \hbar/2$ for the Gaussian packet of Section 2.4 was a first example, and an equation for how expectation values change in time.

Hermitian Operators

The operators we have met are linear, which means that $\hat{Q}(c_1 f + c_2 g) = c_1 \hat{Q}f + c_2 \hat{Q}g$ for any functions f and g and any constants c_1 and c_2 . An operator $\hat{Q}$ is called hermitian if

$$\int_{-\infty}^{\infty} f^* (\hat{Q}g) \, dx = \int_{-\infty}^{\infty} (\hat{Q}f)^* g \, dx \quad (3.1)$$

for all normalisable functions f and g . The position operator is hermitian, since multiplying by the real function x can be done to either factor. The momentum operator is hermitian as well, since integrating by parts gives

$$\int f^* \left(-i\hbar \frac{dg}{dx} \right) dx = i\hbar \int \frac{df^*}{dx} g dx = \int \left(-i\hbar \frac{df}{dx} \right)^* g dx, \quad (3.2)$$

where the boundary term has vanished, and the last step uses $i\hbar = (-i\hbar)^*$. Without the factor i the derivative would not be hermitian, since integration by parts changes its sign. Integrating by parts twice shows in the same way that d^2/dx^2 is hermitian, and so the Hamiltonian $-(\hbar^2/2m) d^2/dx^2 + V(x)$, with V real, is hermitian too. The ladder operators of Section 2.6 are not hermitian, since the property you proved in Problem 6 of Section 2.6 moves $\hat{a}_+$ from one factor to the other as $\hat{a}_-$.

The expectation value of a hermitian operator is real. Taking the complex conjugate of $\langle Q \rangle$, and then using the definition of a hermitian operator with $f = g = \Psi$, gives

$$\langle Q \rangle^* = \int (\hat{Q}\Psi)^* \Psi dx = \int \Psi^* (\hat{Q}\Psi) dx = \langle Q \rangle. \quad (3.3)$$

The result of a measurement is a real number, so the operators that represent measurable quantities, which are called observables, must be hermitian. This requirement is one of the general principles of quantum mechanics.

Eigenvalues and Eigenfunctions

Let $\hat{Q}$ be a hermitian operator, and let f and g be normalisable eigenfunctions of $\hat{Q}$, so that

$$\hat{Q}f = qf, \quad \hat{Q}g = q'g, \quad (3.4)$$

where the constants q and q' are the corresponding eigenvalues. We show first that every eigenvalue is real. Setting $g = f$ in the definition of a hermitian operator, the left-hand side becomes $\int f^*(qf) dx = q \int |f|^2 dx$, and the right-hand side becomes $\int (qf)^* f dx = q^* \int |f|^2 dx$. The integral of $|f|^2$ is positive, so $q = q^*$, and the eigenvalues of a hermitian operator are therefore real.

We now take f and g to have different eigenvalues, $q \neq q'$. Substituting $\hat{Q}g = q'g$ on the left-hand side of the definition and $\hat{Q}f = qf$

on the right-hand side gives

$$q' \int f^* g \, dx = \int f^* (\hat{Q}g) \, dx = \int (\hat{Q}f)^* g \, dx = q \int f^* g \, dx, \quad (3.5)$$

where the last step uses $(qf)^* = q^* f^* = q f^*$, since q is real. Since $q \neq q'$, the integral of $f^* g$ must vanish, and eigenfunctions of a hermitian operator that belong to different eigenvalues are therefore orthogonal. The orthogonality of the stationary states of the infinite well and of the oscillator, which we proved in Sections 2.2 and 2.6 by direct calculation, is a case of this result.

The eigenfunctions of an observable are the states in which that observable has a definite value. Let $\sigma_Q^2 = \langle (Q - \langle Q \rangle)^2 \rangle$ be the variance of Q in the state Ψ . This is the variance $\langle Q^2 \rangle - \langle Q \rangle^2$ of Section 2.1 written in a different form, since expanding the square and using the fact that $\langle Q \rangle$ is a constant gives $\langle Q^2 \rangle - 2\langle Q \rangle^2 + \langle Q \rangle^2$. The operator $\hat{Q} - \langle Q \rangle$ is hermitian, since $\langle Q \rangle$ is a real number, so writing $(\hat{Q} - \langle Q \rangle)^2$ as the product of two factors $\hat{Q} - \langle Q \rangle$ and moving one of them across gives

$$\begin{aligned} \sigma_Q^2 &= \int [(\hat{Q} - \langle Q \rangle) \Psi]^* [(\hat{Q} - \langle Q \rangle) \Psi] \, dx \\ &= \int |(\hat{Q} - \langle Q \rangle) \Psi|^2 \, dx. \end{aligned} \quad (3.6)$$

This vanishes only if $(\hat{Q} - \langle Q \rangle)\Psi = 0$, that is, only if Ψ is an eigenfunction of $\hat{Q}$. A state in which every measurement of Q gives the same result is therefore an eigenfunction of $\hat{Q}$, and the result is the eigenvalue. The stationary states of Section 2.2, in which every measurement of the energy gives the value E , are the case of this result for the energy.

The statement in Section 2.2 about measurements of the energy, that a measurement gives one of the eigenvalues E_n with probability $|c_n|^2$, extends to any observable whose eigenvalues form a discrete set $q_1, q_2, \dots$, with normalised eigenfunctions $f_1, f_2, \dots$. A measurement of Q gives one of the eigenvalues. If the state is expanded as $\Psi = \sum_n c_n f_n$, the probability of the result q_n is $|c_n|^2$, and the orthogonality of the eigenfunctions gives $c_n = \int f_n^* \Psi \, dx$. When the eigenvalues form a continuous range, as they do for the momentum, whose eigenfunctions e^{ikx} cannot be normalised, the sum over n is replaced by an integral over k . As in Section 2.4, the

state at a given instant is written as $\Psi(x) = (2\pi)^{-1/2} \int g(k) e^{ikx} dk$, so the Fourier transform $g(k)$ takes the place of the coefficients c_n , and $|g(k)|^2 dk$ takes the place of $|c_n|^2$ as the probability that a measurement of the momentum gives a value between $\hbar k$ and $\hbar(k+dk)$.

Commutators

Section 2.6 defined the commutator $[\hat{A}, \hat{B}] = \hat{A}\hat{B} - \hat{B}\hat{A}$ and found $[\hat{x}, \hat{p}] = i\hbar$. Commutators obey a few simple rules, which follow from the definition:

1. $[\hat{A}, \hat{B}] = -[\hat{B}, \hat{A}]$;
2. $[\hat{A}, \hat{B} + \hat{C}] = [\hat{A}, \hat{B}] + [\hat{A}, \hat{C}]$;
3. $[\hat{A}, \hat{B}\hat{C}] = [\hat{A}, \hat{B}]\hat{C} + \hat{B}[\hat{A}, \hat{C}]$.

The third rule is proved by adding and subtracting $\hat{B}\hat{A}\hat{C}$:

$$\hat{A}\hat{B}\hat{C} - \hat{B}\hat{C}\hat{A} = (\hat{A}\hat{B} - \hat{B}\hat{A})\hat{C} + \hat{B}(\hat{A}\hat{C} - \hat{C}\hat{A}). \quad (3.7)$$

As an example, the first rule gives $[\hat{p}, \hat{x}] = -i\hbar$, and the third rule with $\hat{A} = \hat{p}$ and $\hat{B} = \hat{C} = \hat{x}$ then gives $[\hat{p}, \hat{x}^2] = -2i\hbar\hat{x}$. More generally, acting on a function f ,

$$[\hat{p}, V(x)]f = -i\hbar \frac{d}{dx}(Vf) + i\hbar V \frac{df}{dx} = -i\hbar \frac{dV}{dx}f, \quad (3.8)$$

so $[\hat{p}, V(x)] = -i\hbar dV/dx$ for any function V of position.

The Schwarz Inequality

The uncertainty relation rests on an inequality between integrals. For any functions f and g whose squared moduli can be integrated,

$$\left| \int f^* g dx \right|^2 \leq \int |f|^2 dx \int |g|^2 dx, \quad (3.9)$$

which is called the Schwarz inequality. It is the analogue for functions of the statement that the scalar product of two vectors is at most the product of their lengths, $|\mathbf{f} \cdot \mathbf{g}| \leq |\mathbf{f}||\mathbf{g}|$, and the proof follows the geometry of that statement, which is drawn in Figure 21.

Let $N_f = \int |f|^2 dx$, $N_g = \int |g|^2 dx$, and $J = \int g^* f dx$, so that $\int f^* g dx = J^*$. If $N_g = 0$ then $g = 0$ and there is nothing to prove,

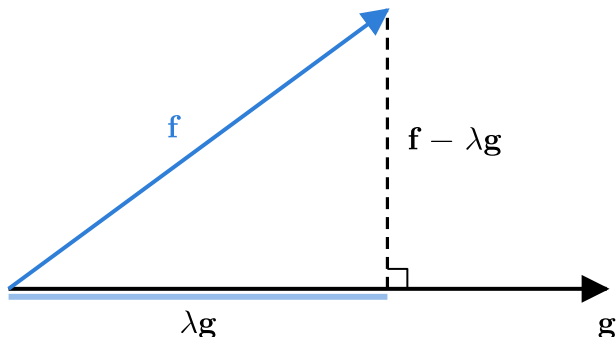


Figure 21. The geometry behind the Schwarz inequality, drawn for ordinary vectors. Subtracting from $\mathbf{f}$ its projection $\lambda\mathbf{g}$ onto $\mathbf{g}$ leaves a vector perpendicular to $\mathbf{g}$, whose squared length cannot be negative.

so we suppose $N_g > 0$. For any complex number λ , the integral of $|f - \lambda g|^2$ cannot be negative, and multiplying out gives

$$0 \leq \int (f - \lambda g)^* (f - \lambda g) dx = N_f - \lambda J^* - \lambda^* J + |\lambda|^2 N_g. \quad (3.10)$$

Let us choose $\lambda = J/N_g$. For ordinary vectors J is the scalar product $\mathbf{g} \cdot \mathbf{f}$ and N_g is $|\mathbf{g}|^2$, so $\lambda\mathbf{g}$ is the projection of $\mathbf{f}$ onto $\mathbf{g}$, which is the construction drawn in Figure 21. The last three terms are then $-|J|^2/N_g$, $-|J|^2/N_g$, and $+|J|^2/N_g$, and the inequality becomes $0 \leq N_f - |J|^2/N_g$, which can be rewritten as $|J|^2 \leq N_f N_g$. Since $|J^*| = |J|$, this is the Schwarz inequality.

The Uncertainty Relation

Let $\hat{A}$ and $\hat{B}$ be the hermitian operators of two observables, and let the system be in the normalised state Ψ . Let

$$f = (\hat{A} - \langle A \rangle) \Psi, \quad g = (\hat{B} - \langle B \rangle) \Psi. \quad (3.11)$$

By the calculation of the variance above, $\sigma_A^2 = \int |f|^2 dx$ and $\sigma_B^2 = \int |g|^2 dx$, and the Schwarz inequality gives

$$\sigma_A^2 \sigma_B^2 \geq |z|^2, \quad z = \int f^* g dx. \quad (3.12)$$

The squared modulus of a complex number is at least the square of its imaginary part, and $\text{Im } z = (z - z^*)/2i$, so $|z|^2 \geq [(z - z^*)/2i]^2$. Moving the operator in f onto g , which is allowed because it is hermitian, and multiplying out gives

$$z = \int \Psi^* (\hat{A} - \langle A \rangle) (\hat{B} - \langle B \rangle) \Psi \, dx = \langle AB \rangle - \langle A \rangle \langle B \rangle, \quad (3.13)$$

where $\langle AB \rangle$ is the expectation value of the operator $\hat{A}\hat{B}$. A similar calculation gives $z^* = \langle BA \rangle - \langle A \rangle \langle B \rangle$, so $z - z^*$ is the expectation value of the commutator. Therefore

$$\sigma_A \sigma_B \geq \frac{1}{2} |\langle [\hat{A}, \hat{B}] \rangle|, \quad (3.14)$$

which is called the generalised uncertainty relation. For position and momentum the commutator is the constant $i\hbar$, whose expectation value in any normalised state is $i\hbar$, and

$$\sigma_x \sigma_p \geq \frac{\hbar}{2}, \quad (3.15)$$

which is called the Heisenberg uncertainty relation. It holds in every state, and the Gaussian wave packet of Section 2.4 and the ground state of the oscillator, for which you found $\sigma_x \sigma_p = \frac{1}{2}\hbar$ in Problem 1 of Section 2.6, reach the lower limit. The relation concerns the spreads of the results obtained when the position is measured on many particles prepared in the same state, and when the momentum is measured on many others prepared in the same way. It says that no state can make both spreads small at once.

Compatible Observables

When two operators commute, the right-hand side of the uncertainty relation is zero, and nothing prevents σ_A and σ_B from vanishing together. Observables whose operators commute are called compatible, and those whose operators do not commute are called incompatible. It is possible for compatible observables to have definite values in the same state. We show this for the case where the eigenvalue concerned belongs to a single eigenfunction. Let $[\hat{A}, \hat{B}] = 0$ and $\hat{A}f = af$. Then

$$\hat{A}(\hat{B}f) = \hat{B}(\hat{A}f) = a(\hat{B}f), \quad (3.16)$$

so $\hat{B}f$ is also an eigenfunction of $\hat{A}$ with the eigenvalue a . If a belongs to only one eigenfunction, apart from a constant factor, then

$\hat{B}f$ must be a multiple of f , say $\hat{B}f = bf$, and f is an eigenfunction of $\hat{B}$ as well.

An eigenvalue that belongs to more than one independent eigenfunction is called degenerate, and in that case the argument fails. As an example, consider the free particle, for which $\hat{H} = \hat{p}^2/2m$ commutes with $\hat{p}$. The functions e^{ikx} and e^{-ikx} are both eigenfunctions of $\hat{H}$ with energy $\hbar^2 k^2/2m$, so this energy is degenerate. Each of e^{ikx} and e^{-ikx} is also an eigenfunction of $\hat{p}$, but their sum, $2 \cos kx$, is an eigenfunction of $\hat{H}$ while it is no longer an eigenfunction of $\hat{p}$, since $\hat{p} \cos kx = i\hbar k \sin kx$, which is not a multiple of $\cos kx$. For a degenerate eigenvalue, the eigenfunctions can always be chosen so that they are eigenfunctions of both $\hat{A}$ and $\hat{B}$, a result whose proof is beyond this course, although not every eigenfunction of one operator is an eigenfunction of the other.

How Expectation Values Change in Time

Let $\hat{Q}$ be an operator, which may depend on time explicitly. Differentiating $\langle Q \rangle = \int \Psi^* \hat{Q} \Psi dx$ with the product rule gives three terms,

$$\frac{d\langle Q \rangle}{dt} = \int \frac{\partial \Psi^*}{\partial t} \hat{Q} \Psi dx + \int \Psi^* \frac{\partial \hat{Q}}{\partial t} \Psi dx + \int \Psi^* \hat{Q} \frac{\partial \Psi}{\partial t} dx. \quad (3.17)$$

The Schrödinger equation gives $\partial \Psi / \partial t = -(i/\hbar) \hat{H} \Psi$, and its complex conjugate gives $\partial \Psi^* / \partial t = (i/\hbar) (\hat{H} \Psi)^*$. In the first term, the hermitian operator $\hat{H}$ can be moved onto $\hat{Q} \Psi$, and the first and third terms combine to give

$$\frac{d\langle Q \rangle}{dt} = \frac{i}{\hbar} \langle [\hat{H}, \hat{Q}] \rangle + \left\langle \frac{\partial \hat{Q}}{\partial t} \right\rangle. \quad (3.18)$$

An observable that does not depend on time explicitly, and whose operator commutes with the Hamiltonian, therefore has a constant expectation value, and it is said to be conserved. The energy itself is conserved, since $\hat{H}$ commutes with itself, and for a free particle the momentum is conserved, since $\hat{p}$ commutes with $\hat{p}^2/2m$.

We now apply this result to $\hat{Q} = \hat{x}$ and $\hat{Q} = \hat{p}$ for a particle in a potential $V(x)$. Using the rules for commutators with $[\hat{p}, \hat{x}] = -i\hbar$

gives $[\hat{p}^2, \hat{x}] = -2i\hbar\hat{p}$, so $[\hat{H}, \hat{x}] = -i\hbar\hat{p}/m$, and the commutator of $\hat{p}$ with $V(x)$ found above gives $[\hat{H}, \hat{p}] = [V, \hat{p}] = i\hbar dV/dx$. Therefore

$$\frac{d\langle x \rangle}{dt} = \frac{\langle p \rangle}{m}, \quad \frac{d\langle p \rangle}{dt} = -\left\langle \frac{dV}{dx} \right\rangle. \quad (3.19)$$

The first equation agrees with the definition of $\langle p \rangle$ in Section 2.1, and the second has the form of Newton's second law, with the force averaged over the wave function. These two results together are called Ehrenfest's theorem. They do not in general mean that $\langle x \rangle$ follows a classical path, since the mean of the force $-dV/dx$ over the wave function is usually different from the force at the mean position. The mean force and the force at the mean position are equal when the force depends linearly on position, as it does for the harmonic oscillator, and they are nearly equal when the wave function is narrow compared with the distance over which the force changes.

The Energy–Time Relation

Let us apply the uncertainty relation with $\hat{A} = \hat{H}$ and $\hat{B} = \hat{Q}$, for an observable that does not depend on time explicitly. The expectation value of the commutator is $-i\hbar d\langle Q \rangle/dt$, by the result above, so

$$\sigma_H \sigma_Q \geq \frac{\hbar}{2} \left| \frac{d\langle Q \rangle}{dt} \right|. \quad (3.20)$$

Let $\tau_Q = \sigma_Q/|d\langle Q \rangle/dt|$, which is the time that the expectation value of Q would take to change by one standard deviation at its present rate of change. The inequality becomes

$$\sigma_H \tau_Q \geq \frac{\hbar}{2}, \quad (3.21)$$

which is called the energy–time uncertainty relation. The time τ_Q is a property of the state and of the observable Q , defined by how quickly the expectation value of Q changes, and the relation says that a state whose properties change quickly must have a large spread of energy. In a stationary state $\sigma_H = 0$ and nothing changes, so τ_Q is infinite for every observable. An excited atom that lasts only a short time before emitting a photon has an energy that is correspondingly uncertain, and the lines it emits have a width in frequency of the order of the reciprocal of its lifetime.

Estimating an Energy

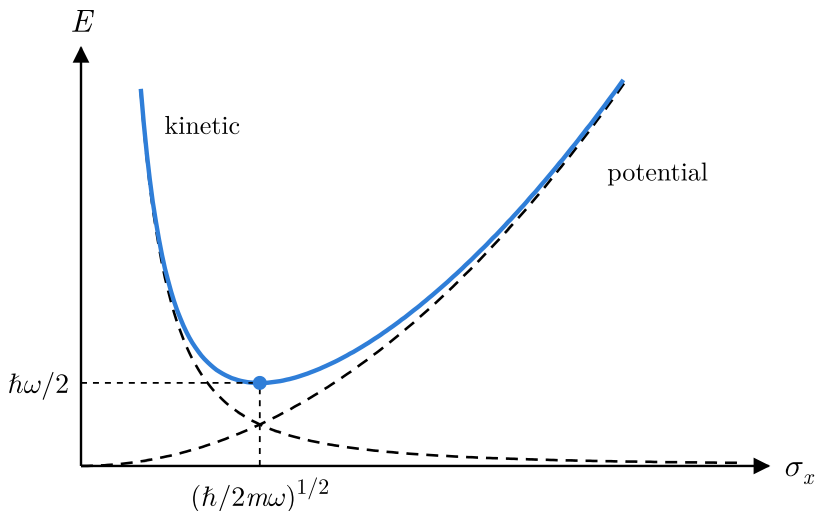


Figure 22. The lower bound on the energy of the harmonic oscillator, $\hbar^2/8m\sigma_x^2 + \frac{1}{2}m\omega^2\sigma_x^2$ (accent), plotted against σ_x , together with its two terms (dashed). The minimum lies at $\sigma_x = \sqrt{\hbar/2m\omega}$, where the bound equals $\frac{1}{2}\hbar\omega$.

The uncertainty relation can be used to estimate the lowest energy of a system. Let us consider the harmonic oscillator. In any state, $\langle p^2 \rangle = \sigma_p^2 + \langle p \rangle^2 \geq \sigma_p^2$, and similarly $\langle x^2 \rangle \geq \sigma_x^2$, so the energy satisfies

$$\langle H \rangle = \frac{\langle p^2 \rangle}{2m} + \frac{1}{2}m\omega^2 \langle x^2 \rangle \geq \frac{\sigma_p^2}{2m} + \frac{1}{2}m\omega^2 \sigma_x^2 \geq \frac{\hbar^2}{8m\sigma_x^2} + \frac{1}{2}m\omega^2 \sigma_x^2, \quad (3.22)$$

where the last step uses $\sigma_p \geq \hbar/2\sigma_x$. The right-hand side is plotted against σ_x in Figure 22. It is large when σ_x is small, because the momentum is then spread widely, and large when σ_x is large, because the particle is then likely to be found far up the potential. Setting its derivative with respect to σ_x to zero gives

$$-\frac{\hbar^2}{4m\sigma_x^3} + m\omega^2 \sigma_x = 0, \quad \sigma_x^2 = \frac{\hbar}{2m\omega}, \quad (3.23)$$

and at this value of σ_x the right-hand side is $\frac{1}{4}\hbar\omega + \frac{1}{4}\hbar\omega = \frac{1}{2}\hbar\omega$. No state of the oscillator can have an energy below $\frac{1}{2}\hbar\omega$, which is

the ground-state energy found in Section 2.6. The estimate is exact in this case because every inequality in the chain above becomes an equality for the ground state, since it has $\langle x \rangle = \langle p \rangle = 0$, it is a Gaussian, which reaches the lower limit of the uncertainty relation, and its spread in position, which you found in Problem 1 of Section 2.6 to be $\sigma_x^2 = \hbar/2m\omega$, is the value at which the bound is smallest. For most systems an estimate of this kind gives only the order of magnitude of the lowest energy.

Problems

§3.1 Operators, Commutators, and the Uncertainty Relations

1. Show that $[\hat{x}, \hat{p}^2] = 2i\hbar\hat{p}$, and that $[\hat{x}^n, \hat{p}] = ni\hbar\hat{x}^{n-1}$ for any positive integer n .
2. Show that if $\hat{A}$ and $\hat{B}$ are hermitian, then $\hat{A}\hat{B} + \hat{B}\hat{A}$ and $i[\hat{A}, \hat{B}]$ are hermitian, and that $\hat{A}\hat{B}$ is hermitian only if $\hat{A}$ and $\hat{B}$ commute.
3. For the equal superposition of ψ_1 and ψ_2 in the infinite square well, treated in Section 2.2, show that $\sigma_H = (E_2 - E_1)/2$. The expectation value $\langle x \rangle$ moves from one extreme to the other in a time $\pi\hbar/(E_2 - E_1)$. Show that the product of this time and σ_H is larger than $\hbar/2$.
4. A particle is confined to the infinite square well of width w . Explain why $\sigma_x \leq w/2$, and use the uncertainty relation to show that its energy is at least $\hbar^2/2mw^2$. Compare this bound with E_1 .
5. Using the results of Problems 2 and 5 of Section 2.2, find $\sigma_x\sigma_p$ in the state ψ_n of the infinite square well, and check that it satisfies the Heisenberg relation for every n .
6. Use Ehrenfest's theorem to show that for the harmonic oscillator $d^2\langle x \rangle/dt^2 = -\omega^2\langle x \rangle$ in any state, so that $\langle x \rangle$ oscillates as a classical oscillator does.

Symmetry, Angular Momentum, and Spin

4.1 Orbital Angular Momentum

In this chapter we turn to angular momentum. This section treats the angular momentum of a particle moving about an origin, which is called orbital angular momentum, and finds its possible values using only the commutation relations of its components. The section that follows treats spin, an angular momentum that the electron and many other particles carry in addition, and then uses the mathematics of spin to describe any system with two states.

The Wave Function in Space

In three dimensions the wave function depends on all three coordinates, $\Psi(x, y, z, t)$, and $|\Psi|^2 dV$ is the probability of finding the particle in the small volume $dV = dx dy dz$. Each component of the momentum is represented by an operator,

$$\hat{p}_x = -i\hbar \frac{\partial}{\partial x}, \quad \hat{p}_y = -i\hbar \frac{\partial}{\partial y}, \quad \hat{p}_z = -i\hbar \frac{\partial}{\partial z}. \quad (4.1)$$

The calculation of Section 2.6 gives $[\hat{x}, \hat{p}_x] = [\hat{y}, \hat{p}_y] = [\hat{z}, \hat{p}_z] = i\hbar$. A position coordinate commutes with a momentum component along a different axis, since $\partial/\partial y$ treats x as a constant, and any two

position coordinates commute, as do any two momentum components, since the order of partial differentiation does not matter for the smooth functions we deal with.

The Components of Angular Momentum

The angular momentum of a particle about the origin is $\mathbf{L} = \mathbf{r} \times \mathbf{p}$, whose components are

$$L_x = yp_z - zp_y, \quad L_y = zp_x - xp_z, \quad L_z = xp_y - yp_x. \quad (4.2)$$

The operators are formed by replacing each quantity by its operator. No question of ordering arises, since each product contains a coordinate and a momentum component along a different axis, and these commute. Let us find the commutator of $\hat{L}_x$ and $\hat{L}_y$. Expanding it gives four terms,

$$[\hat{L}_x, \hat{L}_y] = [\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] - [\hat{y}\hat{p}_z, \hat{x}\hat{p}_z] - [\hat{z}\hat{p}_y, \hat{z}\hat{p}_x] + [\hat{z}\hat{p}_y, \hat{x}\hat{p}_z]. \quad (4.3)$$

In the second and third terms every operator in one product commutes with every operator in the other, so those terms vanish. In the first term the only pair that fails to commute is $\hat{p}_z$ and $\hat{z}$, and the rules for commutators give $[\hat{y}\hat{p}_z, \hat{z}\hat{p}_x] = \hat{y}[\hat{p}_z, \hat{z}]\hat{p}_x = -i\hbar\hat{y}\hat{p}_x$. In the same way the fourth term is $\hat{x}[\hat{z}, \hat{p}_z]\hat{p}_y = i\hbar\hat{x}\hat{p}_y$. Therefore

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \quad (4.4)$$

and the other two relations, $[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x$ and $[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y$, follow by relabelling the axes cyclically. The components of angular momentum are incompatible observables. By the uncertainty relation of Section 3.1, a state in which L_z has a definite non-zero value cannot have a definite value of L_x or of L_y , since the product of their spreads is then at least $\frac{1}{2}\hbar|\langle L_z \rangle|$, which is not zero.

The squared magnitude, $\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2$, does commute with each component. For $\hat{L}_z$, the rules for commutators give

$$[\hat{L}_x^2, \hat{L}_z] = \hat{L}_x[\hat{L}_x, \hat{L}_z] + [\hat{L}_x, \hat{L}_z]\hat{L}_x = -i\hbar(\hat{L}_x\hat{L}_y + \hat{L}_y\hat{L}_x), \quad (4.5)$$

and a similar calculation gives $[\hat{L}_y^2, \hat{L}_z] = i\hbar(\hat{L}_y\hat{L}_x + \hat{L}_x\hat{L}_y)$. These two commutators cancel when they are added, and $\hat{L}_z^2$ commutes with $\hat{L}_z$, so $[\hat{L}^2, \hat{L}_z] = 0$. The squared magnitude and one component are therefore compatible, and by Section 3.1 we may look

for functions that are eigenfunctions of both. It is conventional to choose the component along z .

Ladder Operators for Angular Momentum

As for the oscillator, the eigenvalues can be found with ladder operators. Let

$$\hat{L}_{\pm} = \hat{L}_x \pm i\hat{L}_y. \quad (4.6)$$

Their commutators with $\hat{L}_z$ follow from the relations above,

$$[\hat{L}_z, \hat{L}_{\pm}] = [\hat{L}_z, \hat{L}_x] \pm i[\hat{L}_z, \hat{L}_y] = i\hbar\hat{L}_y \pm \hbar\hat{L}_x = \pm\hbar\hat{L}_{\pm}, \quad (4.7)$$

and since $\hat{L}^2$ commutes with $\hat{L}_x$ and $\hat{L}_y$, it commutes with $\hat{L}_{\pm}$ as well. Let f be an eigenfunction of both $\hat{L}^2$ and $\hat{L}_z$, with $\hat{L}^2 f = \hbar^2 q f$ and $\hat{L}_z f = \hbar m_l f$, where q and m_l are numbers to be determined and the factors of $\hbar$ have been taken out for convenience. Then $\hat{L}^2(\hat{L}_{\pm} f) = \hat{L}_{\pm} \hat{L}^2 f = \hbar^2 q(\hat{L}_{\pm} f)$, and

$$\hat{L}_z(\hat{L}_{\pm} f) = \hat{L}_{\pm} \hat{L}_z f \pm \hbar \hat{L}_{\pm} f = \hbar(m_l \pm 1)(\hat{L}_{\pm} f). \quad (4.8)$$

The raising operator $\hat{L}_+$ leaves the eigenvalue of $\hat{L}^2$ unchanged and raises the eigenvalue of $\hat{L}_z$ by $\hbar$, and the lowering operator $\hat{L}_-$ lowers it by $\hbar$.

The ladder has a top and a bottom. For a given q the z -component cannot exceed the magnitude, since $\langle L^2 \rangle - \langle L_z^2 \rangle = \langle L_x^2 \rangle + \langle L_y^2 \rangle$, and neither term on the right can be negative. The operators $\hat{L}_x$ and $\hat{L}_y$ are hermitian, since each is a sum of products of commuting hermitian operators, which you showed to be hermitian in Problem 2 of Section 3.1, and for a hermitian operator $\hat{Q}$ the expectation value of $\hat{Q}^2$ is $\int (\hat{Q}\Psi)^*(\hat{Q}\Psi) dV = \int |\hat{Q}\Psi|^2 dV$, by the same step that gave the variance in Section 3.1, and this integral cannot be negative. In an eigenstate this gives $m_l^2 \leq q$. There must therefore be a top rung, a function f_t with $\hat{L}_+ f_t = 0$, and let l be its value of m_l . To use this condition we need

$$\begin{aligned} \hat{L}_- \hat{L}_+ &= (\hat{L}_x - i\hat{L}_y)(\hat{L}_x + i\hat{L}_y) = \hat{L}_x^2 + \hat{L}_y^2 + i[\hat{L}_x, \hat{L}_y] \\ &= \hat{L}^2 - \hat{L}_z^2 - \hbar\hat{L}_z. \end{aligned} \quad (4.9)$$

Applying this operator to f_t gives $0 = \hbar^2(q - l^2 - l)f_t$, so

$$q = l(l + 1). \quad (4.10)$$

In the same way there is a bottom rung f_b , with $\hat{L}_-f_b = 0$ and some value $\bar{l}$ of m_l . The product $\hat{L}_+\hat{L}_- = \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z$, found by the same calculation, gives $q = \bar{l}(\bar{l} - 1)$. Equating the two expressions for q gives $\bar{l} = -l$ or $\bar{l} = l + 1$, and $\bar{l} = l + 1$ is impossible, since the bottom rung cannot lie above the top. The eigenvalue m_l therefore runs from $-l$ to l in steps of one. Since the difference $2l$ between the top and the bottom is a whole number of steps, l must be a whole number or a half-integer. In summary,

$$\begin{aligned} \hat{L}^2 f &= \hbar^2 l(l + 1) f, & \hat{L}_z f &= \hbar m_l f, \\ m_l &= -l, -l + 1, \dots, l - 1, l, \end{aligned} \quad (4.11)$$

with $l = 0, \frac{1}{2}, 1, \frac{3}{2}, \dots$ allowed by the commutation relations. For a given l there are $2l + 1$ values of m_l . The number l is called the angular momentum quantum number and m_l the magnetic quantum number.

The magnitude of the angular momentum is $\hbar\sqrt{l(l + 1)}$, which for $l \geq 1$ is always larger than the largest z -component, $\hbar l$. An angular momentum with $l \geq 1$ can never point exactly along the z -axis, since the x - and y -components would then both be zero with certainty, which the uncertainty relation forbids. A useful picture, shown in Figure 23, draws the angular momentum as a vector of length $\hbar\sqrt{l(l + 1)}$ lying somewhere on a cone about the z -axis, with a definite z -component and an undetermined direction around the cone.

Spherical Polar Coordinates

To find the eigenfunctions we need the operators in the coordinates suited to rotations. The spherical polar coordinates of a point are its distance r from the origin, the angle θ between the position vector and the z -axis, and the angle ϕ between the xz -plane and the plane that contains both the z -axis and the position vector, as shown in Figure 24. They are related to the Cartesian coordinates by

$$x = r \sin \theta \cos \phi, \quad y = r \sin \theta \sin \phi, \quad z = r \cos \theta, \quad (4.12)$$

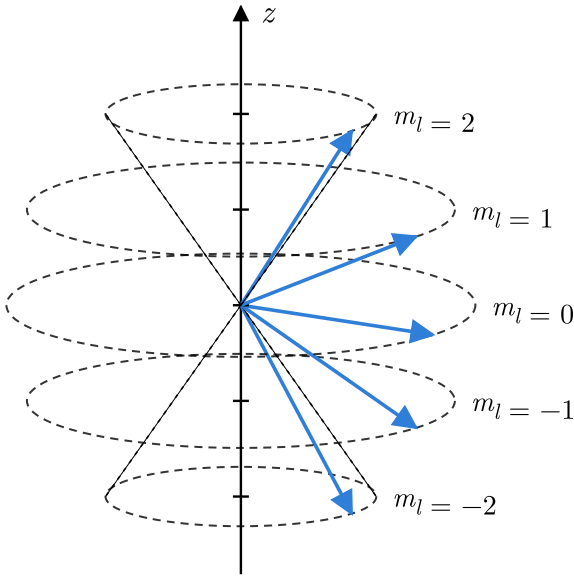


Figure 23. The allowed z -components of an angular momentum with $l = 2$. Each state has $L_z = m_l \hbar$ and a magnitude $\sqrt{6} \hbar$, drawn as a vector on a cone about the z -axis, whose sides are shown dotted for $m_l = \pm 2$. The components L_x and L_y are not definite, and the vector is pictured as lying anywhere on its cone.

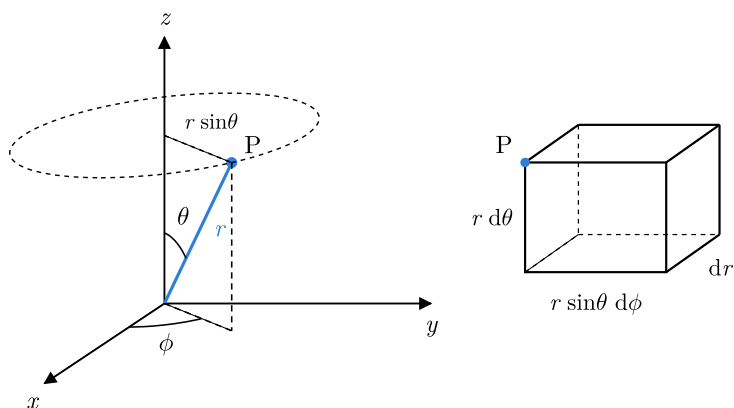


Figure 24. Left: the spherical polar coordinates r , θ , and ϕ of a point P . The angle θ is measured from the z -axis and the angle ϕ is measured in the xy -plane from the x -axis to the projection of the position vector. The dashed ellipse is the horizontal circle through P , whose radius is $r \sin \theta$. Right: the small volume swept out when r , θ , and ϕ change by dr , $d\theta$, and $d\phi$, drawn at a larger scale, with its three edges dr , $r d\theta$, and $r \sin \theta d\phi$.

with $0 \leq r < \infty$, $0 \leq \theta \leq \pi$, and $0 \leq \phi < 2\pi$. Changing r , θ , and ϕ by small amounts moves the point through distances dr , $r d\theta$, and $r \sin \theta d\phi$ in three perpendicular directions, as the right-hand part of Figure 24 shows, the last because the point moves round a circle of radius $r \sin \theta$, so the element of volume is

$$dV = r^2 \sin \theta dr d\theta d\phi. \quad (4.13)$$

The factor $\sin \theta d\theta d\phi$ is the area of a small patch of a sphere of unit radius, and it is called the element of solid angle, $d\Omega$. Integrating it over all directions gives $2\pi \int_0^\pi \sin \theta d\theta = 4\pi$.

The operator $\hat{L}_z$ takes a simple form in these coordinates. A function of position can be written in terms of either set of coordinates, and its derivative with respect to ϕ , holding r and θ fixed, is found from the chain rule for a function of several variables, which adds one term for each Cartesian coordinate,

$$\frac{\partial f}{\partial \phi} = \frac{\partial x}{\partial \phi} \frac{\partial f}{\partial x} + \frac{\partial y}{\partial \phi} \frac{\partial f}{\partial y} + \frac{\partial z}{\partial \phi} \frac{\partial f}{\partial z} = -y \frac{\partial f}{\partial x} + x \frac{\partial f}{\partial y}, \quad (4.14)$$

since $\partial x / \partial \phi = -y$, $\partial y / \partial \phi = x$, and z does not depend on ϕ . Multiplying by $-i\hbar$ gives $\hat{x}\hat{p}_y - \hat{y}\hat{p}_x$, and therefore

$$\hat{L}_z = -i\hbar \frac{\partial}{\partial \phi}. \quad (4.15)$$

A calculation of the same kind gives the ladder operators,

$$\hat{L}_\pm = \pm \hbar e^{\pm i\phi} \left(\frac{\partial}{\partial \theta} \pm i \cot \theta \frac{\partial}{\partial \phi} \right), \quad (4.16)$$

and you will check this result in one case in Problem 3. Substituting these expressions into $\hat{L}^2 = \hat{L}_- \hat{L}_+ + \hat{L}_z^2 + \hbar \hat{L}_z$ gives

$$\hat{L}^2 = -\hbar^2 \left[\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial}{\partial \theta} \right) + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \phi^2} \right]. \quad (4.17)$$

None of these operators involves r , since a rotation about the origin does not change the distance from it.

Spherical Harmonics

The eigenfunctions of $\hat{L}_z$ satisfy $-i\hbar \partial f / \partial \phi = \hbar m_l f$, so they depend on ϕ as $e^{im_l \phi}$. The angles ϕ and $\phi + 2\pi$ describe the same point,

and a wave function must have a single value at each point, so $e^{2\pi i m_l} = 1$, and m_l must be a whole number. Since the values of m_l differ from l by whole numbers, l must be a whole number too. Orbital angular momentum therefore has $l = 0, 1, 2, \dots$, and the half-integer values allowed by the commutation relations do not occur for it. They occur for spin, as the next section shows.

The eigenfunctions of both $\hat{L}^2$ and $\hat{L}_z$ are called spherical harmonics, and are written $Y_l^{m_l}(\theta, \phi)$. The one at the top of the ladder has the form $F(\theta) e^{il\phi}$ and satisfies $\hat{L}_+ Y_l^l = 0$. With the expression for $\hat{L}_+$ this condition reads

$$\hbar e^{i(l+1)\phi} \left(\frac{dF}{d\theta} - l \cot \theta F \right) = 0, \quad (4.18)$$

so $dF/d\theta = l \cot \theta F$. Dividing by F and integrating gives $\ln F = l \ln \sin \theta$ plus a constant, and therefore

$$Y_l^l(\theta, \phi) = C_l \sin^l \theta e^{il\phi}, \quad (4.19)$$

where C_l is fixed by normalising over all directions, $\int |Y_l^l|^2 d\Omega = 1$. The other spherical harmonics with the same l follow by applying $\hat{L}_-$ repeatedly. For $l = 0$ and $l = 1$ the results are

$$Y_0^0 = \frac{1}{\sqrt{4\pi}}, \quad Y_1^{\pm 1} = \mp \sqrt{\frac{3}{8\pi}} \sin \theta e^{\pm i\phi}, \quad Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos \theta, \quad (4.20)$$

and one of the harmonics with $l = 2$ is

$$Y_2^0 = \sqrt{\frac{5}{16\pi}} (3 \cos^2 \theta - 1). \quad (4.21)$$

The minus sign in Y_1^1 is a convention, and it fixes the signs of Y_1^0 and Y_1^{-1} , which are obtained from Y_1^1 with the lowering operator. As an example of the lowering, let us apply $\hat{L}_-$ to Y_1^1 . The derivative with respect to ϕ brings down a factor i , and

$$\begin{aligned} \hat{L}_- Y_1^1 &= -\hbar e^{-i\phi} \left(\frac{\partial}{\partial \theta} - i \cot \theta \frac{\partial}{\partial \phi} \right) \left(-\sqrt{\frac{3}{8\pi}} \sin \theta e^{i\phi} \right) \\ &= 2\hbar \sqrt{\frac{3}{8\pi}} \cos \theta, \end{aligned} \quad (4.22)$$

which is $\sqrt{2} \hbar Y_1^0$, in agreement with the factor

$$\hbar \sqrt{l(l+1) - m_l(m_l - 1)} \quad (4.23)$$

that you will find for the lowering operator in Problem 2.

A spherical harmonic has the form of a function of θ multiplied by $e^{im_l\phi}$. For $m_l = 0$ the function of θ is a constant multiple of a polynomial in $\cos\theta$, which is called a Legendre polynomial and written $P_l(\cos\theta)$. The first three are $P_0 = 1$, $P_1 = \cos\theta$, and $P_2 = \frac{1}{2}(3\cos^2\theta - 1)$, and for $m_l \neq 0$ the functions of θ are called associated Legendre functions. Since $\hat{L}^2$ and $\hat{L}_z$ are hermitian, spherical harmonics with different l or different m_l are orthogonal, by the result of Section 3.1, and with the normalisation above

$$\int Y_l^{m_l*} Y_{l'}^{m_{l'}} d\Omega = \delta_{ll'} \delta_{m_l m_{l'}}. \quad (4.24)$$

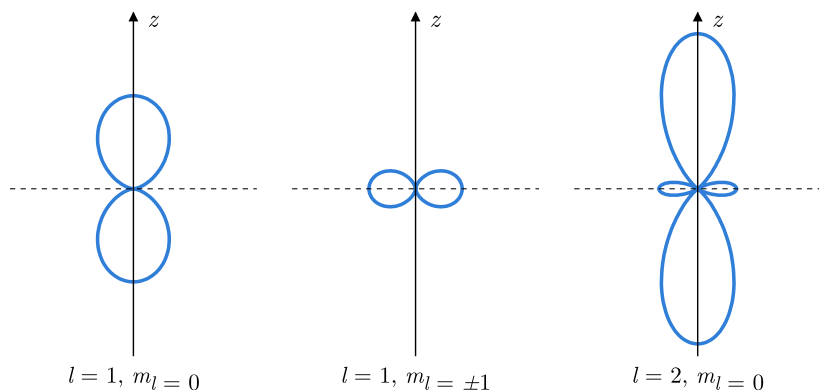


Figure 25. Polar plots of $|Y_l^{m_l}(\theta, \phi)|^2$ in a plane containing the z -axis, for $l = 1, m_l = 0$ (left), $l = 1, m_l = \pm 1$ (centre), and $l = 2, m_l = 0$ (right). The distance from the origin to the curve in the direction θ is proportional to $|Y_l^{m_l}|^2$, and each plot is symmetric about the z -axis.

The probability of finding the particle in a given direction is proportional to $|Y_l^{m_l}|^2$, which does not depend on ϕ . Figure 25 shows this dependence on θ for three of the harmonics.

Rotations

Angular momentum is connected with rotations in the following way. Let us rotate a wave function about the z -axis through a small angle ϵ . The rotated function has at the point with coordinate ϕ

the value that the original had at $\phi - \epsilon$, and a Taylor expansion gives

$$\psi(r, \theta, \phi - \epsilon) \approx \psi - \epsilon \frac{\partial \psi}{\partial \phi} = \left(1 - \frac{i\epsilon}{\hbar} \hat{L}_z\right) \psi. \quad (4.25)$$

For this reason $\hat{L}_z$ is called the generator of rotations about the z -axis. A rotation through a finite angle ϕ_0 can be built from $\mathcal{N}$ small rotations, each through $\phi_0/\mathcal{N}$, and letting $\mathcal{N} \rightarrow \infty$ gives

$$\lim_{\mathcal{N} \rightarrow \infty} \left(1 - \frac{i\phi_0}{\mathcal{N}\hbar} \hat{L}_z\right)^{\mathcal{N}} = \exp\left(-\frac{i\phi_0}{\hbar} \hat{L}_z\right), \quad (4.26)$$

where the exponential of an operator is defined by the Taylor series of the exponential, and the limit is the same one that defines e^x as the limit of $(1 + x/\mathcal{N})^{\mathcal{N}}$. Acting on a spherical harmonic, the exponential gives $e^{-im_l\phi_0} Y_l^{m_l}$, which is $Y_l^{m_l}$ evaluated at $\phi - \phi_0$, as a rotation should.

The connection between rotations and angular momentum leads to a conservation law. A potential that depends only on the distance r is called a central potential, and it is unchanged by any rotation about the origin. The Hamiltonian is $\hat{H} = (\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2)/2m + V(r)$, which is the energy $p^2/2m + V$ written in terms of the three momentum operators of this section, and for a central potential it commutes with $\hat{L}_z$. The potential term commutes with $-i\hbar \partial/\partial \phi$ because $V(r)$ does not depend on ϕ . For the kinetic term, expanding the commutator with the rules of Section 3.1 gives

$$[\hat{p}_x^2 + \hat{p}_y^2, \hat{x}\hat{p}_y - \hat{y}\hat{p}_x] = [\hat{p}_x^2, \hat{x}]\hat{p}_y - [\hat{p}_y^2, \hat{y}]\hat{p}_x = -2i\hbar\hat{p}_x\hat{p}_y + 2i\hbar\hat{p}_y\hat{p}_x = 0, \quad (4.27)$$

and $\hat{p}_z^2$ commutes with $\hat{L}_z$, which contains no z and no $\hat{p}_z$. The z -axis is not special, so $\hat{H}$ commutes with $\hat{L}_x$ and $\hat{L}_y$ as well, and therefore with $\hat{L}^2$. By the result of Section 3.1 that an observable whose operator commutes with the Hamiltonian has a constant expectation value, every component of the angular momentum is conserved in a central potential, in the same way that a central force conserves the angular momentum of a planet, as shown in Vol. 1 of the Fermi Notes on Physics.

4.2 Spin One-Half and Two-Level Systems

Electrons, protons, and neutrons carry an angular momentum that has nothing to do with their motion through space, and which is called spin. This section begins with the experiment that first revealed it, then sets up the mathematics of spin one-half using column vectors and matrices, and finally works out the motion of a spin in a magnetic field. The same mathematics describes any system that has two states, and the section ends with the example of the ammonia molecule.

The Stern–Gerlach Experiment

A current loop, or any circulating charge, has a magnetic dipole moment $\boldsymbol{\mu}$, and in a magnetic field $\mathbf{B}$ its energy is $-\boldsymbol{\mu} \cdot \mathbf{B}$, a result from electromagnetism. In a uniform field the dipole experiences a torque but no net force. In a field along z whose component B_z varies with z , the energy $-\mu_z B_z$ depends on position, and since a force is minus the rate of change of the potential energy with position, the dipole experiences a force

$$F_z = \mu_z \frac{\partial B_z}{\partial z} \quad (4.28)$$

along the direction in which the field changes.

In 1922 Stern and Gerlach sent a beam of silver atoms from an oven through a magnet shaped to give a strong field gradient, and collected the atoms on a glass plate, as shown in Figure 26. The atoms left the oven with their magnetic moments pointing in all directions. If μ_z could take any value between $-\mu$ and μ , the beam would be spread into a continuous band. Instead the atoms arrived in two separate spots, one deflected upwards and one downwards, so that μ_z took only two values.

An angular momentum with quantum number l has $2l + 1$ possible z -components, which is an odd number for orbital angular momentum. A pair of values requires $2l + 1 = 2$, or an angular momentum quantum number of one half, which Section 4.1 showed is allowed by the commutation relations but cannot arise from motion in space. Silver has a single outer electron whose orbital angular momentum is zero, a fact established from the spectrum of silver, and the mag-

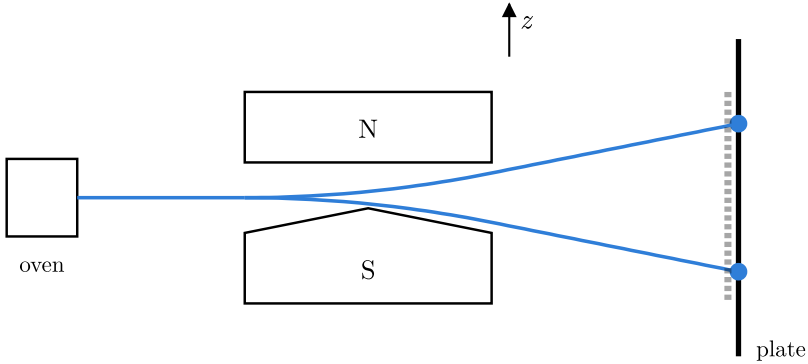


Figure 26. The Stern–Gerlach experiment. A beam of silver atoms passes between the shaped poles N and S of a magnet, where the field is strongest near the pointed pole, so that B_z varies with z , and is split into two beams that strike the plate at two separate spots. A classical distribution of magnetic moments would give the continuous band shown dashed.

netic moment of the atom comes from an angular momentum of the electron itself. Uhlenbeck and Goudsmit proposed in 1925, to explain details of atomic spectra, that the electron has such an intrinsic angular momentum, and the name spin comes from their picture of the electron as a spinning ball of charge. That literal picture cannot be made to work in detail, and spin is treated as an intrinsic property of the electron in the same way as its charge and its mass.

The spin of a particle is written $\mathbf{S}$, and the magnetic moment associated with the spin is proportional to it,

$$\boldsymbol{\mu} = \gamma \mathbf{S}, \quad (4.29)$$

where γ is called the gyromagnetic ratio. For the electron γ is very close to $-e/m_e$, which is twice the value that classical electromagnetism gives for a charge $-e$ of mass m_e moving in an orbit. The classical value follows from the magnetic moment of a current loop, which is the current multiplied by the area of the loop, a result from electromagnetism. An electron moving with speed v round a circle of radius r passes any point $v/2\pi r$ times a second, so it is a current $-ev/2\pi r$ round a loop of area πr^2 , and its magnetic moment is

$-e\hbar/2$. Its angular momentum is $m_e \hbar$, so the magnetic moment is $-e/2m_e$ times the angular momentum.

Spin Operators

The components of spin obey the same commutation relations as the components of orbital angular momentum,

$$[\hat{S}_x, \hat{S}_y] = i\hbar\hat{S}_z, \quad [\hat{S}_y, \hat{S}_z] = i\hbar\hat{S}_x, \quad [\hat{S}_z, \hat{S}_x] = i\hbar\hat{S}_y. \quad (4.30)$$

The argument of Section 4.1 used only these relations, so it applies unchanged. The eigenvalues of $\hat{S}^2$ are $\hbar^2 s(s+1)$ and those of $\hat{S}_z$ are $\hbar m_s$, with m_s running from $-s$ to s in steps of one. The electron, the proton, and the neutron all have $s = \frac{1}{2}$, so S^2 always has the value $\frac{3}{4}\hbar^2$ and S_z has the two values $\pm\frac{1}{2}\hbar$. The states with $m_s = \frac{1}{2}$ and $m_s = -\frac{1}{2}$ are called spin up and spin down.

Spin has no wave function in space, since it is not built from $\mathbf{r}$ and $\mathbf{p}$. The spin state of a particle is instead described by two complex numbers, which give the amounts of spin up and spin down it contains, and these are arranged as a column,

$$\chi = \begin{pmatrix} a \\ b \end{pmatrix} = a\chi_+ + b\chi_-, \quad \chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \chi_- = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (4.31)$$

An object of this kind is called a spinor. If S_z is measured, the result is $+\frac{1}{2}\hbar$ with probability $|a|^2$ and $-\frac{1}{2}\hbar$ with probability $|b|^2$, so a normalised spinor has $|a|^2 + |b|^2 = 1$.

Matrices

Operators on spinors are represented by matrices. A two-by-two matrix acts on a column by the rule

$$\begin{pmatrix} M_{11} & M_{12} \\ M_{21} & M_{22} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \begin{pmatrix} M_{11}a + M_{12}b \\ M_{21}a + M_{22}b \end{pmatrix}, \quad (4.32)$$

and the product of two matrices is the matrix that acts as the two do in succession, which gives $(MN)_{ij} = M_{i1}N_{1j} + M_{i2}N_{2j}$. As with operators, MN and NM are in general different. The conjugate transpose of a matrix, written $M^\dagger$, is obtained by interchanging rows and columns and taking the complex conjugate of every entry, and the conjugate transpose of the column χ is the row $\chi^\dagger = (a^* \ b^*)$.

The product of a row and a column is a number, formed by multiplying corresponding entries and adding, so that the row $(a^* \ b^*)$ times the column with entries c and d is $a^*c + b^*d$, and for two spinors χ and χ' the number $\chi^\dagger\chi'$ plays the part of the integral $\int f^*g \, dx$ for wave functions. In particular, $\chi^\dagger\chi = |a|^2 + |b|^2$, and the expectation value of the observable represented by M is $\langle M \rangle = \chi^\dagger M \chi$. Multiplying out shows that $(M\chi)^\dagger = \chi^\dagger M^\dagger$ for any matrix M and column χ . A matrix with $M^\dagger = M$ is called hermitian, and for such a matrix $(M\chi)^\dagger\chi' = \chi^\dagger(M\chi')$, which is the definition of a hermitian operator in Section 3.1 with the integral replaced by the product of a row and a column. A hermitian matrix therefore has real expectation values, real eigenvalues, and orthogonal eigenvectors for different eigenvalues, by the arguments of Section 3.1.

The eigenvalues of a matrix are found as follows. The equation $M\chi = q\chi$ can be rewritten as $(M - q\mathbf{1})\chi = 0$, where $\mathbf{1}$ is the identity matrix, with ones on the diagonal and zeros elsewhere. Writing out the two components, the equations $(M_{11} - q)a + M_{12}b = 0$ and $M_{21}a + (M_{22} - q)b = 0$ have a solution other than $a = b = 0$ only if the ratio a/b given by the first equation agrees with the one given by the second, which requires

$$(M_{11} - q)(M_{22} - q) - M_{12}M_{21} = 0. \quad (4.33)$$

The left-hand side is called the determinant of $M - qI$, and this quadratic equation for q is called the characteristic equation. Its roots are the eigenvalues, and substituting each root back into either component equation gives the ratio a/b of the corresponding eigenvector.

The Pauli Matrices

The spin operators for $s = \frac{1}{2}$ can now be written down. The spinors $\chi_\pm$ are eigenvectors of $\hat{S}_z$ with eigenvalues $\pm\frac{1}{2}\hbar$, so $\hat{S}_z$ is represented by a diagonal matrix. The raising operator $\hat{S}_+ = \hat{S}_x + i\hat{S}_y$ turns χ_- into $\hbar\sqrt{s(s+1) - m_s(m_s+1)}\chi_+ = \hbar\chi_+$, by the result you found in Problem 2 of Section 4.1 with $s = \frac{1}{2}$ and $m_s = -\frac{1}{2}$, and it turns χ_+ into zero, since χ_+ is the top of the ladder. The lowering

operator turns χ_+ into $\hbar\chi_-$ and χ_- into zero. Therefore

$$\hat{S}_z = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \hat{S}_+ = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \quad \hat{S}_- = \hbar \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}. \quad (4.34)$$

Since $\hat{S}_x = (\hat{S}_+ + \hat{S}_-)/2$ and $\hat{S}_y = (\hat{S}_+ - \hat{S}_-)/2i$, the three components are $\hat{\mathbf{S}} = \frac{1}{2}\hbar\boldsymbol{\sigma}$, where $\boldsymbol{\sigma}$ has the components σ_1 , σ_2 , and σ_3 , given by

$$\sigma_1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad (4.35)$$

are called the Pauli matrices. Multiplying them out gives $\sigma_1\sigma_2 = i\sigma_3$ and $\sigma_2\sigma_1 = -i\sigma_3$, so $[\hat{S}_x, \hat{S}_y] = \frac{1}{4}\hbar^2(2i\sigma_3) = i\hbar\hat{S}_z$, as required, and, as you will show in Problem 1, the square of each Pauli matrix is the identity, so $\hat{S}^2 = \frac{3}{4}\hbar^2\mathbf{1}$.

Consider the following example. Suppose that a particle is in the state χ_+ , with its spin up along z , and S_x is measured. The characteristic equation of $\hat{S}_x$ is $q^2 - \hbar^2/4 = 0$, so the possible results are $\pm\frac{1}{2}\hbar$, as they must be for a spin one-half along any axis. The first component equation with $q = \pm\frac{1}{2}\hbar$ gives $b = \pm a$, and the normalised eigenvectors are

$$\chi_{\pm}^{(x)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}. \quad (4.36)$$

To find the probabilities of the two results, we expand the state χ_+ in the eigenvectors of $\hat{S}_x$, in the same way as a wave function was expanded in the eigenfunctions of the measured observable in Section 3.1. Adding the two eigenvectors gives

$$\chi_+ = \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \frac{1}{\sqrt{2}}\chi_+^{(x)} + \frac{1}{\sqrt{2}}\chi_-^{(x)}. \quad (4.37)$$

The coefficient of each eigenvector is $1/\sqrt{2}$. It can also be found as the product $\chi_{\pm}^{(x)\dagger}\chi_+$, which is the spinor version of the formula $c_n = \int f_n^*\Psi dx$ of Section 3.1. The probability of each result is the squared modulus of its coefficient, which is one half, so the results $+\frac{1}{2}\hbar$ and $-\frac{1}{2}\hbar$ are equally likely. A spin known to point up along z is therefore equally likely to be found pointing in either direction along x .

Spin in a Magnetic Field

A particle at rest in a uniform magnetic field B_0 along z has the energy $-\boldsymbol{\mu} \cdot \mathbf{B} = -\gamma B_0 S_z$, so its Hamiltonian is

$$\hat{H} = -\gamma B_0 \hat{S}_z = -\frac{\gamma B_0 \hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (4.38)$$

whose stationary states are χ_+ and χ_- , with energies $\mp \frac{1}{2} \gamma B_0 \hbar$. The Schrödinger equation for a spinor, $i\hbar d\chi/dt = \hat{H}\chi$, is a pair of equations for the two components, and here each involves one component only. Let $\omega_L = \gamma B_0$. The solution is

$$\chi(t) = \begin{pmatrix} a e^{i\omega_L t/2} \\ b e^{-i\omega_L t/2} \end{pmatrix}, \quad (4.39)$$

and to see what it describes we take $a = \cos(\theta_0/2)$ and $b = \sin(\theta_0/2)$, with θ_0 a fixed angle. You will show in Problem 3 that at $t = 0$ this is the state whose spin points along a direction at an angle θ_0 to the z -axis, in the xz -plane. The expectation values of the components of spin are found by multiplying out $\chi^\dagger \hat{S} \chi$. For the x -component,

$$\begin{aligned} \langle S_x \rangle &= \frac{\hbar}{2} \begin{pmatrix} a e^{-i\omega_L t/2} & b e^{i\omega_L t/2} \end{pmatrix} \begin{pmatrix} b e^{-i\omega_L t/2} \\ a e^{i\omega_L t/2} \end{pmatrix} \\ &= \frac{\hbar}{2} ab (e^{-i\omega_L t} + e^{i\omega_L t}) = \frac{\hbar}{2} \sin \theta_0 \cos \omega_L t, \end{aligned} \quad (4.40)$$

where $2ab = \sin \theta_0$ has been used. Similar calculations give

$$\langle S_y \rangle = -\frac{\hbar}{2} \sin \theta_0 \sin \omega_L t, \quad \langle S_z \rangle = \frac{\hbar}{2} \cos \theta_0. \quad (4.41)$$

The expectation value of the spin keeps a fixed angle θ_0 to the field and rotates about it at the angular frequency $\omega_L = \gamma B_0$, as shown in Figure 27. This motion is called Larmor precession, and ω_L is called the Larmor frequency. A spinning top in a gravitational field precesses in a similar way, and a classical magnetic moment proportional to its angular momentum, acted on by the torque $\boldsymbol{\mu} \times \mathbf{B}$, precesses at the same frequency. Protons in a magnetic field precess at a Larmor frequency set by their gyromagnetic ratio, and the detection of this precession is the basis of nuclear magnetic resonance and of magnetic resonance imaging.

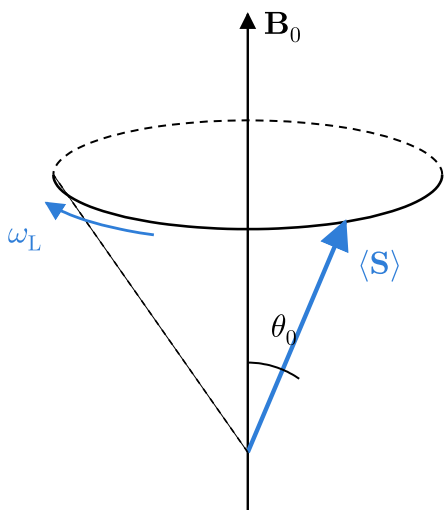


Figure 27. Larmor precession. The expectation value $\langle \mathbf{S} \rangle$ of a spin in a field B_0 along z keeps a fixed angle θ_0 to the field and turns about it at the angular frequency ω_L . The arrow on the circle shows the sense of rotation for positive γ .

Two-Level Systems

Many systems have two states whose energies are close together, while all their other states are far away in energy, and any such system can be described by spinors and two-by-two matrices, whether or not spin is involved. Let the two states be represented by $\chi_1 = (1 \ 0)^T$ and $\chi_2 = (0 \ 1)^T$, where the superscript T denotes the transpose, which turns the row into a column. The most general hermitian Hamiltonian has real numbers E_1 and E_2 on the diagonal and complex conjugate numbers off it, since $\hat{H}^\dagger = \hat{H}$ requires each diagonal entry to equal its complex conjugate and the two off-diagonal entries to be complex conjugates of each other, and by choosing the phases of the two states the off-diagonal entries can be made real, so we take

$$\hat{H} = \begin{pmatrix} E_1 & -\Delta \\ -\Delta & E_2 \end{pmatrix}, \quad (4.42)$$

with Δ real. If $\Delta = 0$, the states χ_1 and χ_2 are stationary with energies E_1 and E_2 , and the constant Δ is called the coupling between them. The characteristic equation is $(E_1 - q)(E_2 - q) - \Delta^2 = 0$, whose roots are

$$E_{\pm} = \bar{E} \pm \sqrt{\frac{(E_1 - E_2)^2}{4} + \Delta^2}, \quad \bar{E} = \frac{E_1 + E_2}{2}. \quad (4.43)$$

The coupling pushes the two levels apart. Figure 28 shows the energies as $E_1 - E_2$ is varied with Δ fixed. Without coupling the two levels would cross where $E_1 = E_2$, and with coupling their separation never falls below $2|\Delta|$. This behaviour is called an avoided crossing.

The Ammonia Molecule

The ammonia molecule, NH_3 , is a pyramid with the three hydrogen atoms at the base and the nitrogen atom at the apex. The nitrogen can sit on either side of the plane of the hydrogen atoms, as shown in Figure 29, and the two arrangements, which we label 1 and 2, have the same energy E_0 . To pass from one arrangement to the other, the nitrogen has to cross the plane of the hydrogen atoms, where its potential energy is higher, and classically it could not do so at low energies. Quantum mechanically it can tunnel through this barrier, as in Section 2.5, and the tunnelling gives a small coupling

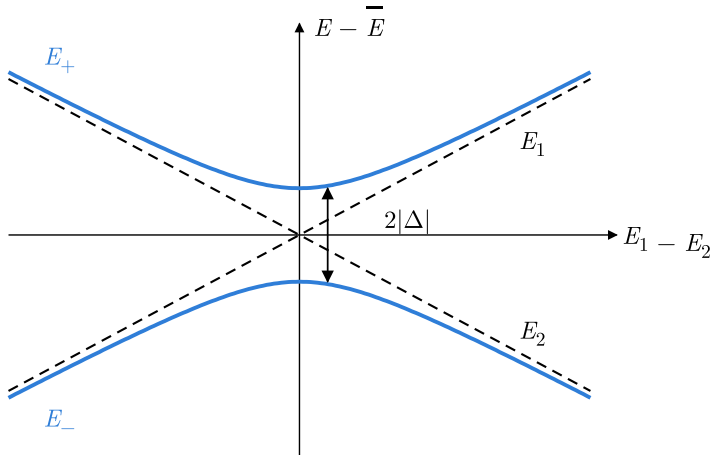


Figure 28. The energies $E_{\pm}$ of a two-level system, measured from $\bar{E}$, plotted against $E_1 - E_2$ for a fixed coupling Δ (accent). The dashed lines show the uncoupled energies E_1 and E_2 , which cross. With coupling the levels are separated by at least $2|\Delta|$.

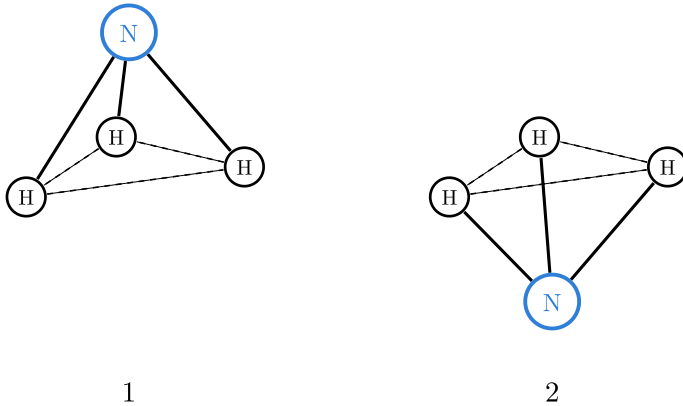


Figure 29. The two arrangements of the ammonia molecule. The nitrogen atom N lies above the plane of the three hydrogen atoms H in arrangement 1 and below it in arrangement 2. Tunnelling through the plane couples the two arrangements.

$\Delta > 0$ between the two arrangements. The Hamiltonian is that of the previous subsection with $E_1 = E_2 = E_0$.

With $E_1 = E_2$ the energies are $E_0 \mp \Delta$, and the eigenvectors, found from the component equations, are

$$\chi_s = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \chi_a = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad (4.44)$$

with energies $E_0 - \Delta$ and $E_0 + \Delta$. In neither stationary state is the nitrogen on a definite side of the plane. Suppose that at $t = 0$ the molecule is in arrangement 1, so that $\chi(0) = \chi_1 = (\chi_s + \chi_a)/\sqrt{2}$. Each stationary state acquires the phase factor belonging to its energy, and

$$\begin{aligned} \chi(t) &= \frac{1}{\sqrt{2}} [\chi_s e^{-i(E_0 - \Delta)t/\hbar} + \chi_a e^{-i(E_0 + \Delta)t/\hbar}] \\ &= e^{-iE_0 t/\hbar} \begin{pmatrix} \cos(\Delta t/\hbar) \\ i \sin(\Delta t/\hbar) \end{pmatrix}, \end{aligned} \quad (4.45)$$

where the last step uses Euler's formula. The probability of finding the nitrogen on the other side is

$$P_2(t) = \sin^2 \frac{\Delta t}{\hbar}, \quad (4.46)$$

so the nitrogen passes back and forth through the plane, and the probability oscillates with angular frequency $2\Delta/\hbar$, which is the difference between the energies of χ_a and χ_s divided by $\hbar$. A molecule that moves from the upper state χ_a to the lower state χ_s emits a photon of frequency $2\Delta/\hbar$, which lies in the microwave region. Townes and his collaborators used this transition in 1954 to build the first maser, a microwave amplifier in which molecules in the upper state are stimulated to emit.

Problems

§4.1 Orbital Angular Momentum

1. Show that $[\hat{L}_z, \hat{x}] = i\hbar\hat{y}$, $[\hat{L}_z, \hat{y}] = -i\hbar\hat{x}$, and $[\hat{L}_z, \hat{p}_x] = i\hbar\hat{p}_y$.
2. Show that $\int (\hat{L}_+ f)^* g \, dV = \int f^* (\hat{L}_- g) \, dV$ for normalisable functions f and g . Use this result, and the expression for $\hat{L}_- \hat{L}_+$, to show that if f_{l,m_l} is a normalised eigenfunction then $\hat{L}_+ f_{l,m_l} =$

- $\hbar\sqrt{l(l+1) - m_l(m_l+1)} f_{l,m_l+1}$, up to a factor of unit modulus, and find the corresponding result for $\hat{L}_-$.
- Using the Cartesian forms of the operators, show that $\hat{L}_+ z = -\hbar(x + iy)$. Check that the spherical form of $\hat{L}_+$ gives the same result when applied to $z = r \cos \theta$.
 - Show that Y_1^0 is normalised and is orthogonal to Y_1^1 . Find the constant C_2 in $Y_2^2 = C_2 \sin^2 \theta e^{2i\phi}$.
 - A particle has the wave function $\psi = (x + iy) R(r)$, where R is any function of r . Show that it is an eigenfunction of $\hat{L}_z$ with $m_l = 1$ and of $\hat{L}^2$ with $l = 1$.
 - Write $\sin \theta \cos \phi$ as a combination of Y_1^1 and Y_1^{-1} , apply the rotation operator $\exp(-i\phi_0 \hat{L}_z / \hbar)$ to it, and show that the result is $\sin \theta \cos(\phi - \phi_0)$.

§4.2 Spin One-Half and Two-Level Systems

- Show that the Pauli matrices are hermitian, that the square of each is the identity, and that $\sigma_2 \sigma_3 = i\sigma_1$.
- Find the eigenvalues and normalised eigenvectors of $\hat{S}_y$. A particle is in the state χ_+ . Find the probability that a measurement of S_y gives $+\frac{1}{2}\hbar$.
- Let $\mathbf{n} = (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta)$ be a unit vector. Show that $\chi = (\cos \frac{\theta}{2} e^{i\phi} \sin \frac{\theta}{2})^T$ is an eigenvector of $\mathbf{n} \cdot \hat{\mathbf{S}} = \frac{1}{2}\hbar(n_x \sigma_1 + n_y \sigma_2 + n_z \sigma_3)$ with eigenvalue $+\frac{1}{2}\hbar$, and that $\langle \mathbf{S} \rangle = \frac{1}{2}\hbar \mathbf{n}$ in this state.
- Verify the result for $\langle S_y \rangle$ in the Larmor problem, and show that $\langle H \rangle$ does not change in time.
- For the two-level system with $|E_1 - E_2| \gg |\Delta|$, show that the energies are approximately $E_1 + \Delta^2 / (E_1 - E_2)$ and $E_2 - \Delta^2 / (E_1 - E_2)$.
- An ammonia molecule has an electric dipole moment along its axis, which points in opposite directions in the two arrangements. In an electric field the energy of arrangement 1 is raised by an amount ε_f proportional to the field and that of arrangement 2 is lowered by the same amount, so that $E_1 = E_0 + \varepsilon_f$ and $E_2 =$

$E_0 - \varepsilon_f$. Find the energies of the stationary states, and describe how they depend on ε_f when ε_f is much smaller than Δ and when it is much larger.

Central Potentials and Canonical Three-Dimensional Systems

5.1 The Schrödinger Equation in Three Dimensions

In this chapter we solve the Schrödinger equation for a particle moving in three dimensions. This section first sets up the equation, then solves it in Cartesian coordinates for two potentials that are sums of terms in the three coordinates, and finally reduces the problem of a central potential to a single radial equation, which we solve for a spherical well and apply to a rotating molecule. The section that follows solves the radial equation for the hydrogen atom.

The Equation in Three Dimensions

The Hamiltonian of a particle of mass m in a potential $V(\mathbf{r})$ is $\hat{H} = (\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2)/2m + V$, and with the momentum operators of Section 4.1 the time-independent Schrödinger equation becomes

$$-\frac{\hbar^2}{2m}\nabla^2\psi + V(\mathbf{r})\psi = E\psi, \quad \nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}, \quad (5.1)$$

where the operator ∇^2 is called the Laplacian. The wave function is normalised by $\int |\psi|^2 dV = 1$, and the stationary states combine into

the general solution with the factors $e^{-iEt/\hbar}$, as in one dimension.

A Particle in a Box

Let the potential be zero inside a rectangular box with sides w_1 , w_2 , and w_3 along the three axes, and infinite outside it. We look for solutions of the form $\psi = f_1(x)f_2(y)f_3(z)$. Substituting this product into the time-independent equation and dividing by ψ gives

$$-\frac{\hbar^2}{2m} \left(\frac{1}{f_1} \frac{d^2 f_1}{dx^2} + \frac{1}{f_2} \frac{d^2 f_2}{dy^2} + \frac{1}{f_3} \frac{d^2 f_3}{dz^2} \right) = E. \quad (5.2)$$

The first term in the bracket depends only on x , the second only on y , and the third only on z , so by the argument of Section 2.2 each is a constant. Each function therefore satisfies the equation of the infinite square well of Section 2.2, with the width of the box along that axis, and vanishes at the walls. The stationary states are products of the one-dimensional states, and the energies are

$$E = \frac{\pi^2 \hbar^2}{2m} \left(\frac{n_1^2}{w_1^2} + \frac{n_2^2}{w_2^2} + \frac{n_3^2}{w_3^2} \right), \quad n_1, n_2, n_3 = 1, 2, 3, \dots \quad (5.3)$$

For a cube of side w the energy is $(\pi^2 \hbar^2 / 2mw^2)(n_1^2 + n_2^2 + n_3^2)$, and different sets of quantum numbers can give the same energy. The ground state has $(n_1, n_2, n_3) = (1, 1, 1)$, while the three states $(2, 1, 1)$, $(1, 2, 1)$, and $(1, 1, 2)$ share the next energy, so that energy is threefold degenerate. The degeneracy comes from the symmetry of the cube, whose three axes are equivalent. In a box with three different sides the three states have different energies.

The Isotropic Oscillator

As another example, consider the potential $V = \frac{1}{2}m\omega^2(x^2 + y^2 + z^2) = \frac{1}{2}m\omega^2 r^2$. A particle moving in this potential is called an isotropic harmonic oscillator, since the restoring force has the same form in every direction. The potential is a sum of three terms, each depending on one coordinate, and the same separation gives three one-dimensional oscillators, each with the energies of Section 2.6. The energy of the product state is

$$E = \left(n_1 + n_2 + n_3 + \frac{3}{2} \right) \hbar\omega = \left(n + \frac{3}{2} \right) \hbar\omega, \quad (5.4)$$

with $n = n_1 + n_2 + n_3 = 0, 1, 2, \dots$. The degeneracy of the level n is the number of ways of writing n as a sum $n_1 + n_2 + n_3$ of three non-negative integers, with different orders counted separately. For each value of n_1 from 0 to n , the number n_2 can take any of the $n - n_1 + 1$ values from 0 to $n - n_1$, after which n_3 is fixed, so the degeneracy is

$$\sum_{n_1=0}^n (n - n_1 + 1) = (n + 1) + n + \dots + 1 = \frac{(n + 1)(n + 2)}{2}. \quad (5.5)$$

The ground state is not degenerate, the first excited level is threefold degenerate, the second is sixfold degenerate, and so on.

Central Potentials

For a central potential $V(r)$ spherical coordinates are the natural choice. In these coordinates the Laplacian is

$$\nabla^2 = \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) - \frac{\hat{L}^2}{\hbar^2 r^2}, \quad (5.6)$$

where $\hat{L}^2$ is the operator of Section 4.1. This form of the Laplacian is obtained with the chain rule, in the same way as $\hat{L}_z$ was in Section 4.1, and you will check the radial part for a function of r alone in Problem 1. The time-independent equation reads

$$-\frac{\hbar^2}{2m} \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial \psi}{\partial r} \right) + \frac{\hat{L}^2}{2mr^2} \psi + V(r) \psi = E \psi, \quad (5.7)$$

and the term in $\hat{L}^2$ suggests that we look for solutions that are eigenfunctions of $\hat{L}^2$, which is possible by the result of Section 3.1 on compatible observables, since Section 4.1 showed that $\hat{L}^2$ commutes with $\hat{H}$ for a central potential. Let

$$\psi(r, \theta, \phi) = R(r) Y_l^{m_l}(\theta, \phi). \quad (5.8)$$

Since $\hat{L}^2 Y_l^{m_l} = \hbar^2 l(l + 1) Y_l^{m_l}$, the spherical harmonic cancels from every term, and the function $R(r)$ satisfies an ordinary differential equation. The equation is simplest in terms of $U(r) = rR(r)$. Differentiating $R = U/r$ gives $r^2 dR/dr = r dU/dr - U$, and differentiating again gives $r d^2U/dr^2$, so that the first term becomes $-(\hbar^2/2m)(1/r) d^2U/dr^2$. Multiplying the whole equation by r gives

$$-\frac{\hbar^2}{2m} \frac{d^2U}{dr^2} + V_{\text{eff}}(r) U = EU, \quad V_{\text{eff}}(r) = V(r) + \frac{\hbar^2 l(l + 1)}{2mr^2}, \quad (5.9)$$

which is called the radial equation. It has the form of the one-dimensional Schrödinger equation for a particle moving on the half-line $r > 0$ in the potential V_{eff} , which is called the effective potential. The extra term is called the centrifugal term. It is the quantum form of the term $L^2/2mr^2$ that appeared when the orbit problem was reduced to a radial problem in Vol. 1 of the Fermi Notes on Physics, and it keeps a particle with angular momentum away from the origin.

The radial equation needs a boundary condition at the origin, and the condition is that $U(0) = 0$. For $l \geq 1$ the centrifugal term dominates the radial equation near the origin, where the equation becomes $d^2U/dr^2 = l(l+1)U/r^2$, and substituting $U = r^\nu$ shows that its two solutions there behave as r^{l+1} and r^{-l} . The second solution makes $\int |U|^2 dr$ diverge at the origin, so it cannot be normalised and is excluded. For $l = 0$ the two solutions near the origin behave as $U \propto r$ and as U tending to a non-zero constant, and the second corresponds to $R \propto 1/r$. This function can be normalised, since $|U|^2$ stays finite, and it is excluded for a different reason. The radial part of the Laplacian of $1/r$, which is $r^{-2} d(r^2 d(1/r)/dr)/dr$, is zero for every $r > 0$, but at the origin the Laplacian of $1/r$ behaves as a three-dimensional form of the delta function of Section 2.3, a result whose proof is beyond this course, so $R \propto 1/r$ would satisfy the Schrödinger equation only if the potential contained a delta function at the origin. The normalisation condition becomes, using the element of volume of Section 4.1 and the normalisation of the spherical harmonics,

$$\int |\psi|^2 dV = \int_0^\infty |R|^2 r^2 dr = \int_0^\infty |U|^2 dr = 1. \quad (5.10)$$

The magnetic quantum number m_l does not appear in the radial equation. For any central potential, therefore, each energy belongs to at least $2l + 1$ states, one for each value of m_l , which is a consequence of the rotational symmetry of the potential. The energies depend on l through the centrifugal term, and for each l the radial equation usually has a set of solutions with different energies.

The Infinite Spherical Well

Let the potential be zero for $r < r_w$ and infinite for $r > r_w$. For $l = 0$ the effective potential is zero inside the well, and the radial equation is $d^2U/dr^2 = -k^2U$ with $k = \sqrt{2mE}/\hbar$. This is the equation of the infinite square well of Section 2.2, with the same conditions, $U(0) = 0$ and $U(r_w) = 0$, so the solutions are the same,

$$U(r) = \sqrt{\frac{2}{r_w}} \sin \frac{n\pi r}{r_w}, \quad E = \frac{n^2\pi^2\hbar^2}{2mr_w^2}, \quad n = 1, 2, 3, \dots \quad (5.11)$$

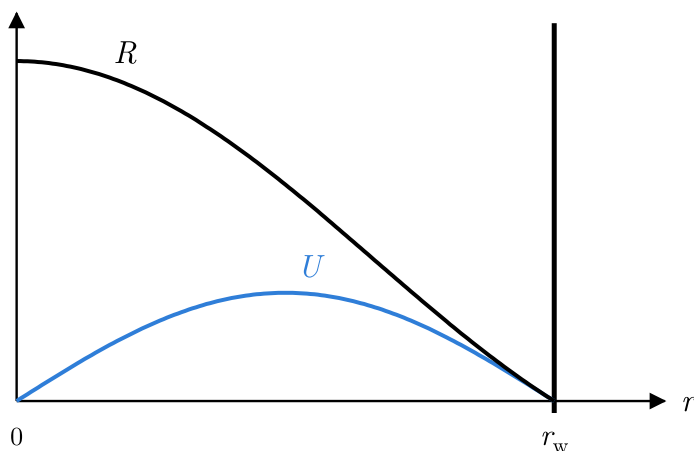


Figure 30. The ground state of the infinite spherical well of radius r_w , drawn with r_w as the unit of length. The function $U = rR$ (accent) is the ground state of the one-dimensional well, and the radial function $R = U/r$ (solid) is largest at the centre and falls to zero at the wall.

The wave function is $\psi = (U/r) Y_0^0 = \sin(n\pi r/r_w)/(r\sqrt{2\pi r_w})$, and its radial part R is drawn for the ground state in Figure 30, together with U . The function $R = U/r$ stays finite at the origin, where $\sin(n\pi r/r_w)/r$ tends to $n\pi/r_w$. For $l > 0$ the centrifugal term makes the radial equation harder to solve, and you will treat the case $l = 1$ in Problem 4.

The Rigid Rotor

A diatomic molecule can rotate about its centre of mass. If the separation of the two atoms is fixed, which is a valid approximation for the lowest rotational states, the molecule is called a rigid rotor, and in the frame of its centre of mass its energy is all rotational. The classical energy of a body rotating with angular momentum L about an axis through its centre of mass is $L^2/2I$, where I is the moment of inertia, and for a diatomic molecule $I = m_r r_e^2$, where r_e is the separation of the atoms and m_r their reduced mass, since you showed in Problem 2 of Section 1.3 that two masses turning about their centre of mass at angular velocity ω and separation r have angular momentum $m_r \omega r^2$. The Hamiltonian is therefore

$$\hat{H} = \frac{\hat{L}^2}{2I}, \quad (5.12)$$

whose eigenfunctions are the spherical harmonics, with the direction of the axis of the molecule given by the angles θ and ϕ . The energies are

$$E_l = \frac{\hbar^2 l(l+1)}{2I}, \quad l = 0, 1, 2, \dots, \quad (5.13)$$

each with a degeneracy $2l+1$, and they are the rotational energies used in Section 5.2 of Vol. 2 of the Fermi Notes on Physics.

A molecule with an electric dipole moment, such as hydrogen chloride, can absorb a photon and move to a higher rotational level. The absorption is found to occur only between neighbouring levels, from l to $l+1$, a result called a selection rule, whose derivation is beyond this course. The photon energies are

$$E_{l+1} - E_l = \frac{\hbar^2}{2I} [(l+1)(l+2) - l(l+1)] = \frac{\hbar^2}{I} (l+1), \quad (5.14)$$

so the photon energies increase in equal steps of $\hbar^2/I$, even though the levels themselves spread further apart as l grows. Each line of the spectrum comes from one of these transitions, and the rotational absorption spectrum, which lies in the far infrared and microwave regions, is therefore a series of equally spaced lines, as shown in Figure 31. Measuring the spacing $\hbar^2/I$ gives the moment of inertia, and hence the separation of the atoms.

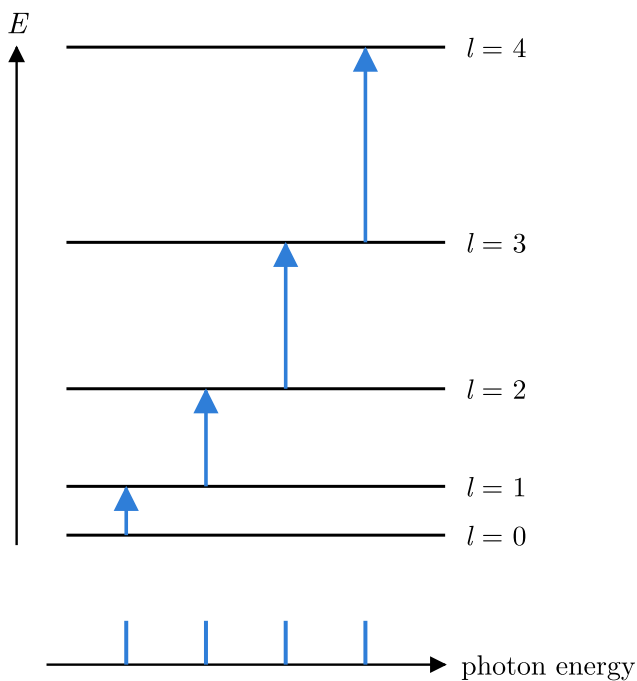


Figure 31. The rotational energy levels $E_l = \hbar^2 l(l+1)/2I$ of a rigid rotor, and the absorption transitions between neighbouring levels. The levels spread apart as l grows, and the photon energies $(l+1)\hbar^2/I$ increase in equal steps. Below, the four lines of the absorption spectrum that these transitions produce, each placed under its transition and equally spaced by $\hbar^2/I$.

5.2 The Hydrogen Atom

The hydrogen atom is a proton and an electron held together by the Coulomb attraction. Its spectrum was the subject of Section 1.3, where Bohr's model reproduced the energy levels by assuming that the angular momentum of the electron is a whole multiple of $\hbar$, an assumption for which the model gave no justification. In this section we first solve the radial equation of Section 5.1 for the Coulomb potential, then show that it gives the same energy levels, and finally find the wave functions and the angular momentum of each state.

The Radial Equation for Hydrogen

The atom is a two-body system. In Vol. 1 of the Fermi Notes on Physics the motion of two bodies was separated into the motion of their centre of mass and the relative motion of a single body with the reduced mass, and the same separation can be carried out for the Hamiltonian. The relative motion is that of a particle of reduced mass m_r in the potential of a fixed centre. As in Section 1.3, the reduced mass differs from the mass of the electron by about one part in two thousand, and we write m_e in what follows, with the understanding that the more accurate results are obtained by replacing m_e with m_r . The potential is

$$V(r) = -\frac{e^2}{4\pi\epsilon_0 r}, \quad (5.15)$$

and the radial equation of Section 5.1 becomes

$$-\frac{\hbar^2}{2m_e} \frac{d^2 U}{dr^2} + \left[-\frac{e^2}{4\pi\epsilon_0 r} + \frac{\hbar^2 l(l+1)}{2m_e r^2} \right] U = EU. \quad (5.16)$$

The effective potential is drawn in Figure 32. For $l = 0$ it is the Coulomb potential itself, and for $l > 0$ the centrifugal term dominates at small r , so that the effective potential has a minimum, which lies at $r = l(l+1)a_0$ and moves outwards as l grows. We look for bound states, with $E < 0$.

To simplify the subsequent expressions, let $\kappa = \sqrt{-2m_e E}/\hbar$, which is real for a bound state, and introduce the dimensionless variable

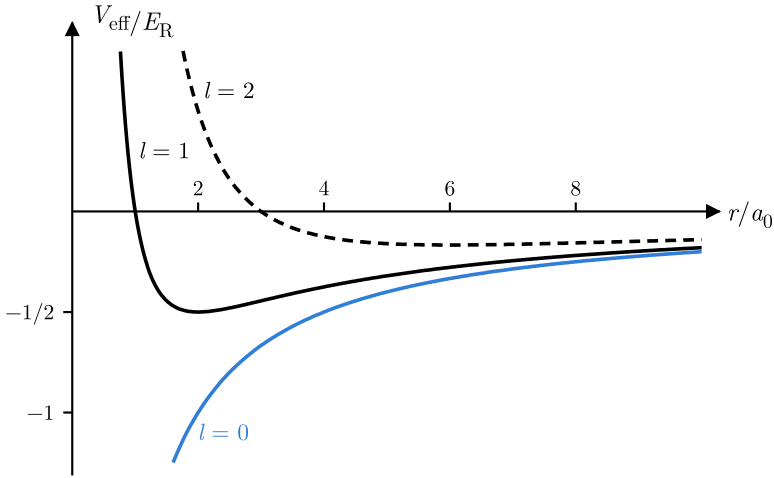


Figure 32. The effective potential of hydrogen, $-e^2/4\pi\epsilon_0 r + \hbar^2 l(l+1)/2m_e r^2$, for $l = 0$ (accent), $l = 1$ (solid), and $l = 2$ (dashed), with r in units of the Bohr radius a_0 and the energy in units of E_R . The centrifugal term keeps the particle away from the origin when $l > 0$.

$\rho = \kappa r$. Dividing the radial equation by $-\hbar^2 \kappa^2 / 2m_e$ gives

$$\frac{d^2 U}{d\rho^2} = \left[1 - \frac{\rho_0}{\rho} + \frac{l(l+1)}{\rho^2} \right] U, \quad \rho_0 = \frac{m_e e^2}{2\pi\epsilon_0 \hbar^2 \kappa}. \quad (5.17)$$

The energy is contained in ρ_0 , and the task is to find the values of ρ_0 for which the equation has an acceptable solution.

The Behaviour at Large and Small Distances

We proceed as for the oscillator in Section 2.6, by first finding how the solution behaves at the ends of its range. For large ρ the bracket tends to one, and the equation becomes $U'' = U$, where primes denote derivatives with respect to ρ . Its solutions are $e^{-\rho}$ and e^{ρ} , and only the first can be normalised. For small ρ the argument of Section 5.1 applies unchanged, since ρ is a constant multiple of r : the acceptable solution behaves as ρ^{l+1} , and the solution behaving as ρ^{-l} is excluded for every l by the boundary condition $U(0) = 0$ of Section 5.1. These two limits suggest that we write

$$U(\rho) = \rho^{l+1} e^{-\rho} v(\rho), \quad (5.18)$$

and look for an equation for v . By the product rule,

$$\frac{dU}{d\rho} = \rho^l e^{-\rho} \left[(l+1-\rho)v + \rho \frac{dv}{d\rho} \right], \quad (5.19)$$

and differentiating once more gives

$$\begin{aligned} \frac{d^2 U}{d\rho^2} = \rho^l e^{-\rho} \left[\left(-2l - 2 + \rho + \frac{l(l+1)}{\rho} \right) v \right. \\ \left. + 2(l+1-\rho) \frac{dv}{d\rho} + \rho \frac{d^2 v}{d\rho^2} \right]. \end{aligned} \quad (5.20)$$

The right-hand side of the radial equation is $\rho^l e^{-\rho} [\rho - \rho_0 + l(l+1)/\rho] v$. Equating this with the expression for $d^2 U/d\rho^2$ above and dividing by the common factor $\rho^l e^{-\rho}$, the terms in ρv and in $l(l+1)v/\rho$ cancel, and we are left with

$$\rho \frac{d^2 v}{d\rho^2} + 2(l+1-\rho) \frac{dv}{d\rho} + [\rho_0 - 2(l+1)] v = 0. \quad (5.21)$$

The Series Solution

We solve this equation by the power-series method of Section 2.6. Let $v = \sum_j c_j \rho^j$. Substituting the series and its derivatives, and shifting the label of summation in the terms $\rho d^2v/d\rho^2$ and $2(l+1)dv/d\rho$ so that every term carries the power ρ^j , gives

$$\sum_{j=0}^{\infty} [j(j+1)c_{j+1} + 2(l+1)(j+1)c_{j+1} - 2jc_j + (\rho_0 - 2(l+1))c_j] \rho^j = 0, \quad (5.22)$$

and setting each coefficient to zero gives the recursion relation

$$c_{j+1} = \frac{2(j+l+1) - \rho_0}{(j+1)(j+2l+2)} c_j. \quad (5.23)$$

If the series does not end, then for large j the relation gives $c_{j+1} \approx 2c_j/j$. The series for $e^{2\rho} = \sum_j (2\rho)^j/j!$ has coefficients whose ratio is $2/(j+1)$, which behaves in the same way, so a series that does not end grows like $e^{2\rho}$, and U grows like $\rho^{l+1}e^\rho$, which cannot be normalised. The series must therefore end. If $c_{j_{\max}}$ is the last non-zero coefficient, the numerator of the recursion relation must vanish for $j = j_{\max}$, which requires

$$\rho_0 = 2(j_{\max} + l + 1) = 2n, \quad n = j_{\max} + l + 1. \quad (5.24)$$

The integer n is called the principal quantum number. Since $j_{\max}$ and l are whole numbers that are not negative, n takes the values $1, 2, 3, \dots$, and for a given n the angular momentum quantum number can take the values $l = 0, 1, \dots, n-1$.

The Energy Levels

The condition $\rho_0 = 2n$ fixes κ . From the definition of ρ_0 ,

$$\kappa = \frac{m_e e^2}{2\pi\epsilon_0 \hbar^2 \rho_0} = \frac{m_e e^2}{4\pi\epsilon_0 \hbar^2} \frac{1}{n} = \frac{1}{na_0}, \quad (5.25)$$

where $a_0 = 4\pi\epsilon_0 \hbar^2 / m_e e^2$ is the Bohr radius of Section 1.3. The energy is $E = -\hbar^2 \kappa^2 / 2m_e$, and therefore

$$E_n = -\frac{\hbar^2}{2m_e a_0^2} \frac{1}{n^2} = -\frac{E_R}{n^2}, \quad E_R = \frac{m_e e^4}{2(4\pi\epsilon_0)^2 \hbar^2}. \quad (5.26)$$

These are the energies of the Bohr model, which here follow from the Schrödinger equation with no further assumption, and they give the Rydberg formula of Section 1.3 for the spectrum. The Lyman, Balmer, and Paschen series are the transitions that end on the levels $n = 1, 2$, and 3 . For a hydrogen-like ion with nuclear charge Ze , replacing e^2 by Ze^2 in the potential gives energies $-Z^2 E_R/n^2$ and a length scale a_0/Z , as in Section 1.3.

A stationary state is labelled by three quantum numbers, n , l , and m_l , and its wave function is written $\psi_{nlm_l} = R_{nl}(r) Y_l^{m_l}(\theta, \phi)$. The energy depends on n alone. For a given n there are $2l + 1$ values of m_l for each of the n values of l , so the degeneracy of the level is

$$\sum_{l=0}^{n-1} (2l + 1) = n^2. \quad (5.27)$$

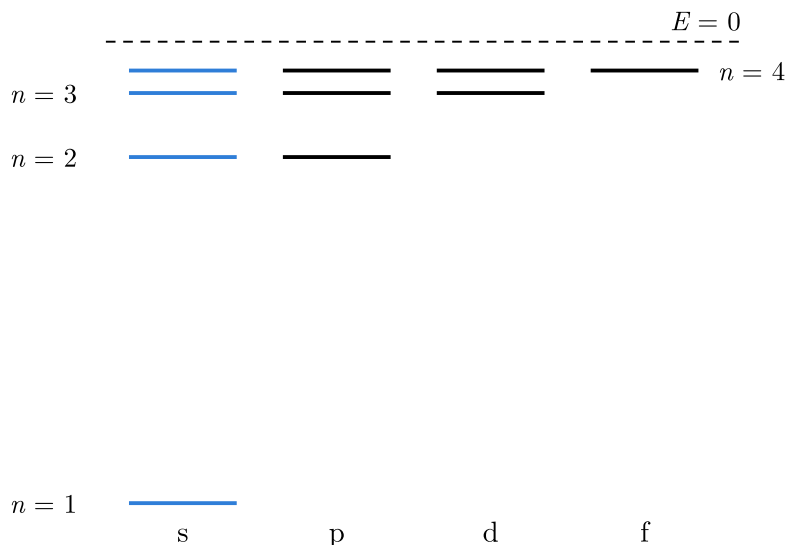


Figure 33. The energy levels of hydrogen arranged in columns by the angular momentum quantum number l , drawn to scale. Each level with principal quantum number n has one line in each of the columns $l = 0, 1, \dots, n - 1$, and each line stands for the $2l + 1$ states with that value of l , one for each value of m_l , so the level has n^2 states in all.

The degeneracy in m_l is present for every central potential, as Sec-

tion 5.1 showed. The degeneracy in l is a special feature of the potential $1/r$. In most other central potentials, states with different l have different energies, as the different effective potentials in Figure 32 would suggest, although the isotropic oscillator of Section 5.1 is another exception: its second excited level holds six states, and the states belonging to a single value of l and a single radial function number only $2l+1$, which is odd. The classical counterpart of the degeneracy is that the bound orbits in an inverse-square force are closed ellipses whose orientation does not change, which is not the case for most other central forces. By a convention taken from spectroscopy, states with $l = 0, 1, 2, 3$ are labelled by the letters s, p, d, f, so the state with $n = 2$ and $l = 1$ is called a 2p state. The levels are drawn in this way in Figure 33.

The solution also accounts for a failure of the Bohr model listed in Section 1.3, which assigns the ground state of hydrogen an angular momentum $\hbar$, whereas experiment finds it to be zero. The ground state has $n = 1$, which allows only $l = 0$, so its angular momentum is zero, in agreement with experiment.

The Radial Wave Functions

The radial functions follow from the recursion relation, with $\rho = r/na_0$, and $R = U/r$. For $n = 1$ we have $l = 0$ and $j_{\max} = 0$, so $v = c_0$ and $U = c_0\rho e^{-\rho}$. For $n = 2$ and $l = 0$, $j_{\max} = 1$, and the recursion relation with $\rho_0 = 4$ gives $c_1 = (2 - 4)c_0/2 = -c_0$, so $v = c_0(1 - \rho)$. For $n = 2$ and $l = 1$, $j_{\max} = 0$ and $v = c_0$. The polynomials v are called associated Laguerre polynomials. Normalising with $\int_0^\infty R^2 r^2 dr = 1$ gives

$$\begin{aligned} R_{10} &= \frac{2}{a_0^{3/2}} e^{-r/a_0}, & R_{20} &= \frac{1}{\sqrt{2} a_0^{3/2}} \left(1 - \frac{r}{2a_0}\right) e^{-r/2a_0}, \\ R_{21} &= \frac{1}{\sqrt{24} a_0^{3/2}} \frac{r}{a_0} e^{-r/2a_0}. \end{aligned} \quad (5.28)$$

The normalisation integrals are all of the form

$$\int_0^\infty x^N e^{-x} dx = N!, \quad (5.29)$$

which is proved by integrating by parts, since each integration by parts, differentiating the power and integrating the exponential,

turns the integral with x^N into N times the integral with x^{N-1} , and the integral with x^0 is one. As an example, for the ground state the substitution $x = 2r/a_0$ gives

$$\int_0^\infty R_{10}^2 r^2 dr = \frac{4}{a_0^3} \left(\frac{a_0}{2}\right)^3 \int_0^\infty x^2 e^{-x} dx = \frac{1}{2} \cdot 2! = 1. \quad (5.30)$$

The ground state wave function is therefore

$$\psi_{100} = R_{10} Y_0^0 = \frac{1}{\sqrt{\pi a_0^3}} e^{-r/a_0}. \quad (5.31)$$

Where the Electron Is Found

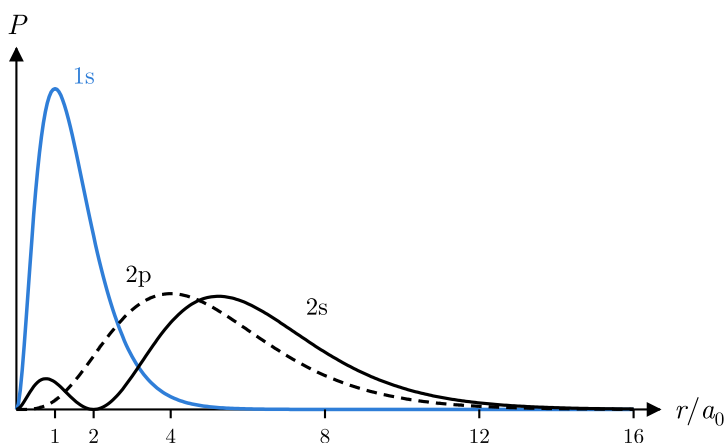


Figure 34. The radial probability densities $P(r) = r^2 |R_{nl}|^2$ of the 1s (accent), 2s (solid), and 2p (dashed) states of hydrogen, with r in units of the Bohr radius a_0 . The 1s density peaks at a_0 , the 2p density at $4a_0$, and the 2s density has a node at $2a_0$.

The probability of finding the electron at a distance between r and $r + dr$ from the proton, in any direction, is found by integrating $|\psi|^2$ over a thin spherical shell. Since the spherical harmonics are normalised, the result is

$$P(r) dr = |R_{nl}(r)|^2 r^2 dr, \quad (5.32)$$

and $P(r)$ is called the radial probability density. It is drawn for the

three states above in Figure 34. For the ground state,

$$P(r) = \frac{4r^2}{a_0^3} e^{-2r/a_0}. \quad (5.33)$$

Setting the derivative to zero gives $2r - 2r^2/a_0 = 0$, so the most probable distance is $r = a_0$, the radius of the first Bohr orbit. The mean distance is larger, since the distribution has a long tail:

$$\langle r \rangle = \int_0^\infty r P(r) dr = \frac{4}{a_0^3} \left(\frac{a_0}{2} \right)^4 3! = \frac{3}{2} a_0. \quad (5.34)$$

Unlike the electron of the Bohr model, the electron here has no definite orbit. The probability density $|\psi_{100}|^2$ per unit volume is largest at the proton itself, while the probability per unit distance, $P(r)$, is largest at a_0 , because the volume of a shell grows as r^2 . The 2s state has a node at $r = 2a_0$, where R_{20} changes sign, and in the 2p state, whose radial function has no node, the electron is most likely to be found at $r = 4a_0$, which is the radius of the second Bohr orbit, as you will show in Problem 2.

Problems

§5.1 The Schrödinger Equation in Three Dimensions

1. For a function $f(r)$ that depends only on the distance from the origin, use $\partial r / \partial x = x/r$ and the chain rule to show that $\nabla^2 f = d^2 f / dr^2 + (2/r) df / dr$, and check that this agrees with the radial part of the Laplacian quoted in the text.
2. Find the four lowest energies of a particle in a cubic box of side w , with their degeneracies.
3. The first excited level of the isotropic oscillator has the three states $x e^{-\beta r^2/2}$, $y e^{-\beta r^2/2}$, and $z e^{-\beta r^2/2}$, where $\beta = m\omega/\hbar$, apart from normalisation. Show that suitable combinations of them are eigenfunctions of $\hat{L}^2$ with $l = 1$ and of $\hat{L}_z$ with $m_l = 1, 0, -1$.
4. Show that $U(r) = \sin(kr)/(kr) - \cos kr$ satisfies the radial equation for the spherical well with $l = 1$, and that it vanishes at the origin. Show that the energies are fixed by $\tan(kr_w) = kr_w$, an equation that is usually solved numerically.

5. Find the probability that a particle in the ground state of the infinite spherical well is found within a distance $r_w/2$ of the centre.
6. The molecules H^{35}Cl and D^{35}Cl have the same separation of the atoms, since the chemical bond depends hardly at all on the mass of the nucleus. Find the ratio of the spacings of the lines in their rotational spectra in terms of the masses of the atoms.

§5.2 The Hydrogen Atom

1. Show that R_{20} and R_{21} are normalised.
2. For the 2p state, find $\langle r \rangle$ and the most probable value of r .
3. In the ground state, find $\langle 1/r \rangle$, and hence show that the mean potential energy is $2E_1$ and the mean kinetic energy is $-E_1$.
4. Find the probability that the electron in the ground state is found closer to the proton than a_0 .
5. For $n = 3$ and $l = 2$, show that the radial function is proportional to $r^2 e^{-r/3a_0}$, and find the most probable value of r .
6. List the states with $n = 3$ by their quantum numbers, and check that there are nine. The electron also has spin one-half, with two possible values of m_s for each state. How many states does the level n have when spin is included?
7. Positronium is a bound state of an electron and a positron, which has the same mass as the electron and the opposite charge. Find its energy levels in terms of E_R .

Solutions

Chapter 1. Origins of Quantum Theory

§1.1 Quantum Physics at School

1. The energy is $E = hc/\lambda$ joules, and dividing by the number of joules in an electronvolt, which is e in coulombs times one volt, gives $hc/(e\lambda)$.
2. Einstein's equation gives $eV_1 = hf_1 - W$ and $eV_2 = hf_2 - W$. Subtracting the first equation from the second gives $h = e(V_2 - V_1)/(f_2 - f_1)$. Substituting this value of h back into the first equation gives $W = e(V_2f_1 - V_1f_2)/(f_2 - f_1)$.
3. The possible transitions are from level 3 to level 2, from 3 to 1, and from 2 to 1, with frequencies $(E_3 - E_2)/h$, $(E_3 - E_1)/h$, and $(E_2 - E_1)/h$. The second of these is the sum of the other two, since $E_3 - E_1 = (E_3 - E_2) + (E_2 - E_1)$.
4. For a slow particle $E_k = p^2/2m$, so $p = \sqrt{2mE_k}$ and $\lambda = h/p = h/\sqrt{2mE_k}$. At equal kinetic energy the wavelength is inversely proportional to the square root of the mass, so $\lambda_e/\lambda_p = \sqrt{m_p/m_e}$, where m_p is the mass of the proton. The electron has the longer wavelength.

§1.2 Thermal Radiation and the Photon

1. When $hf \gg k_B T$ the exponential is much larger than 1, so $e^{hf/k_B T} - 1 \approx e^{hf/k_B T}$, which gives the stated form. In terms of $\eta = hf/k_B T$ it is proportional to $\eta^3 e^{-\eta}$, whose derivative, by the product rule, is $(3\eta^2 - \eta^3)e^{-\eta}$. This vanishes at $\eta = 3$, so the maximum is at $f = 3k_B T/h$. The peak of Planck's law is at η_p , slightly below 3, since the approximation neglects the 1 in the denominator and so underestimates the spectrum most at small η .
2. With $f = c/\lambda$ we have $|df| = (c/\lambda^2)|d\lambda|$, so $u_\lambda(\lambda) = u(c/\lambda) c/\lambda^2 = (8\pi c^2/\lambda^2 c^3)(hc/\lambda)(c/\lambda^2)/(e^{hc/\lambda k_B T} - 1)$, which simplifies to the stated form. Let $\zeta = hc/\lambda k_B T$, so that u_λ is proportional to $\zeta^5/(e^\zeta - 1)$ at fixed temperature. By the quotient rule its derivative with respect to ζ vanishes when $5(e^\zeta - 1) = \zeta e^\zeta$, that is when $\zeta = 5(1 - e^{-\zeta})$, and since ζ and λ are inversely related, the maximum in λ is at $\lambda_{\max} = hc/(\zeta_p k_B T)$. The peak of u and the peak of u_λ fall at different wavelengths because the factor c/λ^2 that converts one spectrum into the other depends on λ , so it shifts the position of the maximum. The product $\lambda_{\max} T = hc/(\zeta_p k_B)$ is still a constant, which is the school form of Wien's law.
3. Write $E = mc^2 \sqrt{1 + p^2/m^2 c^2}$ and expand the square root by the binomial theorem. Since $p^2/m^2 c^2$ is small, keeping the first two terms of the expansion gives $\sqrt{1 + p^2/m^2 c^2} \approx 1 + p^2/2m^2 c^2$, and therefore $E \approx mc^2 + p^2/2m$.
4. With $\Theta = \pi$ the Compton formula gives $\lambda' = \lambda + 2\lambda_C$. The kinetic energy of the electron is the energy lost by the photon, $hc/\lambda - hc/\lambda' = hc(\lambda' - \lambda)/(\lambda\lambda')$, which is $2hc\lambda_C/[\lambda(\lambda + 2\lambda_C)]$.
5. The shift is at most $2\lambda_C$, and the Compton wavelength of the electron is some five orders of magnitude shorter than the wavelength of visible light, so the fractional change in wavelength is far too small to measure. For X-rays, whose wavelengths are not much longer than λ_C , the fractional change is appreciable. In addition, the electrons in matter are bound with energies comparable to the energy of a visible photon, so they cannot be treated as free.

§1.3 The Bohr Model and Matter Waves

1. From the quantisation condition, $v = n\hbar/(m_e r)$, and with $r = n^2 a_0$ this is $\hbar/(m_e n a_0)$. Substituting $a_0 = 4\pi\epsilon_0 \hbar^2/(m_e e^2)$ gives $v_n = e^2/(4\pi\epsilon_0 \hbar n)$. The ratio $v_1/c = e^2/(4\pi\epsilon_0 \hbar c)$ has no dimensions, and its value is about $1/137$; it is called the fine-structure constant. The speed of the electron is therefore a small fraction of the speed of light, and relativistic corrections to the energy, which are of order $(v/c)^2$ relative to the kinetic energy, are small.
2. The electron is at a distance $Mr/(M + m_e)$ from the centre of mass and the nucleus at $m_e r/(M + m_e)$. The total angular momentum is $m_e \omega [Mr/(M + m_e)]^2 + M \omega [m_e r/(M + m_e)]^2$, which simplifies to $m_e M \omega r^2/(M + m_e) = m_r \omega r^2$. The equation of motion of the electron, $m_e \omega^2 Mr/(M + m_e) = e^2/(4\pi\epsilon_0 r^2)$, reads $m_r \omega^2 r = e^2/(4\pi\epsilon_0 r^2)$. The kinetic energy is $\frac{1}{2} m_r \omega^2 r^2$, by a similar calculation to the one for the angular momentum. Every equation of the text is therefore reproduced with m_e replaced by m_r and v by ωr , and the energies become $-(m_r/m_e)E_R/n^2$.
3. The longest wavelength has $n_i = 3$, giving $1/\lambda = \mathcal{R}_H(1/4 - 1/9) = 5\mathcal{R}_H/36$, and the shortest is the limit $n_i \rightarrow \infty$, giving $1/\lambda = \mathcal{R}_H/4$. The ratio of the longest to the shortest wavelength is $(1/4)/(5/36) = 9/5$.
4. The energy is $E = p^2/2m + \frac{1}{2}m\omega^2 x^2$, which can be rewritten as $x^2/(2E/m\omega^2) + p^2/(2mE) = 1$, the equation of an ellipse with the stated semi-axes. The integral $\oint p dx$ over one cycle is the area enclosed by this ellipse, since the upper half, where $p > 0$, is traversed with x increasing and contributes the area between it and the x -axis, and the lower half, where $p < 0$, is traversed with x decreasing and contributes the same area again. The area of an ellipse is π times the product of the semi-axes, $\pi\sqrt{2E/m\omega^2}\sqrt{2mE} = 2\pi E/\omega$. Setting this equal to $n\hbar = 2\pi n\hbar$ gives $E = n\hbar\omega$.
5. The Bragg condition in first order gives $\lambda = 2D \sin \vartheta$. The electrons have $p = \sqrt{2m_e eV}$, so $\lambda = h/\sqrt{2m_e eV}$. Equating the two expressions for λ and solving for V gives $V = \hbar^2/(8m_e e D^2 \sin^2 \vartheta)$.

Chapter 2. Wave Mechanics in One Dimension

§2.1 The Schrödinger Equation and the Wave Function

1. Each term on either side of the Schrödinger equation is proportional to Ψ or to one of its derivatives, and the derivative of a sum is the sum of the derivatives, so substituting $c_1\Psi_1 + c_2\Psi_2$ gives c_1 times the equation for Ψ_1 plus c_2 times the equation for Ψ_2 , which holds. For $\Psi = \cos(kx - \omega t)$, the left-hand side is $i\hbar\omega \sin(kx - \omega t)$ and the right-hand side is $(\hbar^2 k^2 / 2m) \cos(kx - \omega t)$. The two sides are different functions of x and t , and they cannot be equal everywhere unless both coefficients vanish.
2. We have $\partial\Psi/\partial x = ik(Ae^{ikx} - Be^{-ikx})e^{-i\omega t}$, and multiplying by $\Psi^* = (A^*e^{-ikx} + B^*e^{ikx})e^{i\omega t}$ gives $ik(|A|^2 - |B|^2) + ik(AB^*e^{2ikx} - A^*Be^{-2ikx})$. The second bracket is the difference between a complex number and its conjugate, so ik times it is real and contributes nothing to the imaginary part. Therefore $j = (\hbar k/m)(|A|^2 - |B|^2)$.
3. Since $|e^{ik_0x}| = 1$, $|\Psi|^2 = \psi^2$, so Ψ is normalised. By the product rule, $-i\hbar d\Psi/dx = (-i\hbar\psi' + \hbar k_0\psi)e^{ik_0x}$. Multiplying by $\Psi^* = \psi e^{-ik_0x}$ and integrating gives $-i\hbar \int \psi\psi' dx + \hbar k_0 \int \psi^2 dx$. The first integral vanishes, as in the example of this section, and the second is one, so $\langle p \rangle = \hbar k_0$.
4. By definition $\langle p^2 \rangle = -\hbar^2 \int \Psi^* \partial^2\Psi/\partial x^2 dx$. Integrating by parts, with the boundary term vanishing at large $|x|$, gives $\hbar^2 \int (\partial\Psi^*/\partial x)(\partial\Psi/\partial x) dx = \hbar^2 \int |\partial\Psi/\partial x|^2 dx$, which is an integral of a quantity that is never negative.

§2.2 Stationary States and the Infinite Square Well

1. For $E < 0$ let $\kappa = \sqrt{-2mE}/\hbar$, so that the equation becomes $\psi'' = \kappa^2\psi$, whose general solution is $Ae^{\kappa x} + Be^{-\kappa x}$. The condition $\psi(0) = 0$ gives $B = -A$, and then $\psi(w) = A(e^{\kappa w} - e^{-\kappa w})$ vanishes only if $A = 0$. For $E = 0$ the equation is $\psi'' = 0$, with solution $A+Bx$, and vanishing at both walls requires $A = B = 0$.
2. The density ψ_n^2 is symmetric about $x = w/2$, so $\langle x \rangle = w/2$. For

the second moment,

$$\begin{aligned}\langle x^2 \rangle &= (2/w) \int_0^w x^2 \sin^2(n\pi x/w) dx \\ &= (1/w) \int_0^w x^2 [1 - \cos(2n\pi x/w)] dx.\end{aligned}\tag{S.1}$$

The first part gives $w^2/3$. Integrating the second part by parts twice, the boundary terms give $-w^2/(2n^2\pi^2)$ and the remaining integral vanishes, which gives the stated result. Therefore $\sigma_x^2 = w^2(\frac{1}{12} - \frac{1}{2n^2\pi^2})$. As n grows, σ_x approaches $w/\sqrt{12}$, which is the standard deviation of a particle equally likely to be anywhere in the well, as a classical particle bouncing between the walls would be.

3. Since $E_n \geq E_1$ for every n and $|c_n|^2 \geq 0$, we have $\sum_n |c_n|^2 E_n \geq \sum_n |c_n|^2 E_1 = E_1$.
4. Normalisation requires $3|A|^2 = 1$, so $A = 1/\sqrt{3}$. Each of the energies E_1 , $E_2 = 4E_1$, and $E_3 = 9E_1$ is found with probability $\frac{1}{3}$, and $\langle H \rangle = (1 + 4 + 9)E_1/3 = 14E_1/3$.
5. The function ψ_n is real and vanishes at both walls, so $\int \psi_n \psi'_n dx = \frac{1}{2}[\psi_n^2]_0^w = 0$, and $\langle p \rangle = 0$. Since $\psi''_n = -k_n^2 \psi_n$, we have $\langle p^2 \rangle = -\hbar^2 \int \psi_n \psi''_n dx = \hbar^2 k_n^2 = 2mE_n$. Inside the well the energy is all kinetic, and the mean of $p^2/2m$ is E_n . The particle is equally likely to be moving in either direction, which is why $\langle p \rangle = 0$ while $\langle p^2 \rangle$ is not.
6. Differentiating the result for $\langle x \rangle$ gives $\langle p \rangle = m d\langle x \rangle/dt = (16mw\omega_{21}/9\pi^2) \sin \omega_{21}t$, and substituting $\omega_{21} = 3\pi^2\hbar/2mw^2$ gives $\langle p \rangle = (8\hbar/3w) \sin \omega_{21}t$.

§2.3 Bound States in Wells of Finite Depth

1. An odd solution has the form $C \sin kx$ inside the well and $\pm Ge^{-\kappa|x|}$ outside, with the plus sign for $x > w/2$. Matching at $x = w/2$ gives $C \sin z = Ge^{-\kappa w/2}$ and $kC \cos z = -\kappa Ge^{-\kappa w/2}$, and dividing the second equation by the first gives $k \cot z = -\kappa$, which becomes the stated condition. The right-hand side is positive, so a solution needs $\cot z < 0$, and the first interval in which this happens is $\pi/2 < z < \pi$. Since the right-hand side is zero

- at z_0 , where it ends, a crossing requires $z_0 > \pi/2$.
2. For large z_0 the right-hand side is large, so the crossings lie where $-\cot z$ becomes large and positive, which is just below $z = \pi, 2\pi, 3\pi, \dots$. There $kw = n\pi$ with n even, and $E + V_0 = \hbar^2 k^2/2m = n^2 \pi^2 \hbar^2/2mw^2$.
 3. Take $c > 0$ and substitute $u = cx$ in $\int f(x) \delta(cx) dx$, which gives $(1/c) \int f(u/c) \delta(u) du = f(0)/c$. This is the result of integrating f against $\delta(x)/c$. For $c < 0$ the substitution reverses the limits of integration, which introduces an extra minus sign, so the result is $f(0)/|c|$ for either sign of c .
 4. The probability is $2\kappa \int_0^{1/\kappa} e^{-2\kappa x} dx = 1 - e^{-2}$. From Section 2.1, $\sigma_x = 1/(\sqrt{2}\kappa) = \hbar^2/(\sqrt{2}m\alpha)$.
 5. Between the wells the potential vanishes and $E < 0$, so the solutions are combinations of $e^{\pm\kappa x}$, and the even combination is $\cosh \kappa x$. Outside, normalisability requires $e^{-\kappa|x|}$. Continuity at $x = w/2$ gives $C \cosh(\kappa w/2) = Ge^{-\kappa w/2}$. Integrating across $x = w/2$, as for the single well, gives $\psi'(w/2 + \epsilon) - \psi'(w/2 - \epsilon) = -(2m\alpha/\hbar^2)\psi(w/2)$, which reads $-\kappa Ge^{-\kappa w/2} - \kappa C \sinh(\kappa w/2) = -(2m\alpha/\hbar^2)C \cosh(\kappa w/2)$. Using the continuity condition to replace $Ge^{-\kappa w/2}$ and dividing by $C \cosh(\kappa w/2)$ gives $\kappa[1 + \tanh(\kappa w/2)] = 2m\alpha/\hbar^2$. When κw is large, $\tanh(\kappa w/2) \rightarrow 1$, so $\kappa \rightarrow m\alpha/\hbar^2$, which is the single-well value.

§2.4 The Free Particle and Wave Packets

1. Factoring out $e^{i(k_0 x - \omega_0 t)}$ leaves $e^{i(\Delta k x - \Delta \omega t)} + e^{-i(\Delta k x - \Delta \omega t)}$, which is $2 \cos(\Delta k x - \Delta \omega t)$ by Euler's formula. The cosine varies slowly in space when Δk is small and forms an envelope for the rapidly varying exponential. A point of fixed phase of the envelope has $\Delta k x - \Delta \omega t$ constant, and therefore moves at the speed $\Delta \omega/\Delta k$.
2. Splitting the integral at the origin gives

$$\begin{aligned}
 g(k) &= \sqrt{\kappa/2\pi} \left[\int_{-\infty}^0 e^{(\kappa - ik)x} dx + \int_0^{\infty} e^{-(\kappa + ik)x} dx \right] \\
 &= \sqrt{\kappa/2\pi} [1/(\kappa - ik) + 1/(\kappa + ik)] .
 \end{aligned} \tag{S.2}$$

The two fractions add to $2\kappa/(\kappa^2 + k^2)$, which gives the stated result.

3. From the result for $|g(k)|^2$, the integral is $\sqrt{2/\pi} \sigma$ times the Gaussian integral with $\beta = 2\sigma^2$, which is $\sqrt{\pi/2\sigma^2}$, so the product is one.
4. The width has doubled when $1 + (\hbar t/2m\sigma^2)^2 = 4$, that is at $t = 2\sqrt{3} m\sigma^2/\hbar$.
5. The density at time t is a normal distribution centred on $v_0 t$, so $\langle x \rangle = v_0 t$. Therefore $m d\langle x \rangle/dt = mv_0 = \hbar k_0$. The distribution $|g(k)|^2$ is a normal distribution centred on k_0 , so the mean momentum is also $\hbar k_0$.

§2.5 Potential Steps and Tunnelling

1. The sum is $[(k - k')^2 + 4kk']/(k + k')^2 = (k + k')^2/(k + k')^2 = 1$.
2. The calculation for the upward step with $E > V_0$ applies unchanged, with k' replaced by $k'' = \sqrt{2m(E + V_1)}/\hbar$, since only the value of the wave number on the right enters. Since $k'' \neq k$, the reflection probability $[(k - k'')/(k + k'')]^2$ is not zero.
3. Let $\psi = Ae^{ikx} + Be^{-ikx}$ for $x < 0$ and Fe^{ikx} for $x > 0$. Continuity gives $A + B = F$. Integrating the time-independent equation across the origin, as for the delta-function well of Section 2.3, gives $\psi'(0^+) - \psi'(0^-) = (2m\alpha/\hbar^2)\psi(0)$, where 0^+ and 0^- are points just to the right and just to the left of the origin. This reads $ikF - ik(A - B) = (2m\alpha/\hbar^2)F$. Using $B = F - A$ gives $ik(2F - 2A) = (2m\alpha/\hbar^2)F$, so $F/A = 1/[1 + im\alpha/(\hbar^2 k)]$. Therefore $P_T = |F/A|^2 = [1 + m^2\alpha^2/(\hbar^4 k^2)]^{-1}$, and with $\hbar^2 k^2 = 2mE$ this is the stated result.
4. For $E > V_0$ the solutions inside the barrier are $e^{\pm ik'x}$, which are the exponentials of the text with κ replaced by ik' . Making this replacement throughout the derivation, $\sinh \kappa w$ becomes $i \sin k'w$ and κ^2 becomes $-k'^2$, so the bracket in P_T becomes $1 + (k^2 - k'^2)^2 \sin^2 k'w/(4k^2 k'^2)$. With $k^2 - k'^2 = 2mV_0/\hbar^2$ and $k^2 k'^2 = 4m^2 E(E - V_0)/\hbar^4$ this is the stated result. It equals one when $\sin k'w = 0$, that is when $k'w = n\pi$, or $E = V_0 + n^2 \pi^2 \hbar^2/2mw^2$ with $n = 1, 2, 3, \dots$. At these energies

the barrier holds a whole number of half-wavelengths, and the waves reflected from its two edges cancel.

5. With $V_0 w$ fixed and $w \rightarrow 0$, $V_0 \rightarrow \infty$, so $\kappa \rightarrow \infty$ while $\kappa^2 w$ stays finite and $\kappa w = \sqrt{\kappa^2 w \cdot w} \rightarrow 0$. Then $\sinh \kappa w \approx \kappa w$, and $V_0^2 \kappa^2 w^2 / [4E(V_0 - E)] \approx V_0^2 w^2 (2mV_0 / \hbar^2) / (4EV_0) = mV_0^2 w^2 / (2\hbar^2 E)$. This is the result of Problem 3 with $\alpha = V_0 w$, which is the area under the barrier, as expected for a barrier that is narrow and tall.

§2.6 The Harmonic Oscillator

1. Adding and subtracting the definitions of $\hat{a}_+$ and $\hat{a}_-$ gives $\hat{a}_+ + \hat{a}_- = 2m\omega\hat{x}/\sqrt{2\hbar m\omega}$ and $\hat{a}_+ - \hat{a}_- = -2i\hat{p}/\sqrt{2\hbar m\omega}$, which rearrange to the stated forms. Acting on ψ_n , each ladder operator gives a multiple of $\psi_{n\pm 1}$, which is orthogonal to ψ_n , so $\langle x \rangle = \langle p \rangle = 0$. In $\hat{x}^2 = (\hbar/2m\omega)(\hat{a}_+^2 + \hat{a}_+\hat{a}_- + \hat{a}_-\hat{a}_+ + \hat{a}_-^2)$ the first and last terms change n by two and contribute nothing, while $\hat{a}_+\hat{a}_-\psi_n = n\psi_n$ and $\hat{a}_-\hat{a}_+\psi_n = (n+1)\psi_n$. Therefore $\langle x^2 \rangle = (\hbar/2m\omega)(2n+1)$. Similarly $\hat{p}^2 = -(\hbar m\omega/2)(\hat{a}_+^2 - \hat{a}_+\hat{a}_- - \hat{a}_-\hat{a}_+ + \hat{a}_-^2)$, which gives $\langle p^2 \rangle = (\hbar m\omega/2)(2n+1)$. The mean kinetic energy is $\langle p^2 \rangle / 2m = \frac{1}{2}(n + \frac{1}{2})\hbar\omega$, and the mean potential energy $\frac{1}{2}m\omega^2\langle x^2 \rangle$ has the same value. Since the means vanish, $\sigma_x\sigma_p = \sqrt{\langle x^2 \rangle \langle p^2 \rangle} = (n + \frac{1}{2})\hbar$.
2. With $\psi_1 = \sqrt{2m\omega/\hbar} x \psi_0$, the raising operator gives $\hat{a}_+\psi_1 = (1/\sqrt{2\hbar m\omega})\sqrt{2m\omega/\hbar}(-\hbar d/dx + m\omega x)(x\psi_0)$. By the product rule, $-\hbar d(x\psi_0)/dx = -\hbar\psi_0 + m\omega x^2\psi_0$, so $\hat{a}_+\psi_1 = (1/\hbar)(2m\omega x^2 - \hbar)\psi_0 = (2\xi^2 - 1)\psi_0$. Dividing by $\sqrt{2}$ gives $\psi_2 = (2\xi^2 - 1)\psi_0/\sqrt{2}$. The Hermite formula gives $\psi_2 = (4\xi^2 - 2)\psi_0/\sqrt{8}$, which is the same, and for $n = 1$ it gives $2\xi\psi_0/\sqrt{2} = \sqrt{2}\xi\psi_0 = \sqrt{2m\omega/\hbar} x \psi_0$, as found in the text.
3. The turning points of the ground state are where $\frac{1}{2}m\omega^2 x^2 = \frac{1}{2}\hbar\omega$, that is at $\xi = \pm 1$. The probability of finding the particle beyond them is $2 \int_1^\infty |\psi_0|^2 dx$, and substituting $x = \sqrt{\hbar/m\omega} \xi$ turns this into $(2/\sqrt{\pi}) \int_1^\infty e^{-\xi^2} d\xi$, which is the stated form with y in place of ξ .
4. The wave function must vanish at the wall, and for $x > 0$ it must satisfy the oscillator equation. The odd states of the full

oscillator satisfy both conditions, and the even states do not vanish at the origin. The allowed states are therefore those with odd n , and the energies are $(2j + \frac{3}{2})\hbar\omega$ with $j = 0, 1, 2, \dots$

5. For $n = 0$ we have $K = 1$, the even series has $a_2 = (1 - 1)a_0/2 = 0$ and so ends after its first term, with $h = a_0$, and the odd series is set to zero. Then $\psi_0 = a_0 e^{-\xi^2/2}$, which is the ladder-operator result with $a_0 = (m\omega/\pi\hbar)^{1/4}$.
6. Write $\hat{a}_\pm = (\mp\hbar d/dx + m\omega x)/\sqrt{2\hbar m\omega}$. The term $m\omega x$ is real and can be moved from g to f unchanged. For the derivative term, integration by parts gives $\int f^*(\mp\hbar g') dx = \pm\hbar \int (f^*)' g dx = \int (\pm\hbar f')^* g dx$, since the boundary term vanishes, and $\pm\hbar f'$ is the derivative term of $\hat{a}_\mp$ acting on f . Adding the two terms gives the stated property.

Chapter 3. The Formalism of Quantum Mechanics

§3.1 Operators, Commutators, and the Uncertainty Relations

1. The third rule gives $[\hat{x}, \hat{p}^2] = [\hat{x}, \hat{p}]\hat{p} + \hat{p}[\hat{x}, \hat{p}] = 2i\hbar\hat{p}$. For the second result, the statement holds for $n = 1$. If it holds for n , then $[\hat{x}^{n+1}, \hat{p}] = \hat{x}[\hat{x}^n, \hat{p}] + [\hat{x}, \hat{p}]\hat{x}^n = ni\hbar\hat{x}^n + i\hbar\hat{x}^n = (n+1)i\hbar\hat{x}^n$, using the third rule in the form $[\hat{B}\hat{C}, \hat{A}] = \hat{B}[\hat{C}, \hat{A}] + [\hat{B}, \hat{A}]\hat{C}$, so it holds for every n by induction.
2. Moving $\hat{A}\hat{B}$ across in $\int f^*(\hat{A}\hat{B}g) dx$ moves $\hat{A}$ first and then $\hat{B}$, giving $\int (\hat{B}\hat{A}f)^* g dx$. Let us call the operator that appears on the right-hand side when an operator is moved across in this way the hermitian partner of that operator, so that an operator is hermitian when its partner is the same operator. The hermitian partner of $\hat{A}\hat{B}$ is therefore $\hat{B}\hat{A}$, and $\hat{A}\hat{B}$ is hermitian only if $\hat{A}\hat{B} = \hat{B}\hat{A}$. The partner of $\hat{A}\hat{B} + \hat{B}\hat{A}$ is $\hat{B}\hat{A} + \hat{A}\hat{B}$, which is the same operator. The partner of $i(\hat{A}\hat{B} - \hat{B}\hat{A})$ is $-i(\hat{B}\hat{A} - \hat{A}\hat{B})$, since the factor i is conjugated when it is moved across, and this is again the same operator.
3. The energy is E_1 or E_2 , each with probability one half, so $\langle H \rangle = (E_1 + E_2)/2$ and $\langle H^2 \rangle = (E_1^2 + E_2^2)/2$. The variance $\langle H^2 \rangle - \langle H \rangle^2$ is then $\sigma_H^2 = (E_2 - E_1)^2/4$. The product of σ_H and the time

- $\pi\hbar/(E_2 - E_1)$ is $\pi\hbar/2$, which is larger than $\hbar/2$.
- The particle is certainly found between 0 and w , so every value of x lies within $w/2$ of the centre, and the mean square deviation from the mean is at most the mean square deviation from the centre, since $\langle (x - c)^2 \rangle = \sigma_x^2 + (\langle x \rangle - c)^2$ for any constant c , and the mean square deviation from the centre is at most $(w/2)^2$. The uncertainty relation then gives $\sigma_p \geq \hbar/w$. Inside the well the energy is all kinetic, and $\langle p^2 \rangle \geq \sigma_p^2$, so $\langle H \rangle \geq \hbar^2/2mw^2$. This is smaller than $E_1 = \pi^2\hbar^2/2mw^2$ by a factor π^2 , as it must be, since the bound holds for every state.
 - From the two problems, $\sigma_p^2 = \langle p^2 \rangle = n^2\pi^2\hbar^2/w^2$ and $\sigma_x^2 = w^2(\frac{1}{12} - \frac{1}{2n^2\pi^2})$, so $\sigma_x\sigma_p = \hbar\sqrt{n^2\pi^2/12 - 1/2}$. The expression under the square root increases with n , and for $n = 1$ it is $\pi^2/12 - 1/2$, which is greater than $1/4$ since $\pi^2 > 9$. The product is therefore greater than $\hbar/2$ for every n .
 - Differentiating the first equation of Ehrenfest's theorem and substituting the second gives $m d^2\langle x \rangle/dt^2 = -\langle dV/dx \rangle$. For the oscillator $dV/dx = m\omega^2x$, whose expectation value is $m\omega^2\langle x \rangle$, and this gives the stated equation, which is the equation of simple harmonic motion with angular frequency ω .

Chapter 4. Symmetry, Angular Momentum, and Spin

§4.1 Orbital Angular Momentum

- Using $\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x$, the only non-zero commutator in $[\hat{L}_z, \hat{x}]$ comes from the second product, $-[\hat{y}\hat{p}_x, \hat{x}] = -\hat{y}[\hat{p}_x, \hat{x}] = i\hbar\hat{y}$. Similarly $[\hat{L}_z, \hat{y}] = \hat{x}[\hat{p}_y, \hat{y}] = -i\hbar\hat{x}$, and $[\hat{L}_z, \hat{p}_x] = [\hat{x}, \hat{p}_x]\hat{p}_y = i\hbar\hat{p}_y$.
- The operators $\hat{L}_x$ and $\hat{L}_y$ are hermitian, and moving $\hat{L}_x + i\hat{L}_y$ across conjugates the factor i , which gives $\hat{L}_x - i\hat{L}_y = \hat{L}_-$. Let $\hat{L}_+f_{l,m_l} = cf_{l,m_l+1}$. Then $|c|^2 = \int (\hat{L}_+f)^*(\hat{L}_+f) dV = \int f^*\hat{L}_-\hat{L}_+f dV = \hbar^2[l(l+1) - m_l^2 - m_l]$, which gives the stated result. With $\hat{L}_+\hat{L}_- = \hat{L}^2 - \hat{L}_z^2 + \hbar\hat{L}_z$ the same argument gives $\hat{L}_-f_{l,m_l} = \hbar\sqrt{l(l+1) - m_l(m_l-1)}f_{l,m_l-1}$.
- We have $\hat{L}_xz = -i\hbar(y\partial z/\partial z - z\partial z/\partial y) = -i\hbar y$ and $\hat{L}_yz =$

$-i\hbar(z\partial z/\partial x - x\partial z/\partial z) = i\hbar x$, so $\hat{L}_+z = -i\hbar y + i(i\hbar x) = -\hbar(x + iy)$. In spherical coordinates $z = r\cos\theta$ does not depend on ϕ , so $\hat{L}_+z = \hbar e^{i\phi}\partial(r\cos\theta)/\partial\theta = -\hbar r\sin\theta e^{i\phi}$, which is $-\hbar(x + iy)$.

4. We have $\int |Y_1^0|^2 d\Omega = (3/4\pi) \cdot 2\pi \int_0^\pi \cos^2\theta \sin\theta d\theta = (3/2)(2/3) = 1$, using the substitution $u = \cos\theta$. The product $Y_1^{0*}Y_1^1$ contains $e^{i\phi}$, whose integral over $0 \leq \phi < 2\pi$ is zero. For Y_2^2 we need $\int_0^\pi \sin^5\theta d\theta = \int_{-1}^1 (1-u^2)^2 du = 16/15$, so $|C_2|^2 \cdot 2\pi \cdot 16/15 = 1$ and $C_2 = \sqrt{15/32\pi}$.
5. In spherical coordinates $x + iy = r\sin\theta e^{i\phi}$, which is r times a multiple of Y_1^1 . The function is therefore a function of r multiplied by Y_1^1 , and since $\hat{L}_z$ and $\hat{L}^2$ act only on the angles, it is an eigenfunction of both, with $m_l = 1$ and $l = 1$.
6. By Euler's formula, $\sin\theta \cos\phi = \frac{1}{2}\sin\theta(e^{i\phi} + e^{-i\phi}) = \sqrt{2\pi/3}(Y_1^{-1} - Y_1^1)$. The rotation operator multiplies $Y_1^{\pm 1}$ by $e^{\mp i\phi_0}$, which replaces $e^{\pm i\phi}$ by $e^{\pm i(\phi - \phi_0)}$, and reversing the first step gives $\sin\theta \cos(\phi - \phi_0)$.

§4.2 Spin One-Half and Two-Level Systems

1. Each Pauli matrix equals its conjugate transpose, which for σ_2 follows because interchanging the off-diagonal entries $-i$ and i and conjugating them returns the same matrix. Multiplying out gives $\sigma_1^2 = \sigma_2^2 = \sigma_3^2 = \mathbf{1}$. The product $\sigma_2\sigma_3$ has rows $(0 \ i)$ and $(i \ 0)$, which is $i\sigma_1$.
2. The characteristic equation of $\hat{S}_y$ is $q^2 - (\hbar/2)^2 = 0$, so $q = \pm \frac{1}{2}\hbar$. The first component equation is $-qa - i(\hbar/2)b = 0$, which gives $b = \pm ia$, so the normalised eigenvectors are $(1 \ \pm i)^T/\sqrt{2}$. The coefficient of χ_+ along the eigenvector with $q = +\frac{1}{2}\hbar$ is $(1 - i)(1 \ 0)^T/\sqrt{2} = 1/\sqrt{2}$, so the probability is one half.
3. The matrix $n_x\sigma_1 + n_y\sigma_2 + n_z\sigma_3$ has rows $(\cos\theta \ \sin\theta e^{-i\phi})$ and $(\sin\theta e^{i\phi} \ -\cos\theta)$. Acting on χ , the first row gives $\cos\theta \cos\frac{\theta}{2} + \sin\theta \sin\frac{\theta}{2}$, which is $\cos(\theta - \frac{\theta}{2}) = \cos\frac{\theta}{2}$, and the second gives $e^{i\phi}(\sin\theta \cos\frac{\theta}{2} - \cos\theta \sin\frac{\theta}{2}) = e^{i\phi} \sin\frac{\theta}{2}$, so χ is an eigenvector with eigenvalue one, and of $\mathbf{n} \cdot \hat{\mathbf{S}}$ with eigenvalue $+\frac{1}{2}\hbar$. For the expectation values, $\chi^\dagger\sigma_3\chi = \cos^2\frac{\theta}{2} - \sin^2\frac{\theta}{2} = \cos\theta$, and $\chi^\dagger\sigma_1\chi =$

- $\cos \frac{\theta}{2} \sin \frac{\theta}{2} (e^{i\phi} + e^{-i\phi}) = \sin \theta \cos \phi$, and a similar calculation gives $\chi^\dagger \sigma_2 \chi = \sin \theta \sin \phi$. Multiplying by $\frac{1}{2} \hbar$ gives $\frac{1}{2} \hbar \mathbf{n}$.
4. Here $\sigma_2 \chi$ has components $-ib e^{-i\omega_L t/2}$ and $ia e^{i\omega_L t/2}$, and multiplying by the row $\chi^\dagger$ gives $-iab e^{-i\omega_L t} + iab e^{i\omega_L t} = -2ab \sin \omega_L t$. With $2ab = \sin \theta_0$ this gives the stated $\langle S_y \rangle$. The energy is $\langle H \rangle = -\gamma B_0 \langle S_z \rangle$, which is constant.
5. Let $E_{12} = E_1 - E_2$ and suppose $E_{12} > 0$. The square root is $\frac{1}{2} E_{12} \sqrt{1 + 4\Delta^2/E_{12}^2}$, and the binomial expansion to first order gives $\frac{1}{2} E_{12} + \Delta^2/E_{12}$. The upper energy is then $\bar{E} + \frac{1}{2} E_{12} + \Delta^2/E_{12} = E_1 + \Delta^2/(E_1 - E_2)$, and the lower is $E_2 - \Delta^2/(E_1 - E_2)$. For $E_{12} < 0$ the same expressions follow with the roles of the two roots exchanged.
6. Here $\bar{E} = E_0$ and $E_1 - E_2 = 2\varepsilon_f$, so the energies are $E_0 \pm \sqrt{\varepsilon_f^2 + \Delta^2}$. When $\varepsilon_f \ll \Delta$ the binomial expansion gives $E_0 \pm (\Delta + \varepsilon_f^2/2\Delta)$, so the energies change in proportion to the square of the field. When $\varepsilon_f \gg \Delta$ they are approximately $E_0 \pm \varepsilon_f$, which change in proportion to the field itself, and the stationary states are then close to the two arrangements 1 and 2.

Chapter 5. Central Potentials and Canonical Three-Dimensional Systems

§5.1 The Schrödinger Equation in Three Dimensions

1. By the chain rule, $\partial f / \partial x = f'(r) x/r$. Differentiating again with the product rule gives $\partial^2 f / \partial x^2 = f'' x^2 / r^2 + f' (1/r - x^2 / r^3)$. Adding the corresponding results for y and z , and using $x^2 + y^2 + z^2 = r^2$, gives $f'' + f' (3/r - 1/r) = f'' + 2f'/r$. The radial part of the quoted Laplacian gives $(1/r^2)(r^2 f')' = (1/r^2)(2r f' + r^2 f'') = f'' + 2f'/r$, which is the same.
2. The energy is $(\pi^2 \hbar^2 / 2mw^2)$ times $n_1^2 + n_2^2 + n_3^2$. The four smallest values of this sum are 3, from $(1, 1, 1)$, with degeneracy one; 6, from the permutations of $(2, 1, 1)$, with degeneracy three; 9, from the permutations of $(2, 2, 1)$, with degeneracy three; and 11, from the permutations of $(3, 1, 1)$, with degeneracy three.
3. In spherical coordinates, $x \pm iy = r \sin \theta e^{\pm i\phi}$ and $z = r \cos \theta$,

which are r times multiples of $Y_1^{\pm 1}$ and Y_1^0 . The combinations $(x \pm iy)e^{-\beta r^2/2}$ and $ze^{-\beta r^2/2}$ are therefore functions of r multiplied by spherical harmonics with $l = 1$ and $m_l = \pm 1$ and 0 , and since $\hat{L}^2$ and $\hat{L}_z$ act only on the angles, they are eigenfunctions of both with those quantum numbers.

4. The radial equation with $l = 1$ and $V = 0$ reads $U'' - 2U/r^2 + k^2U = 0$. Let $X = kr$, so that the equation becomes $d^2U/dX^2 - 2U/X^2 + U = 0$, and $U = \sin X/X - \cos X$. Differentiating twice gives $d^2U/dX^2 = -\sin X/X - 2\cos X/X^2 + 2\sin X/X^3 + \cos X$, and adding $U - 2U/X^2$ gives zero. Near the origin, $\sin X/X \approx 1 - X^2/6$ and $\cos X \approx 1 - X^2/2$, so $U \approx X^2/3$, which vanishes at the origin. The condition $U(r_w) = 0$ gives $\sin(kr_w)/(kr_w) = \cos(kr_w)$, which can be rewritten as $\tan(kr_w) = kr_w$.
5. The probability is $\int_0^{r_w/2} |U|^2 dr = (2/r_w) \int_0^{r_w/2} \sin^2(\pi r/r_w) dr$. The function $\sin^2(\pi r/r_w)$ is symmetric about $r = r_w/2$, so the integral is half of the full normalisation integral, and the probability is one half. Although the probability density per unit volume is largest at the centre, the volume within $r_w/2$ is only one eighth of the whole.
6. The spacing is $\hbar^2/I$ with $I = m_r r_e^2$, so the ratio of the spacings for HCl and DCl is the inverse ratio of their reduced masses. With m_{Cl} the mass of the chlorine atom, m_H that of the hydrogen atom, and m_D that of the deuterium atom, the ratio of the spacing for HCl to that for DCl is $m_D(m_{Cl} + m_H)/[m_H(m_{Cl} + m_D)]$, which is a little less than two.

§5.2 The Hydrogen Atom

1. With $x = r/a_0$, the normalisation integral for R_{20} is $\frac{1}{2} \int_0^\infty (1 - x/2)^2 x^2 e^{-x} dx = \frac{1}{2} \int_0^\infty (x^2 - x^3 + x^4/4) e^{-x} dx = \frac{1}{2}(2 - 6 + 6) = 1$. For R_{21} it is $\frac{1}{24} \int_0^\infty x^4 e^{-x} dx = 24/24 = 1$.
2. With $x = r/a_0$, $\langle r \rangle = \int_0^\infty r^3 R_{21}^2 dr = (a_0/24) \int_0^\infty x^5 e^{-x} dx = 5a_0$. The radial probability density goes as $r^4 e^{-r/a_0}$, whose derivative, by the product rule, vanishes when $4r^3 = r^4/a_0$, that is at $r = 4a_0$.
3. We have $\langle 1/r \rangle = \int_0^\infty (1/r) P(r) dr = (4/a_0^3) \int_0^\infty r e^{-2r/a_0} dr =$

$(4/a_0^3)(a_0/2)^2 = 1/a_0$. The mean potential energy is $-e^2/(4\pi\epsilon_0 a_0)$. Since $E_R = \hbar^2/2m_e a_0^2$ and $a_0 = 4\pi\epsilon_0 \hbar^2/m_e e^2$, we have $e^2/(4\pi\epsilon_0 a_0) = 2E_R$, so the mean potential energy is $-2E_R = 2E_1$. The mean kinetic energy is the total minus the potential, $E_1 - 2E_1 = -E_1$.

4. The probability is $\int_0^{a_0} P(r) dr$. With $x = 2r/a_0$ it becomes $\frac{1}{2} \int_0^2 x^2 e^{-x} dx$. Integrating by parts twice gives $\int_0^2 x^2 e^{-x} dx = 2 - e^{-2}(4 + 4 + 2) = 2 - 10e^{-2}$, so the probability is $1 - 5e^{-2}$.
5. For $n = 3$ and $l = 2$ we have $j_{\max} = 0$, so v is a constant, and $U \propto \rho^3 e^{-\rho}$ with $\rho = r/3a_0$. Therefore $R = U/r \propto r^2 e^{-r/3a_0}$. The radial probability density goes as $r^6 e^{-2r/3a_0}$, whose derivative vanishes when $6r^5 = (2/3a_0)r^6$, that is at $r = 9a_0$, the radius of the third Bohr orbit.
6. For $n = 3$ the state with $l = 0$ has $m_l = 0$, the states with $l = 1$ have $m_l = -1, 0, 1$, and the states with $l = 2$ have $m_l = -2, -1, 0, 1, 2$, giving $1 + 3 + 5 = 9$ states. With spin each of these states splits into two, so the level n has $2n^2$ states.
7. The reduced mass is $m_e m_e / (m_e + m_e) = m_e/2$. Since E_R is proportional to the mass, the energies are $-E_R/2n^2$.